Disorders of CVS

Content outlines

- **Introduction**
  - **Anatomy & physiologic review of the heart , the vascular ,& hematologic system**
- **I. Disorders of the heart**
  - Arrhythmias/Dysrrhythmia/
  - Coronary heart diseases (Angina, MI, Arteriosclerosis, atherosclerosis)
  - Valvular heart diseases/Ischemic heart diseases/ (Mitral , Aortic & Tricuspid valve d/s)
  - Inflammatory conditions of the heart (RHD & IE (Infective Endocarditis))
  - Pericardial disorders (Pericarditis, Cardiac Tamponade)
  - Myocardial disorders (Myocarditis, Cardiomyopathy)
  - Heart Failure/CHF/ , Cardiac Arrest, Acute Pulmonary Edema, & Cor-Pulmonale
  - Congenital Heart Diseases (VSD, ASD, PDA, AS, PS, COA, Fallot's Tetralogy)

Content outlines

II. Vascular disorders

- **Arterial diseases** (HPN, Hypotension & shock, Aortic dissection, Buerger's disease, Raynaud's phenomenon)
- **Venous diseases** (DVT, chronic venous insufficiency, venous ulcers, varicose veins, PTE , Superior Vena Caval Syndrome(SVCS))
- **Lymphatic system diseases**

  (Non-hodgkin's/Hodgkin's/ lymphoma, Lymphadenitis, Lympangitis, Burkitt's lymphoma , myelofibrosis & Multiple myeloma)

III. Hematological disorders

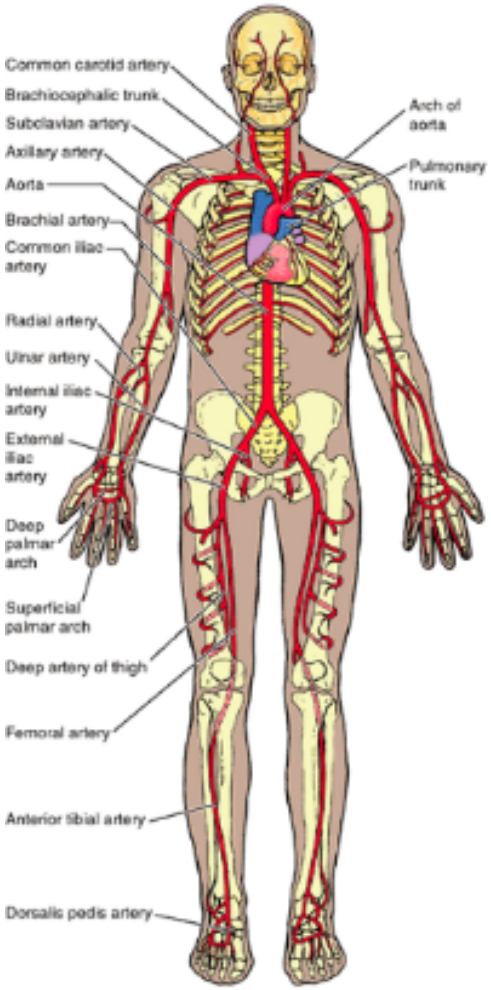
- **RBC disorders** :- Anemia, Polycythemia
- **WBC disorders** :- Leukopenia, leukocytosis,& Leukemia
- **Clotting disorders** :- ITP,TTP,HUS,vWD, Hemophilia A & B, DIC, Vit K deficiency
- **Blood transfusion**:- ABO & Rh system, types of transfusion, transfusion reactions

Anatomy & physiology of circulatory system

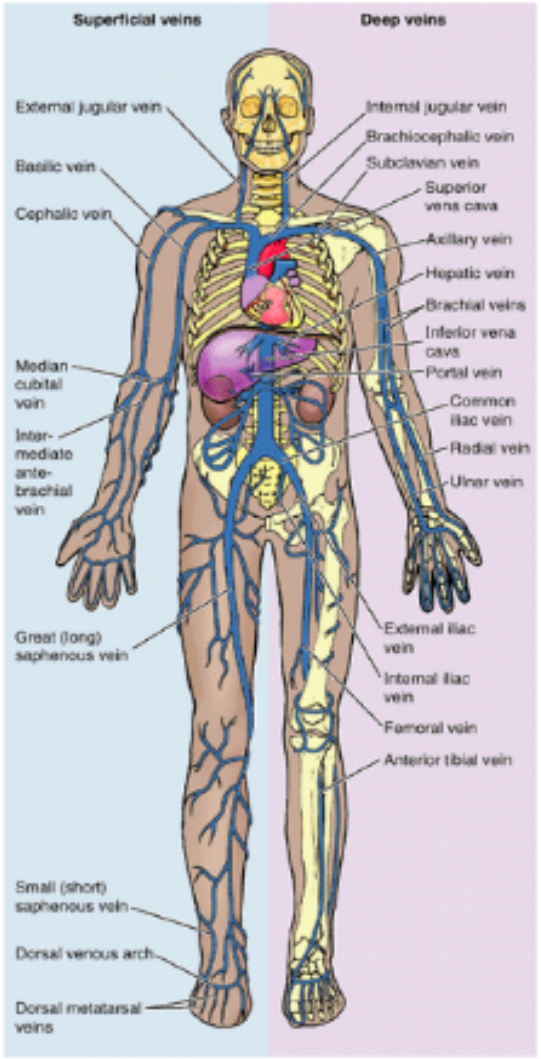
- ✓ Acts as a transport service for the cells
- ✓ Contains two fluids: blood & lymph (made up of two systems)

Major Components of the Circulatory System

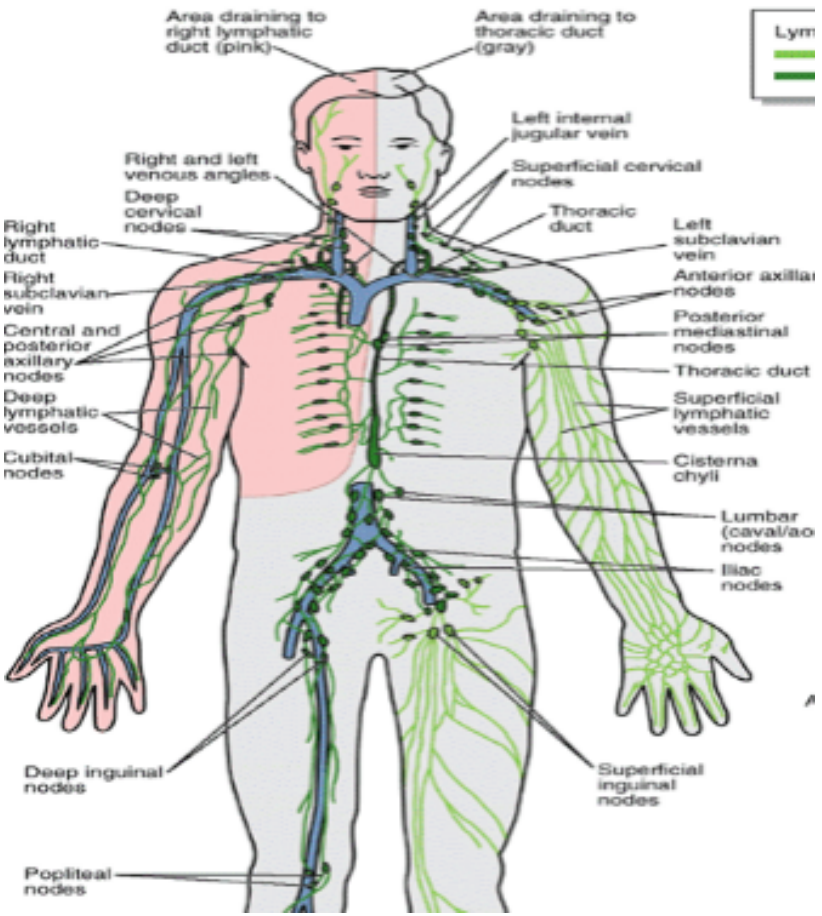
- **Cardiovascular *s* ystem**:- consist of heart and blood vessels (arteries, veins and capillaries) and blood
- **Lymphatic *s* ystem**:- consists of the lymph, lymph nodes and lymph vessels



Principal arteries



Principal veins



Functions of the Circulatory System

- **Transportation**
  - Respiratory: red blood cells carry oxygen. CO [redacted]
  - *Nutritive*
  - Excretory: capillaries in kidney
- **Regulation**- Circulating blood helps maintain homeostasis of all body fluids
    - Blood helps regulate pH through the use of buffers.
    - It also helps adjust body temperature through the heat absorbing and coolant properties of the water
    - The blood carries hormones and other regulatory molecules from their site of origin to distant target tissues.
- **Protection-** Blood can clot, which protects against its excessive loss from the cardiovascular system after an injury. In addition, its white blood cells protect against disease by carrying on

Common diagnostic procedures

A. Health history

- For the patient experiencing an acute MI, the nurse obtains the health history using a few specific questions about the onset and severity of chest discomfort, associated symptoms, current medications, and allergies.
- At the same time, the nurse observes the patient's general appearance and evaluates hemodynamic status(heart rate and rhythm, BP).
- Once the condition of the patient stabilizes, a more extensive history can be obtained.

Health history….

- Patients with cardiovascular disorders commonly have one or more of the following signs and symptoms:-
  - Chest pain or discomfort (angina pectoris, MI, valvular heart disease)
  - Shortness of breath or dyspnea (MI, left ventricular failure, HF)
  - Edema and weight gain (right ventricular failure, HF)
  - Palpitations (dysrhythmias resulting from myocardial ischemia, valvular heart disease, ventricular aneurysm, stress, electrolyte imbalance)
  - Fatigue (earliest symptom associated with several cardiovascular disorders)
  - Dizziness and syncope or loss of consciousness (postural hypotension, dysrhythmias, cerebrovascular disorders)

Health history….

- The following points should be remembered when assessing patients with cardiac symptoms:
  - Women are more likely to present with atypical symptoms of MI than are men
  - There is little correlation between the severity of the chest discomfort and the gravity of its cause.
  - Elderly people and those with diabetes may not have pain with angina or MI because of neuropathies. Fatigue and shortness of breath may be the predominant symptoms in these patients
  - There is poor correlation between the location of chest discomfort and its source
  - The patient may have more than one clinical condition occurring simultaneously
  - In a patient with a history of CAD, the chest discomfort should be assumed to be secondary to ischemia until proven otherwise

Assessment considerations

- General cardiac symptoms
  - Fatigue, Palpitations(racing heart,"pounding")
  - Chest pain, squeezing (MI, myocarditis, pericarditis)
  - **Shortness of breaths**
  - o *Exertional dyspnea* (dyspnea during activity & relieved by rest)
  - o *Orthopnea* (dyspnea in recumbent position from an increase in central venous volume which is consequence of redistribution of body fluids and blood from the periphery especially from the lower extremities)
  - o *PND* (Paroxysmal Nocturnal Dyspnea)– Occurs abruptly 1-5 hrs after the onset of sleep)
  - o *Trepopnea* (dyspnea only in lateral decubitus position)
  - o *Platypnea* (dyspnea in upright position)
  - Edema ,Weight gain ,Dizziness (pre-syncope) & syncope
  - Loss of consciousness

Risk factors for cardiac diseases

- Gerontologic considerations
  - Heart function is adequate at rest; limited ability to respond to stress and takes longer to return to baseline.
  - Decrease sensation of chest pain; tend to be under quantified or even absent.
- Gender considerations
  - Women: contraceptives = increase incidence
    - Smaller hearts and coronary arteries
    - Tend to present with “atypical symptom” of CAD
    - Men > Women ---- CAD
- Other considerations
  - Family hx of HPN,DM,, obesity, sudden death and cardiovascular disease, history of smoking, dietary habit (excess fat &simple sugars)
    - Increased threat; decreased symptoms !!

P/E

- **General appearance, V/S, Pulse pressure , HEENT**
- **Pulse deficit** (apical HR– peripheral HR)
- **CVS (examination of peripheral pulses)**

Arterial pulses (volume, strength, PR,rhythm,vessel quality,& pulse configuration)

Pulse configuration or contour

- **Pulse parvus** (small weak pulse Ex. Low stroke volume, hypovolumia, Lt VHF, mitral stenosis, restrictive pericarditis, narrow pulse pressure, increased peripheral resistance)
- **Bounding pulse** (wide pulse pressure, increased Lt ventricular stroke volume)
- **Bisferiens pulse** (two systolic peaks) – Ex. Aortic regurgitation, hypertrophic cardiomyopathy
- **Dicrotic pulse** (Two palpable pulses one in systole & one in diastole)
- **Pulsus alternans** (weak pulse alternate with strong pulse)
- **Pulsus paradoxus** (presence of decrease SBP greater than 10 mmhg during inspiration,& peripheral pulse may disappear during inspiration, EX. Cardiac tamponade, air way obstruction, superior venacaval obstruction)
- **Collapsing or "Water Hammer" pulse** (Rapidly rising pulse which collapses sudenly)

P/E

Venous pulses

- Jugula venous pressure, CVP
- Hepatojugular reflex test (To Dx RVF or tricuspid valve insufficency)

Others

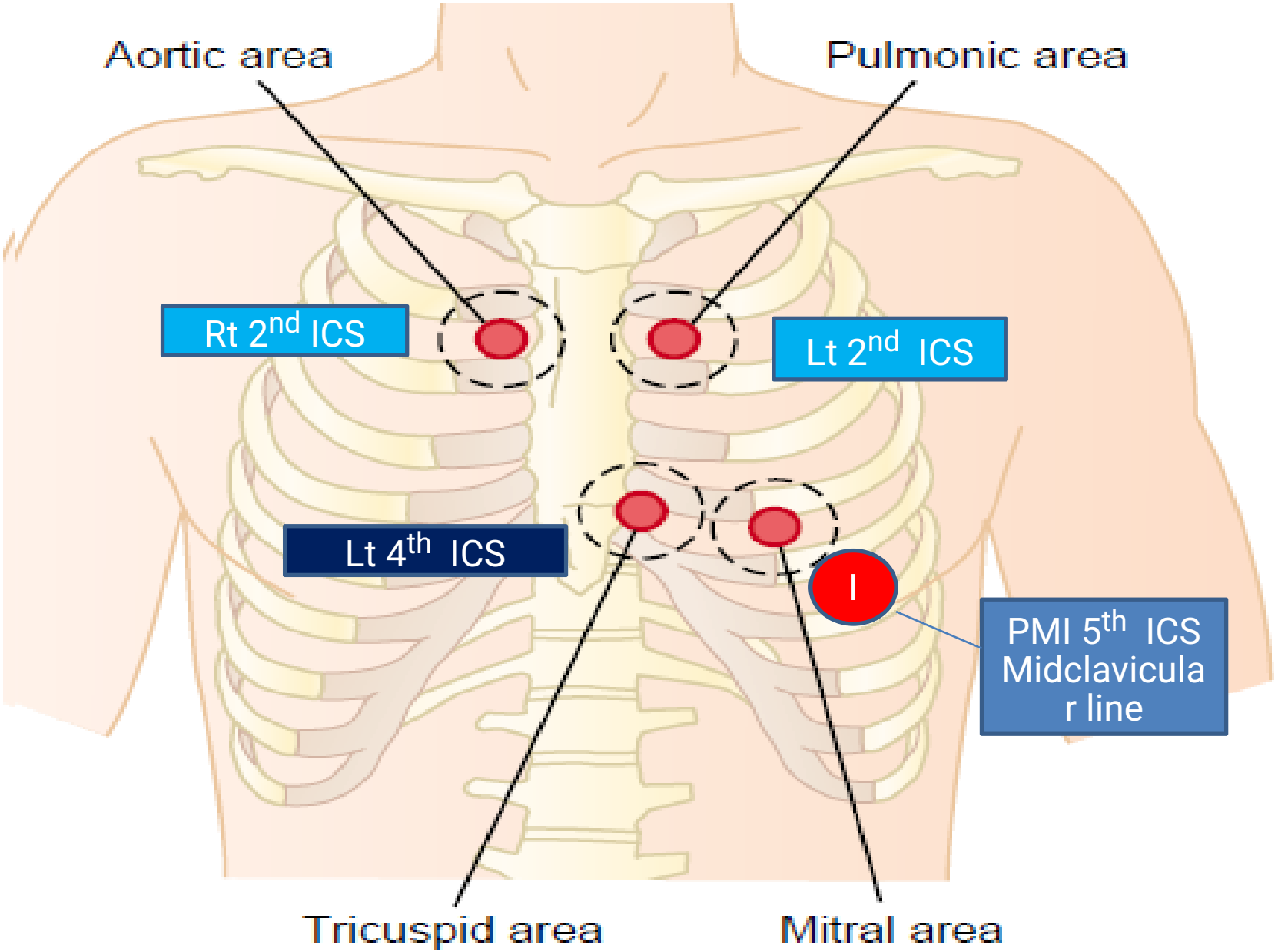
- **Skin pallor-** indicate low cardiac output
- **Cyanosis** (Bluish discoloration of the MM secondary to hypoxia)
- **Central cyanosis** (over tongue & buccal mucosa)
- **Peripheral cyanosis** (over nails, lips, ear lobes, & palms)
- **Xanthelasma** (Yellowish plaques in the nasal portion of the eye lids-indicate CHD)

Respiratory system

- Tachypnea, crackles in CHF & Pul edema

Examination of the six areas of precordium

1. **Aortic area:-** 2nd ICS to the right of the sternum
2. **Pulmonic area :-** 2nd ICS to the left of the sternum
3. **Erb's point :-** 3rd ICS to the left of the sternum
4. **Tricuspid area :-** 4th ICS to the left of the sternum
5. **Apical area :-** 5th ICS to the left of the sternum
6. **Epigastric area :-** below the Xiphoid process.


Aortic area
Pulmonic area
Rt 2nd ICS
Lt 2nd ICS
Lt 4th ICS
I
PMI 5th ICS Midclavicular line
Tricuspid area
Mitral area

The areas for listening to the different heart sounds are not directly over the valves themselves.

- The aortic area is upward along the aorta because of sound transmission up the aorta, and
- The pulmonic area is upward along the pulmonary artery.
- The tricuspid area is over the right ventricle, and
- The mitral area is over the apex of the left ventricle, which is the portion of the heart nearest the surface of the chest; the heart is rotated so that the remainder of the left ventricle lies more posteriorly.

Heart sounds

- ‘Lubb’ (1st sound) - Closure of A-V valves

S 

A = S 

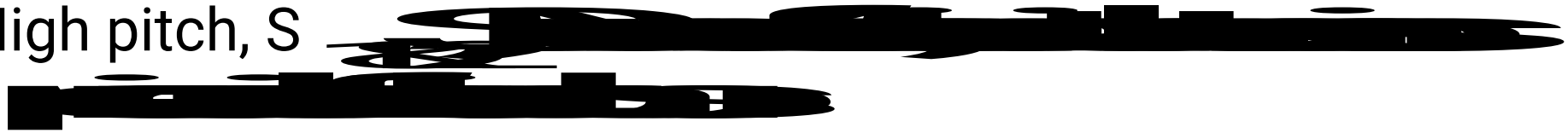
P = S 

T = S 

M = S 

- ‘Dupp’ (2nd sound) - Closure of S-L valves

Diaphragm(2nd, MT)

- High pitch, S 

Bell (1st, APMT)

- Low pitch, S 

Heart Sounds

- 'Lubb' (1st sound) - Closure of A-V valves
- 'Dupp' (2nd sound) - Closure of S-L valves

Caused by Turbulence on closing.

Anything extra ⇒ ***'Murmur'*** (swishing of blood)

Could be due to:-

- Stenosis of Valves (calcification)
- Valves not closing properly

(Incompetence, Insufficiency)

} Increases Pressure on heart

Heart Sounds(S1-Lub/S2-Dub)

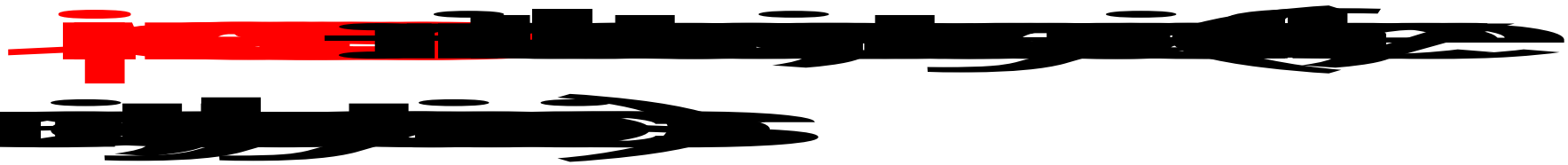
The normal heart sounds, S





- **First heart sound** (S1-**has low pitch,** loudest and longest) is best heard with the diaphragm

- It is created by the simultaneous closure of the Tricuspid and Mitral valves
- Best heard at the apex of the heart
- S



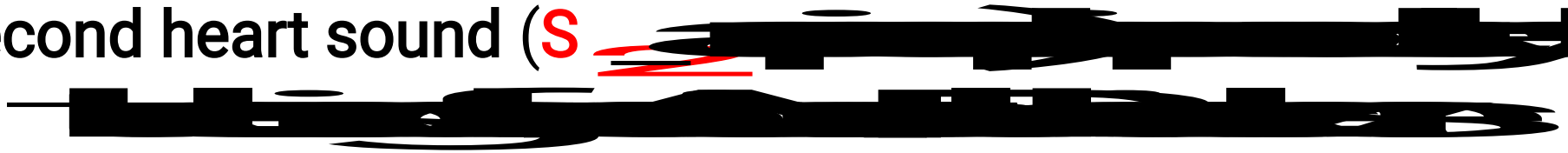
- S



Low frequency = 40 cycle/sec--- ear can detect

Heart Sounds(S1-Lub/S2-Dub)—

- **Second heart sound (S**



-Has a **higher frequency**

(1) The tautness of the semilunar valve

(2) The greater elastic coefficient of the taut arterial walls

- The time between S



The time between S



- Improper closing of a valve results in a heart murmur

- S



-S



- S



Splitting of heart sounds

Normally, S

- **Widening of S**
- **Fixed splitting** (No or little change with inspiration and expiration) :- Atrial septal defect
- Reversed or paradoxical splitting of **S**



- Ex. Left bundle branch block(LBBB),severe aortic outflow obstruction, large aorta to pulmonary artery shunt, systolic HPN, Lt ventricular failure,

Gallop sounds

- Sounds are heard on triplets & resemble sounds of galloping horse
- Are either S3 or S4 heart sounds
- Are low pitched
- Best heard at the apex of the heart

S3

-Is a low pitched heart sound heard immediately after S2

- It is heard during rapid filling of ventricles. May be normal in children & young adults

Occasionally a weak, rumbling third heart sound is heard at the beginning of the *middle third of diastole*

Reason: oscillation of blood back and forth between the walls of the ventricles initiated by inrushing blood from the atria.

The frequency of this sound is usually so low that the ear cannot hear it, yet it can often be recorded in the phonocardiogram.

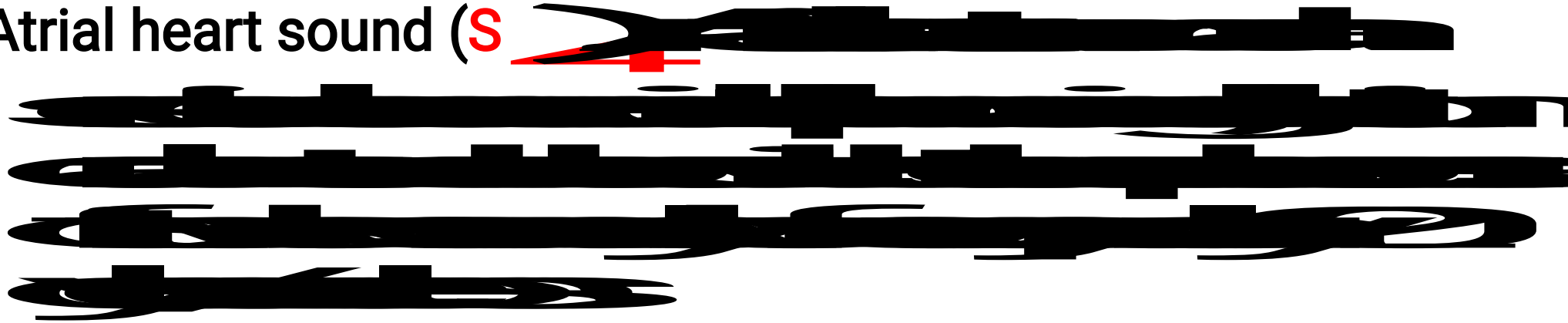
S

S



- It is heard during time of atrial contraction
- It is present when the ventricle is hypertrophid & resistant to filling

Atrial heart sound (S



S



Murmurs: Sounds created by abnormal, turbulent flow of blood in the heart.

Murmurs

- A***pan systolic*** *(holosystolic) murmur* starts with S1 and stops at S2, without a gap between murmur and heart sounds.

Ex. Mitral/tricuspid regurgitation,v entricular septal defect, aorto pulmonary shunts

- A***midsystolic*** *murmur begins after S1* and stops before S2. Brief gaps are audible between the murmur and the heart sounds. Listen carefully for the gap just before S2. It is heard more easily and, if present, usually confirms the murmur as midsystolic, not pansystolic.

EX. Midsystolic murmurs most often are related to blood flow across the semilunar (aortic and pulmonic) valves.

Murmurs

- A***late systolic*** *murmur usually starts* in mid- or late systole and persists up to S2.

Ex. This is the murmur of mitral valve prolapse and is often, but not always, preceded by a systolic click

- An***early diastolic*** *murmur starts* right after S2, without a discernible gap, and then usually fades into silence before the next S1.

EX. Early diastolic murmurs typically accompany regurgitant flow across incompetent semilunar valves

- A***middiastolic*** *murmur starts a short* time after S2. It may fade away or merge into a late diastolic murmur.
- EX. Middiastolic and presystolic murmurs reflect turbulent flow across the atrioventricular valves.
- A***late diastolic*** *(presystolic) murmur* starts late in diastole and typically continues up to S1.

Grades of heart murmurs

Gradations of Murmurs

Grade	Description
Grade 1	Very faint, heard only after listener has "tuned in"; may not be heard in all positions
Grade 2	Quiet, but heard immediately after placing the stethoscope on the chest
Grade 3	Moderately loud
Grade 4	Loud, with palpable thrill
Grade 5	Very loud, with thrill. May be heard when the stethoscope is partly off the chest
Grade 6	Very loud, with thrill. May be heard with stethoscope entirely off the chest

Physical assessment—

Physical examination is performed to confirm the data obtained in the health history

- Effectiveness of the heart as a pump
- Filling volumes and pressures
- Cardiac output
- Compensatory mechanisms

Physical assessment…

- **Effectiveness of the heart as a pump**
  - Indications that the heart is not contracting sufficiently or functioning effectively as a pump include reduced pulse pressure, cardiac enlargement, and murmurs and gallop rhythms (abnormal heart sounds)
- **Filling volumes and pressures**
  - The amount of blood filling the atria and ventricles and the resulting pressures (called filling volumes and pressures)
  - Are estimated by the degree of jugular vein distention and the presence or absence of congestion in the lungs, peripheral edema, and postural changes in BP that occur when the individual sits up or stands.

Physical assessment…

- **Cardiac output**
  - Cardiac output is reflected by congestion, heart rate, pulse pressure, color and texture of the skin, and urine output
- **Compensatory mechanisms**
  - Examples of compensatory mechanisms that help maintain cardiac output are increased filling volumes and elevated heart rate

Diagnosis

➢**Hx , physical examinations & LAB**

- Creatine kinase (CK index) ,N= 0 - 3
- CK-MB fraction ,N = 0 – 3 ng/ml
- Total CK , N= 38 -120 ng/ml

CK-MB fraction percent of total CK ,N= 0 – 4 %

CK, MB2 fraction < 1 U/L

-LDH4 = 3-10%, LDH5 = 2-12% ---- HF, Infarction,CLD

-LDH1 =21-36 % (LDH1/LDH2 > 1 in MI)

-Troponin I ,N = 0.0 - 0.4 ng/mL

(Onset: 4-6 hrs, persist 1-3 wks, has high affinity for myocardial injury)

-Troponin T ,N = 0.0 - 0.2 ng/mL (Onset: 3-4 hrs)

- Total cholesterol (200-239 mg/dL= borderline high, ≥240mg/dl = high)
- Myoglobin (M=10-95ng/ml,F= 10-65ng/ml) (onset: 1-3 hrs) & AST,ALT,etc

Diagnosis—

-LDLs (130-159 mg/dL=bordeline high,160-189= high, ≥190 very high, for CAD < 100)

-HDLs (N= M, 35 to 65 mg/dL; F=35 to 85 mg/dL)

- HDL= < 40 Low/increased risk
- HDL= ≥ 60 High/decreased risk

-Triglycerides (150-199 mg/dL=bordeline high, >200=high)

- Blood urea nitrogen (BUN) N= 10-20mg/dl
- Partial thromboplastin time (PTT) =25- 35 seconds
- Prothrombin time (PT) =25-35 seconds

-Platelet count N= 150,000–450,000/cu mm

Laboratory analysis

- Serum enzymes
- Blood chemistry
  - Lipid studies
  - Electrolytes
  - Renal Function Studies
- Coagulation studies
- Hematologic studies

Serum enzymes: Cardiac

- **Creatine Phosphokinase (Total CK / CPK)**
  - Non-Specific: enzyme elevated with damage to heart or skeletal muscles and brain tissue.
    - Elevates in 4 to 8 hours
    - Peaks in 15 to 24 hours
    - Returns to normal in 3 to 4 days
- **Creatine Phosphokinase Isoenzyme (CPK-MB)**
  - Specific: isoenzyme of CPK; elevated with cardiac muscle damage.
    - Elevates in 4 to 8 hours
    - Peaks in 15 to 24 hours
    - Returns to normal in 3 to 4 days

Cardiac Enzymes

- **Myoglobin**
  - Non-specific: a heme protein found in muscle tissue; elevated with damage to skeletal or cardiac muscle.
    - Elevates in 2 to 3 hours
    - Peaks 6-9 hours
    - Returns to normal 12 hours
- **Lactic Acid Dehydrogenase (LDH)**
  - Non-specific: enzyme elevated with damage to many body tissues. (i.e. heart, liver, skeletal muscle, brain and RBC's); Not frequently used today.
    - Elevates in 1 to 3 days
    - Peaks in 2 to 5 days
    - Returns to normal 10 to 14 days

Cardiac Enzymes–-

- **Troponin I / T**
  - Specific: a contractile protein released with cardiac muscle damage; not normally present in serum.
    - Elevates in 4 to 6 hours
    - Peaks in 10 to 24 hours
    - Returns to normal in 10 to 15 days
  - Sensitivity superior to CK-MB within the first 6 hours of event.
  - Has replaced LDH for client's who delay seeking treatment.

Other Serum Enzymes

- **C-Reactive Protein**
  - Protein marker of acute inflammatory reactions
    - Increased serum levels associated with increased risk of acute cardiovascular events.
- **Homocysteine**
  - Amino acid; presence in serum suggests increased risk of cardio-vascular events.
- **Natriuretic Peptides**
  - Hormone-like substances released into bloodstream with cardiac chamber distention.
    - **Atrial Natriuretic Peptide (ANP)**
    - **Brain or B-type Natriuretic Peptide (BNP)**

Blood Chemistry Analysis

- **Lipoprotein (Lipid) Profile**
  - Total Cholesterol
    - Normal < 200mg/dl
  - Triglyceride
    - Normal < 150 mg/dl
  - Low Density Lipoproteins (LDL)
    - Normal <130 mg/dl / “Optimal” <100mg/dl
  - High Density Lipoproteins (HDL)
    - Normal: > 40 mg/dl

      > 60 mg/dl cardio-protective

Blood Chemistry Analysis---

- Serum Electrolytes
  - i.e. Na, K, Ca and Mg
  - Glucose / Hemoglobin A1C
- Coagulation Studies
  - PTT / aPTT
  - PT / INR
- Hematologic Studies
  - CBC
- Renal Function Studies
  - BUN
  - Creatinine

Diagnostic testing

- Electrocardiography *
  - 12-Lead EKG
  - Continuous bedside monitoring
  - Ambulatory monitoring
- Stress tests
  - Thallium scans
- Echocardiograms
- Cardiac catheterizations

Cardiac stress tests

- Stressing the heart to monitor performance
- Assists in determining
  - Coronary artery disease
  - Cause of chest pain
  - Functional capacity of heart
  - Identify dysrhythmias
  - Effectiveness of medications
  - Establish goals for a physical fitness routine

Cardiac stress tests—
Types of stress tests

- Exercise
  - Treadmill (most common)
  - Bike
  - Arm crank
- Pharmacological
  - Vasodilating agents to mimic the effects of exercise
    - Persantin
    - Adenosine
- Mental / Emotional (new; under investigation)
  - Simulated public speaking
  - Mental arithmetic test

Cardiac stress tests—

- Thallium scan
  - Often combined with stress tests
    - Radiological exam to assess how well the coronary arteries perfuse the myocardium.
    - Images are taken 1 to 2 minutes prior to end of stress test and again 3 hours later.
    - Nursing Considerations
      - NPO
      - IV access

Cardiac stress tests—

- **Nursing considerations**
  - Explain procedure to client
  - Maintain NPO status 4 hour before test
  - Instruct client to avoid stimulants (i.e. chocolate, caffeine and cigarettes)
  - Hold certain medications before testing
    - Exercise: i.e. beta-adrenergic blockers
    - Pharmacologic: i.e. Theophylline (24-48 hours prior)
  - I.V. access must be obtained

Diagnostic tests & procedures
Electrocardiography (**ECG**)

- The ECG is a non invasive diagnostic tool used in assessing the cardiovascular system
- It is a graphic recording of the electrical activity of the heart
- The ECG is obtained by placing disposable electrodes in standard positions on the skin of the chest wall and extremities
- The heart's electrical impulses are recorded as a tracing on special graph paper

Echocardiography

- Echocardiography is a noninvasive ultrasound test that is used to examine the size, shape, and motion of cardiac structures
- It is a particularly useful tool for diagnosing pericardial effusions, determining the etiology of heart murmurs, evaluating the function of prosthetic heart valves, determining chamber size, and evaluating ventricular wall motion
- It involves transmission of high-frequency sound waves into the heart through the chest wall and recording of the return signals
- The ultrasound is generated by a hand-held transducer applied to the front of the chest
- The transducer picks up the echoes, converts them to electrical impulses, and transmits them to the echocardiography machine for display on an oscilloscope and recording on a videotape
- An ECG is recorded simultaneously to assist with interpreting the echocardiogram.

Cardiac catheterization

- Is an invasive diagnostic procedure in which radio-opaque arterial and venous catheters are introduced into selected blood vessels of the right and left sides of the heart
- Catheter advancement is guided by fluoroscopy
- Most commonly, the catheters are inserted percutaneously through the blood vessels, or via a cut down procedure if the patient has poor vascular access.

Purposes

- To measure pressure & oxygen saturation in different chambers of the heart
- To assess patency of coronary artery and extent of atherosclerosis
- To administer fluids, sedatives,& other medications
- To introduce radio-opaque contrast agents into selected arteries for coronary angiography & angiocardiography

Angiocardiography

- Involves injection of radio-opaque contrast agent into the Aorta & Rt or Lt side of the heart and radiographic examination
- It is an invasive diagnostic procedure

Coronary Angiography

- In this case, the opening of the Rt & Lt coronary arteries are selectively cannulated & the contrast is injected for radiographic evaluation
- It is an invasive diagnostic procedure

Phonocardiogram

If a microphone specially designed to detect low-frequency sound is placed on the chest, the heart sounds can be amplified and recorded by a high-speed recording apparatus. The recording is called a *phonocardiogram,* and the heart sounds appear as waves, as shown above. Recording A is an example of normal heart sounds, showing the vibrations of the first, second, and third heart sounds and even the very weak atrial sound. Note specifically that the third and atrial heart sounds are each a very low rumble. The third heart sound can be recorded in only one third to one half of all people, and the atrial heart sound can be recorded in perhaps one fourth of all

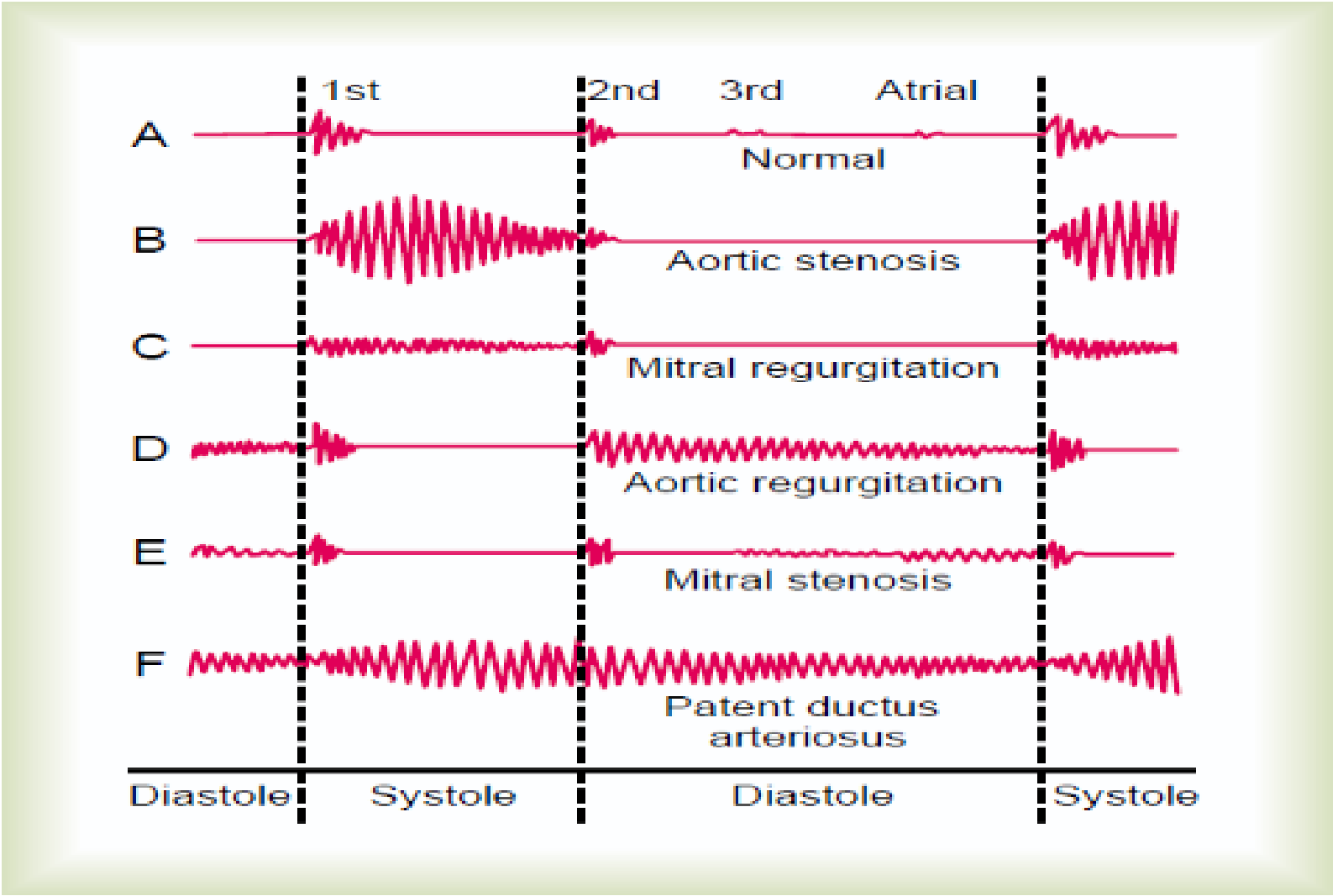


Figure 23–3

Phonocardiograms from normal and abnormal hearts.

Chest X-ray

To determine

- Cardiac size, Ex. Cardiomegally in CHF
- Contour and position of the heart ,Ex. Displacement in pleural effusion ,pneumothorax ,& lung fibrosis, scoliosis, etc
- To see correct placement of cardiac catheters & pacemakers
- To reveal abnormal dilatation of the Aorta, EX. Marfan's syndrome, syphilitic aortitis, post stenotic dilatation,etc

I. Heart arrhythmias/dysarrhythmias/

✓**Sinus arrhythmias** (*usually no Rx)*

1.Sinus bradycardia

2.Sinus tachycardia

✓**Premature beats**

3. Premature Atrial Contractions (PACs)

4. Premature Ventricular Contractions (PVCs)

✓**Supra-ventricular arrhythmias**

5. Atrial Fibrillation

6. Atrial Flutter

7. Paroxysmal Supra-ventricular Tachycardia

Heart arrhythmias/dysarrhythmias/––

- ✓**Ventricular arrhythmias**

8. Ventricular Tachycardia

9. Ventricular Fibrillation

- ✓**AV junctional blocks** (AV nodal blocks)

10. 1st degree AV block

11. 2nd degree AV block, Type I

12. 2nd degree AV block, Type II

13. 3rd degree AV block

Sinus bradycardia



- Rate? 30 bpm
- Regularity? regular
- P waves? normal
- PR interval? 0.12 s
- QRS duration? 0.10 s

Interpretation? *Sinus bradycardia*

Sinus bradycardia–-



- Deviation from *normal sinus rhythm*

- Rate < 60 bpm
  - All aspects of sinus bradycardia are the same as those of normal sinus rhythm, except for the rate

Sinus bradycardia–-



- Etiology: SA node is depolarizing slower than normal, impulse is conducted normally (i.e. normal PR and QRS interval)
- Is common in athletes & during sleep

Causes: B-blockers, digoxin, inferior wall MI, hypothyroidism, ICP, hyperkalemia ,hypothermia, severe pain, vagal maneuver

RX: Rx of underlying cause

- Atropine (0.5 to 1.0 mg – IV bolus is rapidly given blocks vagal stimulation) = if symptomatic & no underlying cause
- Rarely Oxygen

Sinus tachycardia



- Rate? 130 bpm
- Regularity? regular
- P waves? normal
- PR interval? 0.16 s
- QRS duration? 0.08 s

Interpretation? *Sinus tachycardia*

Sinus tachycardia–––



- Deviation from *normal sinus rhythm*

- Rate > 100 bpm
  - All aspects of sinus tachycardia are the same as those of normal sinus rhythm, except for the rate

Sinus tachycardia––



- Etiology: SA node is depolarizing faster than normal, impulse is conducted normally.

Causes: stress, fever, anxiety, pain, anemia, acute CHF, shock, hypovolemia, thyrotoxicosis, atropine, excessive caffeine & alcohol, alcohol withdrawal, acute blood loss, exercise

RX: - Aimed at abolishing the cause

- Use of B-blockers and CCB
- Digoxin

Nursing interventions

1. Monitoring and managing the arrhythmia
   - The nurse regularly evaluates blood pressure, pulse rate and rhythm, rate and depth of respirations, and breath sounds to determine the dysrhythmia's hemodynamic effect
   - The nurse also asks patients about episodes of lightheadedness, dizziness, or fainting as part of the ongoing assessment
   - If a patient with a dysrhythmia is hospitalized, the nurse may obtain a 12-lead ECG, continuously monitor the patient, and analyze rhythm strips to track the dysrhythmia

Nursing interventions…

- Control of the incidence or the effect of the dysrhythmia, or both, is often achieved by the use of antiarrhythmic medications
- The nurse assesses and observes for the beneficial and adverse effects of each of the medications
- The nurse also manages medication administration carefully so that a constant serum blood level of the medication is maintained at all times
- In addition to medication, the nurse assesses for factors that contribute to the dysrhythmia (eg, caffeine, stress, non adherence to the medication regimen) and assists the patient in developing a plan to make lifestyle changes that eliminate or reduce these factors

Nursing interventions…

2. Minimizing anxiety

- When the patient experiences episodes of dysrhythmia, the nurse maintains a calm and reassuring attitude
- This performance fosters a trusting relationship with the patient and assists in reducing anxiety (reducing the sympathetic response)

Nursing interventions…

3. Promoting home and community-based care

Teaching patients self-care

- When teaching patients about dysrhythmias, the nurse presents the information in terms that are understandable and in a manner that is not frightening or threatening
- The nurse explains the importance of maintaining therapeutic serum levels of anti arrhythmic medications so that the patient understands why medications should be taken regularly each day
- If the patient has a potentially lethal dysrhythmia, it is also important to establish with the patient and family a plan of action to take in case of an emergency
- This allows the patient and family to feel in control and prepared for possible events

II. Coronary Artery Disease(CAD)

- Coronary artery disease is the most prevalent type of cardiovascular disease.
- For this reason, it is important for nurses to become familiar with the various types of coronary artery conditions and the methods for assessing, preventing, and treating these disorders medically and surgically.

Coronary atherosclerosis

Definition

- **Atherosclerosis** is an abnormal accumulation of lipid, or fatty, substances and fibrous tissue in the vessel wall
- Create blockages or narrow the vessel in a way that reduces blood flow to the myocardium
- It is the most common heart disease in the United States
- It is a progressive disease

Physiology/Pathophysiology

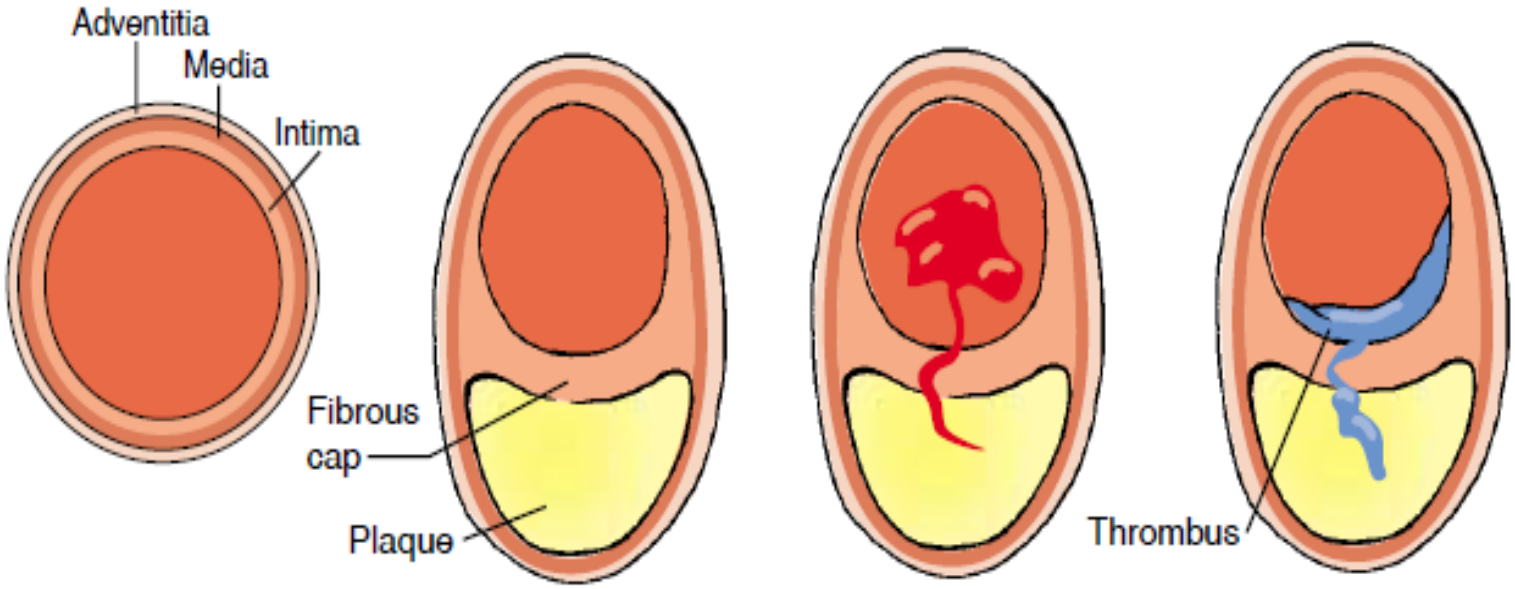


FIGURE 28-1 Atherosclerosis involves a normally patent artery (**A**) and an inflammatory response whereby smooth muscle cells proliferate within the blood vessel to form a fibrous cap (**B**). The proliferation results in deposits, called atheromas or plaques, which protrude into the lumen of the vessel, narrowing it and obstructing blood flow. If the cap ruptures and hemorrhages into the plaque (**C**), a thrombus (**D**) may develop and obstruct blood flow further.

Arteriosclerosis and Atherosclerosis

- **Arteriosclerosis** is the most common disease of the arteries; the term means *hardening of the arteries.* It is a diffuse process whereby the muscle fibers and the endothelial lining of the walls of small arteries and arterioles become thickened.
- **Atherosclerosis** involves a different process, affecting the intima of the large and medium sized arteries. These changes consist of the accumulation of lipids, calcium, blood components, carbohydrates, and fibrous tissue on the intimal layer of the artery. These accumulations are referred to as **atheromas or plaques.**
- Because atherosclerosis is a generalized disease of the arteries, when it is present in the extremities, atherosclerosis is usually present elsewhere in the body.

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Arteriosclerosis and Atherosclerosis––

Pathophysiology

- The most common **direct results of atherosclerosis** in arteries include narrowing (stenosis) of the lumen, obstruction by thrombosis, aneurysm, ulceration, and rupture.
- Its **indirect results are** malnutrition and the subsequent fibrosis of the organs that the sclerotic arteries supply with blood.
- All actively functioning tissue cells require an abundant supply of nutrients and oxygen and are sensitive to any reduction in the supply of these nutrients.
- If such reductions are severe and permanent, the cells undergo ischemic necrosis (death of cells due to deficient blood flow) and are replaced by fibrous tissue, which requires much less blood flow.
- Atherosclerosis can develop at any point in the body, but certain sites are more vulnerable, typically bifurcation or branch areas.
- In the proximal lower extremity, these include the distal abdominal aorta, the common iliac arteries, the orifice of the superficial femoral and profunda femoris arteries, and the superficial femoral artery in the adductor canal. Distal to the knee, atherosclerosis occurs anywhere along the artery.

Arteriosclerosis and Atherosclerosis––

- It may be that there is no single cause or mechanism for the development of atherosclerosis; rather, multiple processes may be involved.
- Morphologically, atherosclerotic lesions are of two types: fatty streaks and fibrous plaque.
- **Fatty streaks** are yellow and smooth, protrude slightly into the lumen of the artery, and are composed of lipids and elongated smooth muscle cells. These lesions have been found in the arteries of people of all age groups, including infants. They do not usually cause clinical symptoms.
- **The fibrous plaque** characteristic of atherosclerosis is composed of smooth muscle cells, collagen fibers, plasma components, and lipids. It is white to whitish yellow and protrudes in various degrees into the arterial lumen, sometimes completely obstructing it. These plaques are found predominantly in the abdominal aorta and the coronary, popliteal, and internal carotid arteries. This plaque is believed to be an irreversible lesion.
- Gradual narrowing of the arterial lumen as the disease process progresses stimulates the development of collateral circulation.

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Atherosclerosis & Arteriosclerosis –-

Pathophysiology

- *The most common direct results of atherosclerosis in arteries:*
- Narrowing (stenosis) of the lumen
- Obstruction by thrombosis
- Aneurysm
- Ulceration & rupture
- Indirectly it results in malnutrition & subsequent fibrosis of the organs
- Death of tissue cells due to deficient blood flow

Atherosclerosis

- Affecting the intima of the large and medium sized arteries and is accumulation of lipids, calcium, blood components, carbohydrates and fibrous tissues
- These accumulation are refereed to as atheromas, or plaques
- Presence of atheromas
  - Plaques
    - Consist of lipids, cells, fibrin, cell debris
  - Lipids usually transported with lipoprotein
-



Etiology (Atherosclerosis)

- Age
- Gender
- Genetic factors
- Obesity, diet high in cholesterol, animal fats
- Cigarette smoking
- Sedentary life style
- Diabetes mellitus
- Poorly controlled hypertension
- Combo of BC pills and smoking

Clinical Manifestations

- The clinical signs and symptoms resulting from atherosclerosis depend on the organ or tissue affected

Diagnosis

- Serum lipid levels
- Exercise stress test
- Radioisotope

Atherosclerosis & Arteriosclerosis –-

Clinical manifestations

- Depends on the organs or tissues affected
- Coronary Atherosclerosis: angina, MI
- Cerebrovascular Disease: transient cerebral ischemic attacks, stroke
- Aorta: aneurysm
- Extremities: gangrene
- Reno vascular Disease: renal artery stenosis, ESRD

Diagnosis

- Medical & family Hx, Risk factors
- Physical exam & diagnostic tests

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Atherosclerosis & Arteriosclerosis –––

Medical & Nursing management:-

- Modification of risk factors
- Medication therapy, & surgical procedures
- Improving peripheral arterial circulation:

- Lower the extremities below the level of the heart (if arterial condition)
- Elevate the extremities above the level of the heart (if venous condition)
- Encourage moderate amount of walking & controlled exercise

- Promoting vasodilatation & preventing vascular compression

- Maintain warm temperature & avoid chilling
- Avoid emotional upsets; stress management
- Avoidance of constrictive clothing
- Avoidance of leg crossing
- Administer vasodilators

- Relief of pain: analgesics, promoting increased circulation
- Maintenance of tissue integrity
- Adherence to self-care program

Atherosclerosis—Treatment

- Decrease cholesterol and LDL
- Decrease sodium ion intake
- Control primary disorders
- Quit smoking
- Oral anticoagulant
- Surgical intervention
  - Percutaneous transluminal coronary angioplasty (PTCA)
  - Cardiac catheterization
  - Laser beam technology
  - Coronary artery bypass grafting

Coronary artery diseases---
(Ischemic Heart Diseases)

A/ Angina pectoris

- Is a clinical symptom characterized by pain or a feeling of pressure in the anterior chest

Pathophysiology

- Pain is ciliated as result of insufficient coronary blood flow resulting in inadequate oxygen supply to the myocardium

Risk factors

- Major risk factors include
  - Use of tobacco
  - Hypertension
  - Elevated blood lipid levels,
  - Family history of premature cardiovascular disease (first-degree relative with cardiovascular disease at age 55 or younger for men and at age 65 or younger for women) and
  - Age (> 45 years for men; >55 years for women)

Types of angina

✓ **Unstable angina/** Pre-infarction angina or crescendo angina / symptoms occur more frequently and last longer than stable angina(over 20 minutes). The threshold for pain is lower, and pain may occur at rest

- Unstable angina occurs with exercise or emotional stress, but it increases in occurrence, severity, and duration over time.

- Angina of recent onset (within 2 months) that markedly limits usual activity
- Angina that increases in severity , frequency, or duration, or that occurs with less provocation over a short time period (i.e., within 2 months)

Stable angina

– Predictable and consistent pain that occurs on exertion and is relieved by **rest** or **nitroglycerin**

Types of angina…

- **Intractable or refractory angina**
  - Severe incapacitating chest pain
- **Variant angina /**Prinzmetal's angina/

_ Pain often occurring at rest or awakens pt from sleep with reversible ST-segment elevation; thought to be caused by coronary artery vasospasm. Recurrent, prolonged attacks of severe ischemic pain

- **Silent ischemia** /Asymptomatic angina/

Objective evidence of ischemia (such as electrocardiographic changes with a stress test), but patient reports no symptoms(no chest pain) Ex. DM

Table 28-3 • Canadian Cardiovascular Society Classification of Angina

CLASS	ACTIVITY EVOKING ANGINA	LIMITS TO ACTIVITY
I	Prolonged exertion	None
II	Walking >2 blocks	Slight
III	Walking <2 blocks	Marked
IV	Minimal or rest	Severe

Adapted from Campeau L. (1976). Grading of angina pectoris. *Circulation 54,*

Factors that produce angina pain

- Physical exertion: precipitate on attack by increasing myocardial demand
- Exposure to cold: vasoconstriction
- Eating a heavy meal: decreases available blood flow to the heart as the mesenteric blood flow increases
- Any emotion provoking situation: causing the release of adrenalin & increase blood pressure, may accelerate the heart rate, thus decreasing the available blood supply

Clinical features of angina

- Ischemia of the coronary arteries cause pain
- Location: usually felt deep in the chest behind the upper and middle third of sternum
- Pressure in the upper chest
- Severe apprehension and a feeling of impending death
- A feeling of weakness in the arms, wrists & hands
- Sensation of pressure, heaviness, or squeezing in the anterior chest area. Sharp pain is not a typical symptom of IHD.
- Pain may radiate to the neck, jaw, shoulder, back, or arm.
- Pain may be accompanied by dyspnea, nausea, vomiting, or **diaphoresis.**
- Symptoms are often provoked by exertion (e.g., walking, climbing stairs, and doing yard or house work) or emotional stress and relieved within minutes by rest or nitroglycerin.

Diagnosis

- Clinical manifestation of pain and patients history
- Abnormal heart sounds, such as paradoxical splitting of the second heart sound, a third heart sound, or a loud fourth heart sound
- CK, CK-MB fraction, troponin I and troponin T) are elevated in MI (ST segment elevation MI and non–ST-segment elevation MI), but normal in chronic stable angina and unstable angina

Management

- The objective of treatment is to decrease the oxygen demand and to increase the oxygen supply of the myocardium
- Medically: through pharmacologic therapy and control of risk factors

Control risk factors

1. Non modifiable risk factors
   - Positive family history
   - Gender (High in men-3x & premenopausal women)
   - Race (higher in African American)
   - Age (M > 45, F > 55)
2. Modifiable risk factors
   - High blood cholesterol
   - Elevated blood pressure
   - Cigarette smoking – produces tachycardia and raises the B/P
   - Elevated blood glucose
   - Obesity
   - Physical inactivity

Pharmacologic therapy

A/ Nitroglycerin

- To reduce myocardial oxygen consumption, which decreases ischemia and relieves pain within 3 minutes, the route is sublingually
- Nitroglycerin dilates primarily the veins and, in higher doses, also dilates the arteries.
- It helps to increase coronary blood flow by preventing vasospasm and increasing perfusion through the collateral vessels.
- Dilation of the veins causes venous pooling of blood throughout the body.
- As a result, less blood returns to the heart, and filling pressure (preload) is reduced.
- If the patient is hypovolemic (does not have adequate circulating blood volume), the decrease in filling pressure can cause a significant decrease in cardiac output and blood pressure.

Pharmacologic therapy…

A/ Nitroglycerin ….

- Nitrates in higher doses also relax the systemic arteriolar bed and lower blood pressure (decreased after load)
- Nitrates may increase blood flow to diseased coronary arteries and through collateral coronary arteries, arteries that have been underused until the body recognizes poorly perfused areas
- These effects decrease myocardial oxygen requirements and increase oxygen supply, bringing about a more favorable balance between supply and demand
- It usually is not given if the systolic blood pressure is 90 mm Hg or less

Pharmacologic therapy–-

B/ Beta-adrenergic blocking agents

- To reduce myocardial oxygen consumption by blocking the beta-adrenergic sympathetic stimulation to the heart. The result is
  - A reduction in heart rate,
  - Slowed conduction of an impulse through the heart,
  - Decreased blood pressure, and
  - Reduced myocardial **contractility (force of contraction) that establishes** a more favorable balance between myocardial oxygen needs (demands) and the amount of oxygen available (supply)
- This helps to control chest pain and delays the onset of ischemia during work or exercise. Beta-blockers reduce the incidence of recurrent angina, infarction, and cardiac mortality.
- The dose can be titrated to achieve a resting heart rate of 50 to 60 beats per minute

Pharmacologic therapy–-

C/ Calcium channel blockers

- Some decrease sinoatrial node automaticity and atrioventricular node conduction, resulting in a slower heart rate and a decrease in the strength of the heart muscle contraction (negative inotropic effect).
- These effects decrease the workload of the heart.
- Calcium channel blockers also relax the blood vessels, causing a decrease in blood pressure and an increase in coronary artery perfusion
- Calcium channel blockers increase myocardial oxygen supply by dilating the smooth muscle wall of the coronary arterioles;
- They decrease myocardial oxygen demand by reducing systemic arterial pressure and the workload of the left ventricle

Pharmacologic therapy–-

D/ Antiplatelet agents and anticoagulant medications

- **Aspirin:** prevents platelet activation and reduces the incidence of MI and death in patients with CAD.
- **Heparin:** prevents the formation of new blood clots

E/ Oxygen administration

Intervention	Rationale
Oxygen	Increases myocardial oxygen supply.
Beta-blocker	Antidysrhythmic, antihypertensive, reduces afterload to decrease myocardial oxygen consumption, decreases infarct size, and improves survival rates.
ACE inhibitor	Decreases afterload and therefore myocardial oxygen consumption.
Stool softener	Reduces straining which could precipitate MI, vasovagal effect.
Nitroglycerin	Reduces preload and afterload and dilates coronary arteries, increases myocardial oxygen supply and reduces oxygen consumption.
Morphine	Reduces preload by venous vasodilation decreasing myocardial oxygen consumption, relieves pain, decreases sympathetic nervous system stimulation.
Thrombolytic agent	Removes clot that is causing ischemia/infarction.

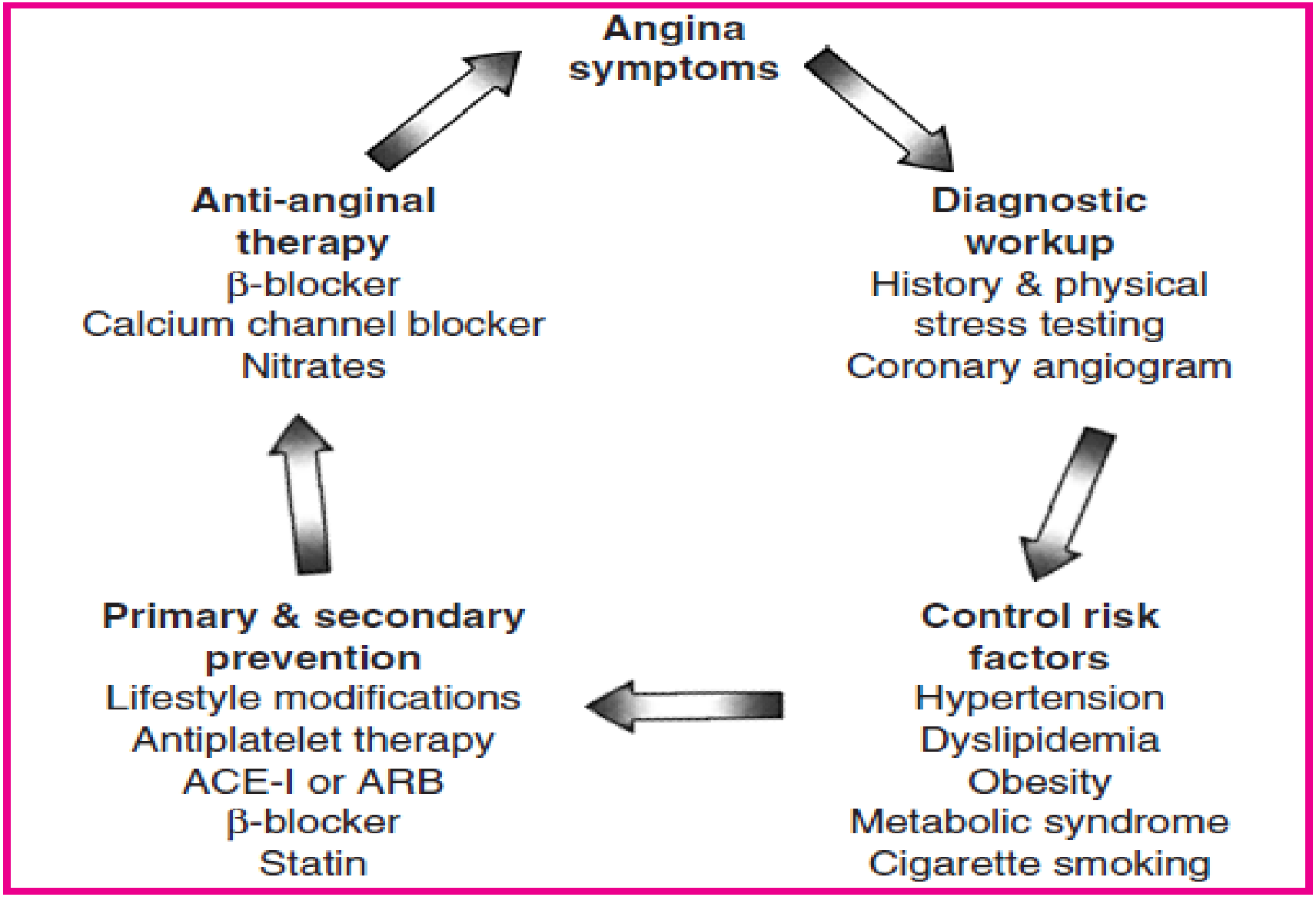


FIGURE 4–4. General treatment strategies for angina follow in clockwise fashion from the top center. ACE-I, angiotensin-converting enzyme inhibitor; ARB, angiotensin receptor blocker.

Myocardial Infarction(MI)

B/ MI

- Is the death of a portion of heart muscle in an area where there is sudden loss of blood supply due to occlusion of a major coronary artery or one of its branches.
- Is sometimes called a heart attack or a coronary thrombosis.
- Refers to the process by which areas of myocardial cells in the heart are permanently destroyed.

Pathophysiology

- Necrosis in myocardial infarction is not completed at once, but the automimc nervous systems attempt to compensate results in a further depressed cardiac performance, resulting in a further imbalance between myocardial oxygen supply and demand.
- Location: the sub endocardial layer of the myocardium is most susceptible to hypoxia.

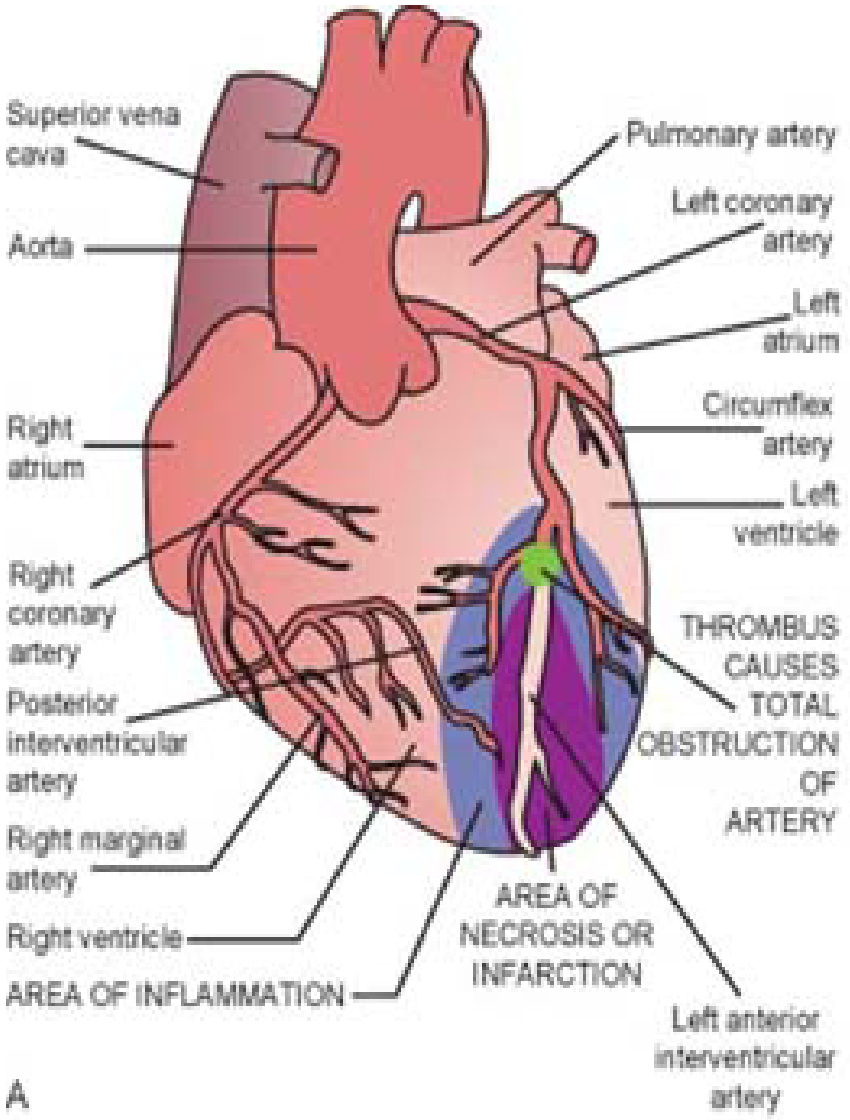
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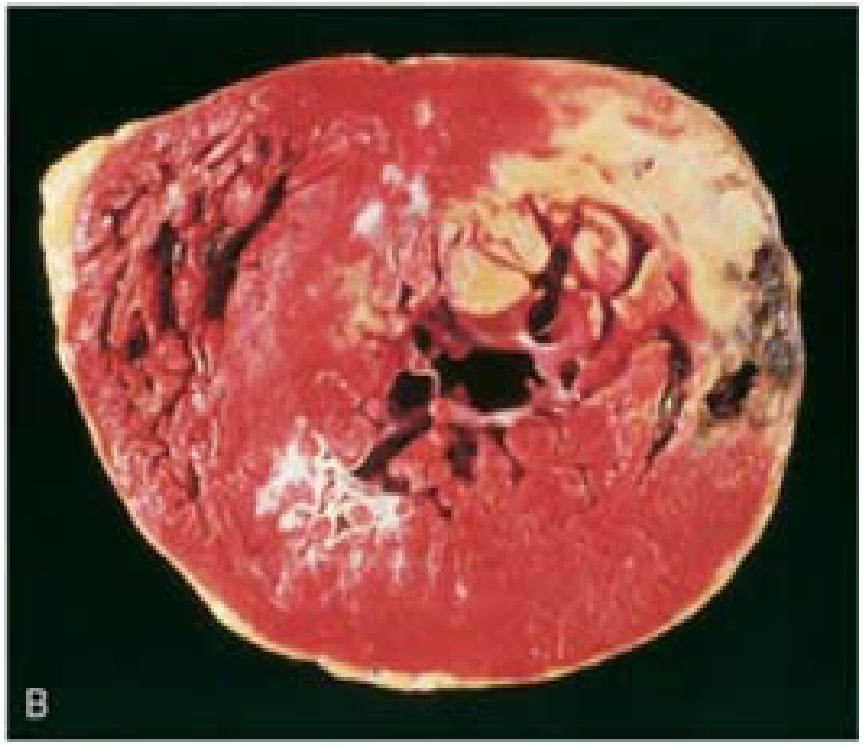
Myocardial Infarction––

Pathophysiology…

- Coronary artery completely obstructed
  - Prolonged ischemia and cell death of myocardium
- Most common cause is atherosclerosis with thrombus
- 3 ways it may develop:
  - Thrombus obstructs artery
  - Vasospasm due to partial occlusion
  - Embolus blocks small branch of coronary artery
- Majority involve Lt ventricle
  - Size and location of infarction determine severity of damage
- Function of myocardium contraction and conduction quickly lost
  - Oxygen supplies depleted
    - 1st 20 minutes critical
- Time Line
  - 1st 20 min critical
  - 48 hrs inflammation begins to subside
  - 7th day necrosis area replaced by fibrous tissue
  - 6-8 weeks scar forms

Myocardial Infarction





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Myocardial Infarction––

- Anterior, inferior , (posterior) or lateral wall of the myocardium are affected
- The left ventricle is the usual site of injury.
- The cause of the reduced blood flow is either a critical narrowing of a coronary artery due to atherosclerosis or a complete occlusion of an artery due to embolus or thrombus
- Decreased coronary blood flow may result from shock and hemorrhage

High risk

- Usually male > 40yrs
- Atherosclerosis of the coronary vessels
- HTN(hypertension)
- Younger women and men (25s & 30s especially women who take oral contraceptives and smoke)

Risk factors for acute MI

Modifiable risk factors

- Hyperlipidemia
- Smoking & alcoholism
- Diabetes mellitus
- Hypertension
- Obesity
- Physical inactivity
- Oral contraceptives

Non modifiable risk factors

- Age
- Gender
- Genetic/family history
- Pre-existing coronary heart diseases

Etiology -MI

- The major cause of MI is coronary artery occlusion by thrombosis or atheroma

- **Uncommon causes**
  - Inflammation of the coronary arteries (rare);
  - A stab wound to the heart;
  - A blood clot forming elsewhere in the body (for example, in a heart chamber) and traveling to a coronary artery where it gets stuck;
  - Cocaine abuse which can cause a coronary artery to go into spasm; complications from heart surgery; and some other rare heart problems.

MI—Signs and Symptoms

- **Pain**
  - Sudden/sub-sternal area/, more severe,& lasts longer than angina pectoris
  - Radiates to Lt arm and neck
  - Less severe in females
- Pallor(conjunctiva), dyspnea, sweating, nausea, dizziness, palpitations ,loss of consciousness
- Anxiety and fear
- Hypotension, rapid and weak pulse (low CO)
- Low grade fever after 12 hrs of infarction
- Sudden death

MI—Signs and Symptoms--

- Chest pain:
  - It is a heavy which may radiate to the shoulder and down the arms, usually the left arm.
  - In some cases the pain may radiate to the jaw & neck.
  - Pain is often accompanied by pallor, diaphoresis, dizziness, nausea and vomiting.

Diagnosis

- Patient Hx, ECG, serum enzymes and isoenzymes
- WBC 12000-15000,high ESR

Cardiac enzymes in MI

Cardiac Enzyme	Normal Levels	Elevated Levels First Detectable Following Myocardial Injury	Expected Duration of Elevated Levels
Creatinine kinase MB isoenzyme (CK-MB) – more sensitive to myocardium	0% of total CK (30-170 units/L)	4-6 hr	3 days
Troponin T	< 0.2 ng/L	3-5 hr	14-21 days
Troponin I	< 0.03 ng/L	3 hr	7-10 days
Myoglobin	< 90 mcg/L	2 hr	24 hr

Management -MI

- **Vasodilators**: Nitroglycerine 0.5mg sublingual Q.5min
- **Anti coagulants:** heparin reducing the probability of thrombus formation and the subsequent diminished blood flow

(Heparin: IV bolus 60-70 u/kg, then 12-15u/kg/hr)

- **Thrombolyti**c - to dissolve any thrombus in a coronary artery (streptokinase is the known agent),not given for unstable angina.
- **High flow 0** 
- **Analgesics** - Morphine sulphate IV 2-4 mg . The need for analgesia is limited to those patients in whom nitrates and anticoagulants are ineffective in relieving pain
- **Cardiac rehabilitation**
  - Prognosis depends on site/size of infarct, presence of collateral circulation, time elapsed before treatment
- Mortality rate in 1st year
  - 30-40% due to complications, recurrences

Management of acute MI

- Assess circulation: Pulse, BP, Capillary refill
  - *Use B-Blockers:* Atenolol 50 mg orally and 12 hours later then 100 mg per day or metoprolol
  - Morphine 2-4mg I/V ,repeat every 5-10 minutes
  - Other analgesics- pethidine, tramadol, diclofenac
  - Aspirin immediately
  - Nitrates – sublingual or skin patches

Nursing care/ interventions/

- Preventing pain: avoid activates known to cause Angina pectoris
- Reducing anxiety: Physical presence of another alleviate fear of death
- Patient education : home care considerations to improve the quality of life and promote health
- Relieving chest pain: vasodilator, anticoagulant, physical rest
- Importing adequate tissue perfusion: keeping the pt on bed or chair that he may rest and administer o 
- Monitoring and managing potential complaisant

Differences b/n angina & MI

Angina	Myocardial Infarction
Precipitated by **exertion** or stress	Occurs without cause, often in the morning after **rest**
Relieved by **rest or nitroglycerine**	Relieved only by **opioids**
< 15 min	> 30 min
Not associated with nausea, epigastric distress, dyspnea, anxiety, diaphoresis	Associated with nausea, epigastric distress, dyspnea, anxiety, diaphoresis

The difference between chest pain of angina pectoris and myocardial infarction

	Character, location and radiation	Duration	Precipitating Events	Relieving Measures
Angina pectoris	Substernal(retrosternal)pain spreading across chest · May radiate spread widely throughout chest	5-15min	·Usually related to exertion, emotion, eating, cold	·Rest ·Nitroglycerin ·Oxygen
Myocardial infarction	·Painful disability of shoulders and hands may be present.	>15min	·Occurs spontaneously but may be sequel to unstable angina	·Morphine sulfate ·Reperfusion of blocked coronary artery

III. Valvular Heart Diseases

Pathophysiology - mitral regurgitation

- Mitral regurgitation may be caused by problems with one or more of the leaflets, the chordae tendineae, the annulus, or the papillary muscles

- A mitral valve leaflet may shorten or tear
- The chordae tendineae may elongate, shorten, or tear
- The annulus may be stretched by heart enlargement or deformed by calcification
- The papillary muscle may rupture, stretch, or be pulled out of position by changes in the ventricular wall (eg, scar from a myocardial infarction or ventricular dilation).
- The papillary muscle may be unable to contract because of ischemia. Regardless of the cause, blood regurgitates back into the Lt. atrium during systole

Mitral regurgitation–-

Causes:- RHD(33%),congenital anomalies,CHD, bacterial endocarditis, valvular calcification, etc

Clinical manifestations

- Chronic mitral regurgitation is often asymptomatic, but acute mitral regurgitation (eg, that resulting from a myocardial infarction) usually manifests as severe congestive heart failure
- Dyspnea(PND), fatigue, chest pain, Wt loss and weakness are the most common symptoms
- Palpitations, shortness of breath on exertion, and cough from pulmonary congestion also occur
- Orthopnea, dyspnea, fatigue, angina, palpitations
- Peripheral edema, jugular vein distention, hepatomegaly (right-sided heart failure)
- Tachycardia, crackles, pulmonary edema
- Auscultation reveals a holosystolic murmur at apex, possible split second heart sound (S2), and an S3

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Mitral regurgitation—

Assessment and diagnostic findings

- A systolic murmur is heard as a high-pitched, blowing sound at the apex. S[redacted]
- The pulse may be regular and of good volume, or it may be irregular as a result of extra systolic beats or atrial fibrillation
- **Echocardiography** is used to visualize valvular lesions & left atrial enlargement
- **ECG**:- may show left atrial and ventricular hypertrophy, sinus tachycardia, and atrial fibrillation
- **X-ray**:- Lt atrial & Rt ventricular enlargement & can show pulmonary congestion
- **Cardiac catheterization**: mitral insufficiency with increased left ventricular end-diastolic volume and pressure, increased atrial pressure and pulmonary artery wedge pressure (PAWP), and decreased cardiac output

Medical management

- Management of mitral regurgitation is the same as that for congestive heart failure
- Surgical intervention consists of mitral valve replacement or valvuloplasty (ie, surgical repair of the heart valve)

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Mitral stenosis

- Mitral stenosis is an obstruction(fibrosis & calcification) of blood flowing from the Lt atrium into the Lt ventricle
- Causes:-rheumatic fever(common), congenital anomalies, atrial myxomas, which progressively **thickens** the mitral valve leaflets and chordae tendineae
- The leaflets often fuse together. Eventually, the mitral valve orifice narrows and progressively obstructs blood flow into the ventricle
- Normally, the mitral valve opening is as wide as the diameter of three fingers(4-6cm^2). In cases of marked stenosis, the opening narrows to (2cm^2)the width of a pencil
- F > M (2:1)

Mitral stenosis—

Pathophysiology

- The left atrium has great difficulty moving blood into the ventricle because of the increased resistance of the narrowed orifice; it dilates (stretches) and hypertrophies (thickens) because of the increased blood volume it holds.
- Because there is no valve to protect the pulmonary veins from the backward flow of blood from the atrium, the pulmonary circulation becomes congested.
- As a result, the right ventricle must contract against an abnormally high pulmonary arterial pressure and is subjected to excessive strain. Eventually, the right ventricle fails
- Lt atrium causes --- P. hypertension--- Pulm.congestion – incompetent pulmonic & tricuspid valve ---- Rt ventricular failure

Mitral stenosis–-

Clinical manifestations

- The first symptom of mitral stenosis is often breathing difficulty (ie, dyspnea (PND,orthopnea)) on exertion as a result of pulmonary venous hypertension
- Patients with mitral stenosis are likely to show progressive fatigue as a result of low cardiac output. They may expectorate blood (ie, hemoptysis -due to pulmonary edema), cough, and experience repeated respiratory infections
- Palpitation – due atrial fibrillation
- Chest pain- due to Lt ventricular hypertrophy & ischemia
- Malar flush over the cheeks (mitral faces)
- Weak & irregular pulse

Consequences of mitral stenosis

- Decrease CO – Lt ventricular failure
- Pulmonary HPN, Rt heart failure, Atrial fibrillation
- Thrombi ma form in the Lt ventricle& they mobilize & travel to brain and kidneys causing infarction

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Mitral stenosis---

Assessment and diagnostic findings

- The pulse is weak and often irregular because of atrial fibrillation (caused by the strain on the atrium)
- Palpation: palpable S [redacted]
- Auscultation: A low-pitched, rumbling, diastolic murmur is heard at the apex, basal cracles- indicate p.edema
- When there is Rt sided H failure & systemic congestion – peripheral edema
- As a result of the increased blood volume and pressure, the atrium dilates, hypertrophies, and becomes electrically unstable, and the patient experiences atrial dysrhythmias
- Echocardiography is used to diagnose mitral stenosis
- Electrocardiography (ECG) and cardiac catheterization with angiography are used to determine the severity of the mitral stenosis

Mitral stenosis––

Investigations

ECG:- Lt atrial elargment, later Rt atrium & ventricle follows

Echocardiography is used to visualize valvular lesions & chamber enlargement,

X-ray:- shows straightening of the Lt heart boarder called mitralization

Medical management

- Antibiotic prophylaxis therapy(B.penicillin-monthly) is instituted to prevent recurrence of infections.
- Anticoagulants to decrease the risk for developing atrial thrombus
- They may also require treatment for anemia
- Salt restriction,digoxin,diuretics,O_2
- Surgical intervention consists of valvuloplasty, usually a commissurotomy to open or rupture the fused commissures of the mitral valve, Percutaneous transluminal valvuloplasty or mitral valve replacement may be performed

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Aortic regurgitation

- Aortic regurgitation is the flow of blood back into the Lt ventricle from the aorta during diastole(M > F ,3:1)
- It may be caused by inflammatory lesions that deform the leaflets of the aortic valve, preventing them from completely closing the aortic valve orifice
- This valvular defect also may result from endocarditis, congenital abnormalities, diseases such as syphilis, a dissecting aneurysm that causes dilation or tearing of the ascending aorta, or deterioration of an aortic valve replacement
- 2/3 rd are caused by RHD

Aortic regurgitation—

Pathophysiology

- In aortic regurgitation, blood from the aorta returns to the Lt ventricle during diastole in addition to the blood normally delivered by the Lt atrium. The Lt ventricle dilates, trying to accommodate the increased volume of blood
- It also hypertrophies, trying to increase muscle strength to expel more blood with above normal force —raising systolic blood pressure
- The arteries attempt to compensate for the higher pressures by reflex vasodilation; the peripheral arterioles relax, reducing peripheral resistance and diastolic blood pressure
- There may be Lt ventricular hyperthrophy & myocardial ischemia

Aortic regurgitation––

Clinical manifestations

- Aortic insufficiency develops without symptoms in most patients
- Some patients are aware of a forceful heartbeat, especially in the head or neck, (Bobbing of the head)
- There may be marked arterial pulsations that are visible or palpable at the carotid or temporal arteries
- This is a result of the increased force and volume of the blood ejected from the hypertrophied left ventricle. Exertional dyspnea and fatigue follow.
- Progressive signs and symptoms of left ventricular failure include breathing difficulties (eg, orthopnea, paroxysmal nocturnal dyspnea), especially at night.
- Palpitation, chest pain in Lt v.failure

Aortic regurgitation––

Assessment and diagnostic findings

- A diastolic murmur is heard as a high-pitched, blowing sound at the third or fourth intercostal space at the Lt sternal border
- Wide pulse pressure, collapsing pulse, water hammer pulse, diastolic thrill at 3rd & 4th ICS of Lt lateral sternal border.
- Diastolic blowing murmur at Lt lateral sternal border
- **Quineke's sign** :-Apply pressure over the pt's nail tip, then, the nail root will be flushed & pale
- **Duroziez's sign**:- Presence of to and fro murmur when the stethoscope is placed over the femoral artery
- Diagnosis may be confirmed by echocardiogram, radionuclide imaging, ECG, magnetic resonance imaging, and cardiac catheterization

Aortic regurgitation––

Investigations

ECG:- Lt ventricular hypertrophy & sinus tachycardia

Echocardiography :- to visualize increased wall motion, dilated Lt ventricle & aortic root & calcified aortic valve

X-ray:- shows dilatation of ascending aorta, Lt ventricular hypertrophy & pulmonary congestion

Medical management

- Before the patient undergoes invasive or dental procedures, antibiotic prophylaxis is needed to prevent endocarditis.
- Treat heart failure and dysrhythmias
- Aortic valvuloplasty or valve replacement is the treatment of choice, preferably performed before left ventricular failure.
- Surgery is recommended for any patient with left ventricular hypertrophy, regardless of the presence or absence of symptoms.

Aortic stenosis

- Aortic valve stenosis is narrowing of the orifice between the left ventricle and the aorta.
- In adults, the stenosis may involve congenital leaflet malformations or an abnormal number of leaflets (ie, one or two rather than three), or it may result from rheumatic endocarditis or cusp calcification of unknown cause.
- The leaflets of the aortic valve may fuse.
- M>F (3:1)

Aortic stenosis––

Pathophysiology

- There is progressive narrowing of the valve orifice, usually over a period of several years to several decades. The left ventricle overcomes the obstruction to circulation by contracting more slowly but with greater energy than normal, forcibly squeezing the blood through the very small orifice.
- The obstruction to left ventricular outflow increases pressure on the left ventricle, which results in thickening of the muscle wall. The heart muscle hypertrophies.
- When these compensatory mechanisms of the heart begin to fail, clinical signs and symptoms develop.

Aortic stenosis––

Causes:- Congenital lesions, atherosclerosis , degenerative calcification

Clinical manifestations

- Many patients with aortic stenosis are asymptomatic. After symptoms develop, patients usually first have excertional dyspnea, caused by Lt ventricular failure. Other signs are dizziness and syncope because of reduced blood flow to the brain
- Angina pectoris is a frequent symptom that results from the increased oxygen demands of the hypertrophied Lt ventricle, the decreased time in diastole for myocardial perfusion, and the decreased blood flow into the coronary arteries.
- Blood pressure can be **low** but is usually normal; there may be a low pulse pressure (30 mm Hg or less) because of diminished blood flow

- Pulsus alternans, diminished carotid pulse

Aortic stenosis––

Assessment and diagnostic findings

- On physical examination, a loud, rough systolic murmur may be heard over the aortic area. The sound to listen for is a systolic rescendo-decrescendo murmur, which may radiate into the carotid arteries and to the apex of the left ventricle
- The murmur is low-pitched, rough, rasping, and vibrating. If the examiner rests a hand over the base of the heart, a vibration may be felt. The vibration is caused by turbulent blood flow across the narrowed valve orifice
- After the stenosis progresses to the point that surgical intervention is considered, left-sided heart catheterization is necessary to measure the severity of the valvular abnormality and evaluate the coronary arteries. Pressure tracingsmare taken from the left ventricle and the base of the aorta.
- The systolic pressure in the left ventricle is considerably higher than that in the aorta during systole
- Decrease B/P & pulse pressure, pulsus alternans,systolic thrill at the base of the heart

Aortic stenosis––

Investigations

ECG:- shows Lt ventricular hypertrophy

Echocardiography :- to visualize Lt ventricular hypertrophy & thickened and narrowed valve

X-ray:- shows Lt ventricular hypertrophy & valvular calcification

Medical management

- Prophylactic antibiotic to prevent infective endocarditis
- Close & periodic follow up
- Treat CHF if it occur
- Definitive treatment for aortic stenosis is surgical replacement of the aortic valve.
- Patients who are symptomatic and are not surgical candidates may benefit from one- or two-balloon percutaneous valvuloplasty procedures

Tricuspid stenosis

- Uncommon disease
- Results from rheumatic fever (mostly), congenital
- Associated with mitral or aortic valve disease
- M>F

S&SXs

- May be symptomatic with dyspnea, fatigue, syncope
- Possibly peripheral edema, jugular vein distention, hepatomegaly, ascites (right-sided heart failure)
- Auscultation reveals mid diastolic murmur at lower left sternal border that increases with inspiration- is diagnostic

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Tricuspid stenosis––

Diagnostic measures

- **Cardiac catheterization**: increased pressure gradient across valve, increased right atrial pressure, and decreased cardiac output

X-ray: right atrial enlargement,pul.congestion

Echocardiography: leaflet abnormality, right atrial enlargement

ECG: Right atrial hypertrophy, right or left ventricular hypertrophy, and atrial fibrillation

Management

Treat CHF if it occurs, Ballon dilatation or valve replacement

Left-sided valve damage results in dyspnea, fatigue, increased pulmonary artery pressure, and decreased cardiac output.

Mitral stenosis	Mitral insufficiency	Aortic stenosis	Aortic insufficiency
• Palpitations • Hemoptysis • Hoarseness • Dysphagia • Jugular vein distention • Orthopnea • Cough • Diastolic murmur • Atrial fibrillation	• Proximal nocturnal Dyspnea • Orthopnea • Palpitations • S_3 and/or S_4 • Crackles in lungs • Systolic murmur • **Atrial fibrillation**	• Angina • Syncope • Decreased SVR • S_3 and/or S_4 • Systolic murmur • Narrowed pulse pressure	• Angina • S_3 • Diastolic murmur • Widened pulse pressure

Right-sided valve damage results in dyspnea, fatigue, increased right atrial pressure, peripheral edema, jugular vein distention, and hepatomegaly

Tricuspid stenosis	Tricuspid insufficiency	Pulmonic stenosis	Pulmonic insufficiency
• Atrial dysrhythmias • Diastolic murmur • Decreased cardiac output	• Conduction delays • Supraventricular tachycardia • Systolic murmur	• Cyanosis • Systolic murmur	• Diastolic murmur

IV. Inflammatory conditions of the heart

A/ Rheumatic endocarditis

Directly attributed to rheumatic fever (group Streptococci)

Clinical features

- Mitral value is most often affected producing left sided heart failure
- The S[redacted]
- Shortness of breath with crackles and wheezes in the lung

RX – directed at eradicating the causative organism

- Antibiotic therapy is initiated (penicillin remains to be choice of drug

Prevention

- Early and adequate treatment of streptococcal infection
- Every nurse should be familiar with the S[redacted]
- Susceptible pts require long term antibiotic

Eg. Penicillin administered before dental checkup is an excellent example

B/ Infective endocarditis

Defn: is a bacterial or fungal infection of endocardium, heart valves, or cardiac prosthesis

Cause: direct infection by bacteria or other organism leading to deformity of the value leaflets causative organisms include. Fungi, bacteria, ricketsiae and streptococcal viridians

Clinical features

- Onset is insidious
- Malaise, cough, back and joint pain, fever is intermittent
- Hemorrhages with pale centers in the eyes/Roth's spot/
- Murmurs: enlargement of the heart or evidence of CHF

Management

- Antibiotics
- Antifungal agent , amphotercin B

Infective endocarditis–-

Complications

- CHF and cerebrovascular accidents such as stroke
- Valvular stenosis , regurgitation
- Myocardial damage

Surgery: Surgical value replacement

Prevention: antibiotic prophylaxis for personas at risk

E.g.

- People undergoing dental procedures
- Tonsillectomy
- Gall bladder surgery
- Vaginal hysterectomy
- Vaginal delivery in the presence of infection
- Surgical operation that involve intestinal or respiratory muscles

C/ Rheumatic Heart Disease (RHD)

V. Pericardial disorders

A/ Pericarditis B/ Cardiac tamponade

Pericarditis

Pericarditis refers to an inflammation and irritation of the pericardium, the fibro-serous sac that envelops(5-30ml), supports, and protects the heart

✓ **Clinically**, pericarditis can be classified as:

1/ Acute pericarditis is characterized by serous, purulent, or hemorrhagic exudates

2/ Chronic (constrictive) pericarditis is characterized by dense, rigid, adherent, ,fibrous ,pericardial thickening that restrict ventricular filling b/ se of chronic inflammation

✓ **Pathologically**, pericarditis can be classified as:

1/ Effusive pericarditis , 2/ Effusive constriction pericarditis, 3/ Constrictive pericarditis & 4/ Adhesive forms

Causes: could be a non specific type

Pericarditis may be idiopathic, or it may result from infection that causes inflammation, connective tissue disorders, immune reactions (Hypersensitivity state), CAD (MI), pneumonia, Tuberculosis , pleural disease, cancer, trauma, uremia or renal failure

Pericarditis---

Signs and symptoms

- Pericarditis may be asymptomatic; when symptoms do occur, the most common is a sharp, piercing, sudden chest pain that typically starts over the sternum and radiates to the neck, shoulders, back, and arms
- Other symptoms include pleuritic pain that increases with deep inspiration and decreases when the patient sits up and leans forward, dyspnea, dry cough, low-grade fever, pericardial friction rub, hypotension, and tachycardia
- Pulsus paradoxus, raised JVP, distant heart sounds, ascites, hepatomegaly, Kussmaul's sign positive(raised JVP during inspiration)

Aggravating factors: the outer part of the heart is inflamed so while bearthing, twisting the body and turning the position on the bed the pain will be aggravated

Relieving factor: sitting up position

Complications include pericardial effusion, cardiac tamponade, and heart failure

Pericarditis—

Diagnosis

Auscultation of precordium reveals friction rub

X-ray: bigger heart, pericardial calcification

Echocardiography: pericardial effusion, thick percardium & small chambers

ECG: ST-segment elevation, T - wave flattening or inversion

Others : WBC count, sedimentation rate, and C-reactive protein, which are all elevated, ASO titre , AFB

Treatment

- Identifying and treating the underlying cause guides therapy
- Bed rest, antibiotics (antiTB),O
- Assess triad Sxs of carardiac tamponade (decrease Bp, rising venous pressure & distant heart sounds)
- Analgesics and non steroidal anti-inflammatory drugs, such as aspirin or ibuprofen , for pain relief during the acute phase
- Diuretics, & salt restriction
- Pericardiocentesis removes some of the pericardial fluid, reduces pressure, and can be cultured to reveal the causative infectious agent
- For recurrent pericarditis ,partial pericardiectomy (to create window to allow fluid to drain in to pleural space)
- For constrictive pericarditis, total pericardiectomy

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NB. Avoid ASA & anticoagulants b/se it may precipitate cardiac tamponade

Pericarditis—

Nursing interventions

- Administer pain medications as needed as well as steroids and other anti-inflammatory agents; give with food to minimize the risk of GI complications
- Administer an antibiotic or antifungal agent based on the underlying causative organism
- Prepare the patient for pericardiocentesis if signs and symptoms of cardiac tamponade develop, which may begin with shortness of breath, chest tightness, or dizziness; developing signs include progressive restlessness and a drop of 10 mm Hg or more in the systolic blood pressure during inspiration (pulsus paradoxus)
- Prepare the patient for pericardectomy or pericardotomy (pericardial window)
- Provide appropriate postoperative care
- Supply oxygen therapy as needed
- Monitor the patient's hemodynamics
- Place the patient upright to relieve dyspnea and chest pain; allow for frequent rest periods, and cluster activities to reduce energy expenditure and oxygen demand
- Encourage the patient to express concerns about the effects of activity restrictions on his normal routines and responsibilities

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Cardiac tamponade

- Presence of excessive fluid and consequent pressure within pericardial cavity sufficient to obstruct ventricular filling is called **cardiac tamponade** (if untreated, cardiogenic shock -- death)
- Pericardial effusion(> 250 ml) refers to the accumulation of fluid in the pericardial sac. This occurrence may accompany pericarditis , advanced HF, metastatic carcinoma, cardiac surgery, trauma, or non traumatic hemorrhage.

Pericardial effusion has the following effects:

- Increased right and left ventricular end-diastolic pressures
- Decreased venous return
- Inability of the ventricles to distend adequately and to fill

- Pericardial fluid may accumulate slowly without causing noticeable symptoms. A rapidly developing effusion, however, can stretch the pericardium to its maximum size and, because of increased pericardial pressure, reduce venous return to the

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Cardiac tamponade---

Causes

- Idiopathic
- Acute pericarditis with effusion
- Trauma (Gunshot, stab wound of the chest)
- Use of anticoagulants in patients with any form of acute pericarditis
- Rupture of the heart or great vessels

Clinical manifestations

- Feeling of fullness within the chest or may have substantial or ill-defined pain. The feeling of pressure in the chest may result from stretching of the pericardial sac.
- Because of increased pressure within the pericardium, venous pressure tends to rise, as evidenced by engorged neck veins.
- Pt prefers sitting up position & leans forward
- Anxiety,restlessness,diaphoresis
- Pallor, cyanosis, neck vein distension,& raised JVP

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Cardiac tamponade –

Clinical manifestations

- Tachycardia,tachypnea,weak and rapid pulses, low BP or shock. Narrow pulse pressure
- Other signs include shortness of breath and a drop and fluctuation in blood pressure. Systolic blood pressure that is detected during exhalation but not heard with inhalation is called **pulsus paradoxus**
- The difference in systolic pressure between the point that it is heard during exhalation and the point that it is heard during inhalation is measured.
- Pulsus paradoxus exceeding 10 mm Hg is abnormal.
- The cardinal signs of cardiac tamponade are falling systolic blood pressure, narrowing pulse pressure, rising venous pressure (increased jugular venous distention), and distant (muffled) heart sounds

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Cardiac tamponade---

Assessment and diagnostic findings

- Pericardial effusion is detected by percussing the chest and noticing an extension of flatness across the anterior aspect of the chest
- **X-ray**: Wide mediastinum & cardiomegaly
- **ECG**: Reduced QRS complex & elevated ST segment
- **An echocardiogram**: Massive effusion,inadequate ventricular filling,& diastolic collapse of Rt ventricle& atrium
- The clinical signs and symptoms and chest x-ray findings are usually sufficient to diagnose pericardial effusion

Management

- If cardiac function becomes seriously impaired: pericardiocentesis ,microbiology& cytology

VI. Myocardial disorders

A/ Myocarditis B/ Cardiomyopathy

Myocarditis

- Myocarditis is a focal or diffuse inflammatory process involving the myocardium; it may be acute or chronic
- The underlying cause is most often an infectious organism(coxsachie viruses group A & B, polio, influenza, rubeola, HIV, bacteria, parasitic infections) that triggers an autoimmune, cellular, and humoral reaction; the heart muscle weakens and contractility decreases; the conduction system can also be affected
- The disorder can result in heart dilation, heart failure, thrombi on the heart wall (mural thrombi),infiltration of circulating blood cells around coronary vessels and between muscle fibers, and degeneration of the muscle fibers themselves
- Most patients with mild signs and symptoms recover completely, but some develop cardiomyopathy, heart failure, and arrhythmias

Myocarditis–-

Signs and symptoms

- The signs and symptoms of acute myocarditis depend on the type of infection, the degree of myocardial damage, and the capacity of the myocardium to recover
- Patients may be asymptomatic, with an infection that resolves on its own
- Initially, flulike signs and symptoms typically occur
- Mild to moderate symptoms include fatigue, dyspnea, palpitations, and occasional discomfort in the chest and upper abdomen
- Severe congestive heart failure can quickly develop, and sudden cardiac death can occur

Myocarditis––

Diagnosis

- P/E: tachycardia, S3 gallop, muffled S1 heart sound
- Laboratory tests include cardiac enzyme levels, including creatine kinase (CK), CK-MB, aspartate aminotransferase, and lactate dehydrogenase, which are elevated; troponin T and I levels are also elevated
- WBC count, C-reactive protein, and erythrocyte sedimentation rate are all elevated
- Antibody titers such as antistreptolysin-O titer in rheumatic fever are elevated
- Stool cultures, throat or pharyngeal washings, and other body fl uid cultures show the causative bacteria or virus
- Diagnostic tests include two-dimensional echocardiography, which may reveal impaired systolic or diastolic ventricular function or both
- A chest X-ray may show cardiomegaly, pulmonary edema, and possible pleural effusions
- Cardiac angiography helps rule out cardiac ischemia as a cause
- MRI reveals the extent of infl ammation and cellular edema
- Biopsy of the endomyocardium can confi rm the diagnosis
- Although electrocardiography can produce highly variable results, it may show sinus tachycardia; diffuse ST-segments; T-wave abnormalities, such as T-wave inversion, ST-segment elevation, and bundle-branch block; conduction defects (prolonged PR interval); and ventricular and supraventricular ectopic arrhythmias

Myocarditis–-

Nursing interventions

- Assess the patient for resolution of tachycardia, fever, and any other clinical manifestations
- Focus your cardiovascular assessment on signs and symptoms of heart failure and arrhythmias
- For a patient with arrhythmias, provide continuous cardiac monitoring, with personnel and equipment readily available to treat life-threatening arrhythmias
- Provide ventricular assistance if needed
- Keep in mind that patients with myocarditis are sensitive to digitalis; closely monitor the patient for indications of digitalis toxicity, such as arrhythmias, anorexia, nausea, vomiting, headache, and malaise
- Use antiembolism stockings and provide passive and active range-of-motion exercises for patients on bed rest to help prevent embolization from venous thrombosis and mural thrombi

Myocarditis––

Complications

Lt sided heart failure , Dilated cardiomyopathy , Arrhythimia , Thromboembolic complications

Treatment

- Treat underlying infections
- Bed rest
- Salt restriction, diuretics, O 
- Avoid NSAIDS(ASA, ibuprofen) can cause further myocardial damage
- Antibiotics, analgesics
- Reassure that it is self limiting condition

Cardiomyopathy

Cardiomyopathy is a disease of the heart muscle, reducing cardiac output and eventually resulting in heart failure

It is classified according to the structural and functional abnormalities of the heart muscle:-

A. Dilated cardiomyopathy(DCM) (*formerly congestive cardiomyopathy)*

B. Hypertrophic cardiomyopathy (HCM)

C. Restrictive or constrictive cardiomyopathy

D. Arrhythmogenic right ventricular cardiomyopathy (ARVC) and

E. Unclassified cardiomyopathy

- *Ischemic cardiomyopathy is a term frequently* used to describe **an enlarged heart** caused by coronary artery disease, which is usually accompanied by heart failure . Regardless of the category and the cause cardiomyopathy may lead to severe heart failure, lethal dysrhythmias, and death

Cardiomyopathy

Causes

- **Dilated cardiomyopathy** include chronic alcoholism, viral or bacterial infection, metabolic and immunologic disorders, and pregnancy and postpartum disorders
- **Hypertrophic cardiomyopathy** include congenital disorders and hypertension
- **Restrictive cardiomyopathy** may be idiopathic, or it may stem from amyloidosis, cancer, or heart transplant
- **Arrhythmogenic right ventricular cardiomyopathy** most likely has a genetic cause and results from the infiltration of fibrous and adipose tissue into the myocardium
- **Unclassified cardiomyopathy** doesn't fit into other categories and can have various causes

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Cardiomyopathy---

Pathophysiology

- The pathophysiology of all cardiomyopathies is a series of progressive events that culminate in impaired cardiac output
- Decreased stroke volume stimulates the sympathetic nervous system and the renin-angiotensin-aldosterone response, resulting in:-

o Increased systemic vascular resistance and

o Increased sodium and fluid retention, which places an increased workload on the heart

- These alterations can lead to heart failure

A/ Dilated Cardiomyopathy (DCM)

- DCM is the most common form of cardiomyopathy
- DCM occurs more often in men and African Americans, who also experience higher mortality rates
- DCM is distinguished by significant dilation of the ventricles without significant concomitant hypertrophy (ie, increased muscle wall thickness) and systolic dysfunction
- DCM was formerly named *congestive cardiomyopathy, but DCM may exist without signs and symptoms* of congestion

Dilated Cardiomyopathy(DCM) –-

- Microscopic examination of the muscle tissue shows diminished contractile elements of the muscle fibers and diffuse necrosis of myocardial cells
- The result is poor systolic function. These structural changes decrease the amount of blood ejected from the ventricle with systole, increasing the amount of blood remaining in the ventricle after contraction
- Less blood is then able to enter the ventricle during diastole, increasing end-diastolic pressure and eventually increasing pulmonary pressures
- Altered valve function can result from the enlarged stretched ventricle, usually resulting in regurgitation. Embolic events caused by ventricular and atrial thrombi as a result of the poor blood flow through the ventricle may also occur

Dilated Cardiomyopathy(DCM) –-

- More than 75 conditions and diseases may cause DCM, including pregnancy, heavy alcohol intake, and viral infection (eg, influenza). When the causative factor cannot be identified, the term used is *idiopathic DCM*
- *Idiopathic* DCM accounts for approximately 25% of all heart failure cases. Early diagnosis and treatment can prevent or delay significant symptoms and sudden death from DCM
- Echocardiography and ECG are used to diagnose DCM and should be conducted for all first-degree relatives (eg, parents, siblings, children) of patients with DCM

B/ Hypertrophic Cardiomyopathy(HCM)

- In HCM, the heart muscle increases in size and mass, especially along the septum. The increased thickness of the heart muscle reduces the size of the ventricular cavities and causes the ventricles to take a longer time to relax, making it more difficult for the ventricles to fill with blood during the first part of diastole and making them more dependent on atrial contraction for filling
- The increased septal size may misalign the papillary muscles so that the septum and mitral valve obstruct the flow of blood from the left ventricle into the aorta during ventricular contraction
- Hence, HCM may be obstructive or non obstructive. Because of the structural changes, HCM had also been called idiopathic hypertrophic sub aortic stenosis (IHSS) or asymmetric septal hypertrophy (ASH).

Hypertrophic Cardiomyopathy(HCM)–-

- Structural changes may also result in a smaller than normal ventricular cavity and a higher velocity flow of blood out of the left ventricle into the aorta, which may be detected by echocardiography
- HCM may cause significant diastolic dysfunction, but systolic function can be normal or high, resulting in a higher than normal ejection fraction
- Because HCM is a genetic disease, family members are observed closely for signs and symptoms indicating development of the disease
- HCM is rare, occurring in men, women, and children (often detected after puberty) with an estimated prevalence rate of 0.05% to 0.2%
- It may also be idiopathic (ie, no cause can be found)

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C/ Restrictive Cardiomyopathy(RCM)

- RCM is characterized by diastolic dysfunction caused by rigid ventricular walls that impair ventricular stretch and diastolic filling
- Systolic function is usually normal. Because RCM is the least common cardiomyopathy, representing approximately 5% of pediatric cardiomyopathies, its pathogenesis is the least understood
- Restrictive cardiomyopathy can be associated with amyloidosis (in which amyloid, a protein substance, is deposited within the cells) and other such infiltrative diseases. However, the cause is unknown in most cases (ie, idiopathic)

D/ Arrhythmogenic Right Ventricular Cardiomyopathy (ARVC)

- ARVC occurs when the myocardium of the right ventricle is progressively infiltrated and replaced by fibrous scar and adipose tiss
- Initially, only localized areas of the right ventricle are affected, but as the disease progresses, the entire heart is affected.
- Eventually, the right ventricle dilates and develops poor contractility, right ventricular wall abnormalities, and dysrhythmias

D/ Arrhythmogenic Right Ventricular Cardiomyopathy (ARVC)

- ARVC should be suspected in patients with ventricular tachycardia originating in the right ventricle (ie, a left bundle branch block configuration on ECG) or sudden death, especially among previously symptom-free athletes
- The disease may be genetic (ie, autosomal dominant)
- Family members should be screened for the disease with a 12-lead ECG, Holter monitor, and echocardiography

E/ Unclassified Cardiomyopathies

- Unclassified cardiomyopathies are different from or have characteristics of more than one of the previously described cardiomyopathies
- Examples of unclassified cardiomyopathies include fibro-elastosis, non compacted myocardium, systolic dysfunction with minimal dilation, and mitochondrial involvement

Cardiomyopathy---

Signs and symptoms

- Tachycardia, S3 and S4 heart sounds, exertional dyspnea, paroxysmal nocturnal dyspnea, cough, fatigue, jugular venous distention, dependent pitting edema, peripheral cyanosis, and hepatomegaly
- Heart murmurs and arrhythmias may also occur
- Frequently, dilated and restrictive cardiomyopathy are first diagnosed when the patient presents with signs and symptoms of heart failure (eg, dyspnea on exertion, fatigue)
- Paroxysmal nocturnal dyspnea, cough (especially with exertion), and orthopnea, which may lead to a misdiagnosis of bronchitis or pneumonia
- Fluid retention, peripheral edema, and nausea, which is caused by poor perfusion of the gastrointestinal system. The patient may experience chest pain, palpitations, dizziness, nausea, and syncope with exertion. However, with HCM, cardiac arrest (ie, sudden cardiac death) may be the initial manifestation in young people, including athletes

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Cardiomyopathy–-

Medical management

- Medical management is directed toward determining and managing possible underlying or precipitating causes; correcting the heart failure with medications, a low-sodium diet, and an exercise rest regimen ; and controlling dysrhythmias with antiarrhythmic medications and possibly with an implanted electronic device, such as an implantable cardioverter-defibrillator
- If patients exhibit signs and symptoms of congestion, their fluid intake may be limited to 2 liters each day. The person with HCM may also have to limit physical activity to avoid a life-threatening dysrhythmia
- A pacemaker may be implanted to alter the electrical stimulation of the muscle and prevent the forceful hyperdynamic contractions that occur with HCM

Cardiomyopathy–-

Surgical management

- Surgical intervention, including heart transplantation, is considered
- However, because of the limited number of organ donors, many patients die waiting for transplantation.
- In some cases, a left ventricular assist device (LVAD) is implanted to support the failing heart until a suitable donor heart becomes available

Cardiomyopathy---

Diagnosis and treatment

- Diagnostic tests include electrocardiogram (ECG), echocardiogram, cardiac catheterization, radionuclide studies, and chest X-ray
- Medications for dilated cardiomyopathy include an angiotensin-converting enzyme (ACE) inhibitor or hydralazine plus a nitrate (the mainstay of therapy), a beta-adrenergic blocker, digoxin, a diuretic, and an anticoagulant
- Medications for hypertrophic cardiomyopathy include a beta-adrenergic blocker and a calcium channel blocker
- No specific medications are used to treat restrictive cardiomyopathy; however, diuretics, digoxin, nitrates, and other vasodilators can worsen the condition and should be avoided
- An antiarrhythmic, a pacemaker, or an implantable cardiac defibrillator may be necessary to control arrhythmias
- Surgery, such as heart transplantation or cardiomyoplasty (for dilated cardiomyopathy) or ventricular myotomy or myectomy (for hypertrophic obstructive cardiomyopathy) may be indicated if medications fail

Cardiomyopathy–––

Nursing interventions

- Monitor ECG results, cardiovascular status, vital signs, and hemodynamic variables to detect heart failure and arrhythmias and assess the patient's response to medications
- If the patient is receiving a diuretic, monitor his serum electrolyte levels to detect abnormalities such as hypokalemia
- Administer oxygen and keep the patient in semi-Fowler's position to promote oxygenation
- Make sure the patient restricts activity if necessary to reduce oxygen demands on the heart
- Teach the patient the signs and symptoms of heart failure he should report to the practitioner
- Explain the importance of checking his weight daily and reporting an increase of 1.4 kg or more
- Encourage the patient to express his feelings such as a fear of dying

VI. Heart Failure(HF)

- HF is the inability of the heart to maintain adequate circulation to meet tissue needs for oxygen and nutrients

- HF occurs when the heart muscle is unable to pump effectively, resulting in inadequate cardiac output, myocardial hypertrophy, and pulmonary/systemic congestion

- HF is the result of an acute or chronic cardiopulmonary problem, such as systemic hypertension, myocardial infarction, pulmonary hypertension, dysrhythmias, valvular heart disease, pericarditis, and cardiomyopathy

HF––

- Severity of HF is graded on the New York Heart Association's functional classification scale indicating how little, or how much, activity it takes to make the client symptomatic (chest pain, shortness-of-breath)

- **Class I**: Client exhibits no symptoms with activity
- **Class II**: Client has symptoms with ordinary exertion
- **Class III**: Client displays symptoms with minimal exertion
- **Class IV**: Client has symptoms at rest

Different forms of heart failure

A. High out put failure

- The cardiac out put is normal or above normal but is unable to meet the body's need
- An uncommon form of heart failure

Causes: Anemia, pregnancy, Hyperthyroidism, atrioventricular fistula ,beriberi

B. Low out put failure

- Cardiac out put are below normal

Causes: Hypertension, MI, arteriosclerosis, dilated cardiomyopathy, valvular & pericardial disease

HF––

- Low output HF can initially occur on either the left or right side of the heart

A. **Left-sided heart (ventricular) failure** results in inadequate left ventricle (cardiac) output and consequently in inadequate tissue perfusion. Forms include:
  - **Systolic** heart (ventricular) failure (ejection fraction below 40%, pulmonary and systemic congestion)
  - **Diastolic** heart (ventricular) failure (inadequate relaxation or "stiffening" prevents ventricular filling), ejection fraction is normal

B. **Right-sided heart (ventricular) failure** results in inadequate right ventricle output and systemic venous congestion (for example, peripheral edema)

➢ Acute Vs chronic HF

HF––

Risk Factors/Causes/

- Left-Sided Heart (Ventricular) Failure
  - Hypertension, CHD
  - Valvular disease (mitral and aortic)
- Right-Sided Heart (Ventricular) Failure
  - Left-sided heart (ventricular) failure
  - Right ventricular myocardial infarction
  - Pulmonary problems (COPD, ARDS)
- High-Output Heart Failure
  - Increased metabolic needs, Septicemia (fever)
  - Anemia, Hyperthyroidism
- Cardiomyopathy
  - Coronary artery disease
  - Infection or inflammation of the heart muscle
  - Various cancer treatments, Prolonged alcohol abuse, Heredity

HF---

Signs and symptoms

Left-sided failure

- Dyspnea on exertion, orthopnea, nocturnal dyspnea,PND
- Fatigue, pallor, cyanosis
- Displaced apical pulse, pulsus alternans
- S3 heart sound (gallop),tachycardia
- Pulmonary congestion (dyspnea, cough, bibasilar crackles)
- Frothy sputum (may be blood-tinged)
- Altered mental status(confusion, disorientation)
- Symptoms of organ failure, such as oliguria

Hemodynamic findings:-

- CVP/right atrial pressure (N = 1 - 8 mm Hg): Normal or elevated
- PAP (N = 15 to 26 mm Hg/5 to 15 mm Hg/): Elevated
- CO (N = 4 to 7 L/min): Decreased

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Table 30-1 • Hemodynamic Parameters

PARAMETER	RIGHT VENTRICLE	LEFT VENTRICLE
Preload		
Normal Values	CVP: 0–8 mm Hg	PAWP: 4–12 mm Hg
Afterload		
Normal Values	PVR: 20–120 dyne/sec/cm^{-5}	SVR: 800–1500 dyne/sec/cm^{-5}
Calculation	$\frac{\text{mean PAP} - \text{PAWP}}{\text{CO}} \times 80$	$\frac{\text{Mean arterial pressure} - \text{CVP}}{\text{CO}} \times 80$
Contractility		
Normal Values	Right ventricular stroke work index: 7–12 g/beat/m^2	Left ventricular stroke work index: 35–85 g/beat/m^2
Calculation	$\frac{(\text{PA systolic pressure} - \text{CVP}) \times \text{SV} \times 0.0136}{\text{Body Surface Area (height and weight)}}$	$\frac{(\text{Systolic BP} - \text{PAWP}) \times \text{SV} \times 0.0136}{\text{Body Surface Area (height and weight)}}$

HF—

Signs and symptoms—

Right-sided failure

- Jugular vein distention
- Ascending dependent edema (legs, ankles, sacrum, buttocs)
- Abdominal distention(bloating), ascites
- Fatigue, weakness
- Nausea and anorexia
- Nocturnal diuresis
- Liver enlargement (hepatomegaly) and tenderness
- Weight gain

Hemodynamic findings

- CVP/right atrial pressure (normal = 1 to 8 mm Hg): Elevated

Cardiomyopathy

- Fatigue, weakness
- Heart failure (left with dilated type, right with restrictive type)
- Dysrhythmias (for example, heart block)
- Cardiomegaly

Framingham's criteria for diagnosis of CHF

Major	Minor
- Neck vein distension - Cardiomyopathy - Acute pulmonary congestion - Increased CVP …..etc	- Peripheral edema - Night cough - Dyspnea on exertion - Hepatomegaly - Pleural effusion -Tachycardia(> 120) ✓At least **One major** & **two minor** criteria

HF––

Diagnostic procedures

- BNP < 100 pg/mL = No HF
- BNP = 100 to 300 pg/mL suggest heart failure is present;

BNP > 300 pg/mL = Mild HF

BNP > 600 pg/mL = Moderate HF

BNP > 900 pg/mL = Severe HF

Chest X-ray :- Cardiomegaly and pleural effusions

Electrocardiogram (ECG), cardiac enzymes, electrolytes, and arterial blood gases

- Assess factors contributing to heart failure and/or the impact of heart failure.

HF––

Diagnostic procedures

- **Ultrasound** to measure both systolic and diastolic function of the heart
- **LVEF** : The volume of blood pumped from the left ventricle into the arteries upon each beat
  Normal is 55 - 70 %
- **RVEF**: The volume of blood pumped from the right ventricle to the lungs upon each beat
  Normal is 45 – 60 %
- CBC, electrolytes ,RF, LF, thyroid function tests

HF––

Assess/monitor

- Oxygen saturation
- Vital signs
- Heart rhythm
- Lung sounds for crackles, wheezes
- Level of dyspnea upon exertion
- Serum electrolytes (especially potassium if receiving diuretics)
- Daily weight
- Changes in level of consciousness
- Intake and output
- For signs of drug toxicity
- Coping ability of client and family

HF––

Nursing interventions

- If a client is experiencing respiratory distress, place the client in high-Fowler's position and administer oxygen as prescribed
- Encourage bed rest until the client is stable
- Encourage energy conservation by assisting with care and activities of daily living
- Maintain dietary restrictions as prescribed (restricted fluid intake, restricted sodium intake)
- Monitor the patient for common signs and symptoms of heart failure, such as chest discomfort, shortness of breath, and paroxysmal nocturnal dyspnea
- Watch for signs and symptoms of left-sided heart failure, such as anxiety, orthopnea, and abnormal breath sounds

Nursing interventions––

- Monitor for signs and symptoms of right-sided heart failure, such as jugular venous distension, hepatomegaly , spleenomegaly , peripheral edema, and bounding peripheral pulses
- Encourage bed rest in semi-Fowler's position for ease of breathing
- Provide rest intervals between periods of activity
- Restrict fluids & salt as prescribed
- Administer medications as prescribed, and monitor for their therapeutic and adverse effects

Nursing interventions–-

- Monitor fluid intake and output
- Administer oxygen as prescribed
- Monitor vital signs carefully, especially when administering vasoactive drugs
- Check the patient’s weight daily
- Frequently assess for cardiac and respiratory signs of heart failure
- Note changes that suggest worsening of heart failure or fluid imbalance
- Explain procedures and provide reassurance to decrease patient and family anxiety
- Teach the patient and family about medications and the importance of careful management of fluids, sodium intake, and weight

HF—

Administer medications as prescribed

- **Diuretics**: To decrease preload
  - Loop diuretics, such as furosemide , bumetanide
  - Thiazide diuretics, such as hydrochlorothiazide
  - Potassium-sparing diuretics, spironolactone
- Teach the client to take foods and drinks that are high in potassium
- Potassium supplementation may be required (Lasix) no faster than 20 mg/min
- **Inotropic agents**, such as digoxin, dopamine, dobutamine , milrinone (Primacor): To increase contractility and thereby improve cardiac output
- **Vasodilators**, such as nitrates: To decrease preload and afterload

HF—

- **Afterload-reducing agents**
  - Angiotensin converting enzyme (ACE) inhibitors, such as enalapril , captopril ; monitor for initial dose hypotension
  - Beta-blockers, such as carvedilol , metoprolol
  - Angiotensin receptor II blockers, such as losartan
- **Anticoagulants:** warfarin , heparin, clopidogrel: To prevent thrombus formation (risk associated with congestion/stasis and associated atrial fibrillation)

HF––

- Teach clients who are self-administering digoxin to:
  - Count pulse for one full minute before taking the medication.
  - If the pulse rate is irregular or the pulse rate is outside of the limitations set by the provider (usually less than 60 or greater than 100), instruct the client to hold the dose and to contact the primary care provider
  - Take digoxin dose at same time each day
  - Do not take digoxin at the same time as antacids separate by 2 hr
  - Report signs of toxicity, including fatigue, muscle weakness, confusion, and loss of appetite
  - Regularly have digoxin and potassium levels checked
  - Provide emotional support to the client and family

HF—

Dosage of digoxin

- Start with 1 mg over 24 – 48 hrs, when rapid effect is required & then 0.25mg po daily. Monitoring of serum digoxin is important

✓ Digoxin is contraindicated in:-

- o Acute MI
- o AV conduction disturbance
- o Hypomagnesemia
- o Chronic lung disease
- o Hypokalemia

Cardiac arrest

- Cardiac arrest occurs when the heart ceases to produce an effective pulse and blood circulation.
- It may be caused by a cardiac electrical event, as when the HR is too fast (especially ventricular tachycardia or ventricular fibrillation) or too slow (bradycardia or AV block) or when there is no heart rate at all (asystole).
- Cardiac arrest may follow respiratory arrest; it may also occur when electrical activity is present but there is ineffective cardiac contraction or circulating volume, which is called pulseless electrical activity (PEA). Formerly called electrical-mechanical dissociation (EMD)
- PEA can be caused by hypovolemia (eg, with excessive bleeding), cardiac tamponade, hypothermia, massive pulmonary embolism, medication overdoses (eg, tricyclic agents, digitalis, beta-blockers, calcium channel blockers), significant acidosis, and massive acute myocardial infarction

Cardiac arrest–-

- Ventricular fibrillation (VF) is the cause of sudden, non-traumatic cardiac arrest in 80 to 90% of victims
- Without sufficient cardiac output, the brain will suffer cell anoxia (cell death) within 4 to 6 min, with death following shortly thereafter
- An interdisciplinary team will provide care during in-hospital cardiac arrests
- This team should include nurses, physicians, respiratory therapists, laboratory personnel, and chaplain services
- Management of cardiac arrest depends on prompt recognition of signs and symptoms and the introduction of therapeutic interventions directed at artificially sustaining circulation and ventilation

Cardiac arrest---

Causes

1/ Tachyarrhythmia (VT,VF,AF,SVT)

2/ Cardiac dysfunction (Asystole,AS,Congenital HD, cor-pulmonale, cardiac tamponade)

3/ Pulseless electrical activity (PEA) or EMD

4/ Electrolyte disorders(hyperkalemia, hypokalemia)

5/ Acidosis

6/ Bradyarrhythmia

7/ Others(electric shock, hypothermia, severe hypovolemia, T. pneumothorax)

Cardiac arrest––

Clinical manifestations

- Consciousness, pulse, and blood pressure are lost immediately
- Ineffective respiratory gasping may occur
- The pupils of the eyes begin dilating within 45 seconds
- Seizures may or may not occur
- The risk of irreversible brain damage and death increases with every minute from the time that circulation ceases
- The interval varies with the age and underlying condition of the patient
- During this period, the diagnosis of cardiac arrest must be made, and measures must be taken immediately to restore circulation
- The most reliable sign of cardiac arrest is the absence of a pulse (In the adult and the child, the carotid pulse)
- In an infant, the brachial pulse is assessed
- Valuable time should not be wasted taking the blood pressure, listening for the heartbeat, or checking proper contact of electrodes

Cardiac arrest–-

The goals for management of a cardiac arrest include:

- Rapid identification of the signs and symptoms of cardiac arrest
- Quick initiation of both circulatory and respiratory support
- Activation of the emergency medical system (EMS)
- Utilization of emergency equipment and cardiac monitoring
- Stabilization of the client following the arrest
- Diagnosis and treatment of the cause of the cardiac arrest

- CPR significantly increases the chances of survival when initiated immediately.
- Pre-hospital care greatly improves the chance of survival for cardiac arrest victims
- CPR is a component of basic life support (BLS) and advanced cardiac life support (ACLS)

Cardiac arrest–-

BLS involves the ABCs of CPR:-

Airway

- Confirm the absence of spontaneous respirations
- Establish a patent airway
- Provide the Heimlich maneuver if the airway is obstructed with a foreign object
- Use abdominal thrusts for unconscious clients

Breathing

- Provide artificial respirations (ventilations) to deliver oxygen into the blood in an attempt to prevent cell anoxia

Circulation

- Confirm the absence or presence of a pulse
- Provide external support of circulation (chest compressions) to transport oxygenated blood to the brain

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Cardiac arrest---

In addition to the ABCs of BLS, ACLS involves:-

- Diagnosis of underlying cardiac dysrhythmias
- Pharmacological interventions are dependent upon the rhythm identified
- Defibrillation may be required
- Insertion of an oropharyngeal or endotracheal airway with bag ventilation and supplemental oxygen
- Administration of IV fluids
- Administration of IV antidysrhythmic drugs
- The chain of survival is a series of interventions directed at the resuscitation of the cardiac arrest victim. It involves:-

- Early activation of the emergency medical services
- Early CPR/early defibrillation
- Early ACLS care

Cardiac arrest–-

Nursing responsibilities include:-

- Maintain airway patency
- Assess the depth and rate of respirations
- Provide chest compressions at the appropriate rate and depth for age
- Assess vital signs for effectiveness of chest compressions and ventilations
- Defibrillate the client when indicated
- Obtain and maintain IV access
- Provide medications as ordered
- Monitor laboratory values (for example, ABGs, CBC, electrolytes)
- Document all interventions and medications

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Cardiac arrest–-

ACLS protocols

1/ VF or Pulseless Ventricular Tachycardia **(VT)**

- Perform CPR (stop when the defibrillator is ready)
- Provide oxygen
- Defibrillate: 200 joules, 300 joules ,360 joules (> CPR)
- Establish IV access
- Administer epinephrine 1 mg IV push every 3 to 5 min OR vasopressin 40 units IV x 1 only (switch to epinephrine if no response)
- Defibrillate: 360 joules within 30 to 60 seconds
- Consider the following medications:-

  Amiodarone, lidocaine, magnesium sulfate, procainamide, bicarbonate

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Cardiac arrest–––

2/ Pulseless Electrical Activity **(PEA)**

- Perform CPR , Provide oxygen
- Defibrillate for VF or pulseless VT, & Establish IV access

Consider the most common causes:-

5 H's

- Hypovolemia, Hypoxia , Hypothermia
- Hydrogen ion accumulation, resulting in acidosis
- Hyperkalemia or hypokalemia

5 T's

- Tables (accidental or deliberate drug overdose)
- Tamponade (cardiac), Tension pneumothorax
- Thrombosis (coronary), Thrombosis (pulmonary)
- Administer epinephrine 1 mg IV push every 3 to 5 min
- If PEA rate is slow (bradycardic), administer atropine 1 mg IV every 3 to 5 min (maximum total dose 0.04 mg/kg)

Cardiac arrest–-

3/ Asystole

- Perform CPR
- Provide oxygen
- Defibrillate for VF or pulseless VT
- Confirm true asystolic rhythm
- Establish IV access
- Begin immediate transcutaneous pacing, if possible.
- Administer epinephrine 1 mg push every 3 to 5 min
- Administer atropine 1 mg IV every 3 to 5 min (maximum total dose 0.04 mg/kg)
- Consider ceasing resuscitation if asystole persists

Cardiac arrest––

EX. For each of the following dysrhythmias, identify the appropriate interventions

Dysrhythmia	Interventions
1. VF or pulseless VT	A, B, C, F, G, H
2. Pulseless Electrical Activity (PEA)	A, B, C, D, E, F, H
3. Asystole	A, B, C, E, F, H

Interventions:-

A. Perform CPR

B. Administer epinephrine 1 mg IV push every 3 to 5 min.

C. Obtain IV access

D. Consider the 5 H's and 5 T's

E. Administer atropine 1 mg IV every 3 to 5 min (max total dose 0.04 mg/kg)

F. Defibrillate up to three times:-

- Defibrillate: 200 joules.
- Defibrillate: 300 joules.
- Defibrillate: 360 joules.

G. Consider administering:-

- Amiodarone , Lidocaine, Magnesium sulfate , Procainamide, Bicarbonate

H. Provide oxygen

Cardiac arrest–-

Answer

EX. For each of the following dysrhythmias, identify the appropriate interventions

Dysrhythmia	Interventions
1. VF or pulseless VT	A, B, C, F, G, H
2. Pulseless Electrical Activity (PEA)	A, B, C, D, E, F, H
3. Asystole	A, B, C, E, F, H

Defibrillation (0-4 minutes)
CPR (4-10 minutes)
Metabolic > 10 minutes

Acute pulmonary edema

- Acute pulmonary edema is a life-threatening medical emergency. Pulmonary edema is the collection of fluid in the interstitium and alveoli of the lungs as pressure rises in the pulmonary vessels
- It can result from ARDS, fluid overload, left-sided heart failure, mitral stenosis, MI, or pulmonary emboli
- With pulmonary edema, the left ventricle can't effectively pump blood from the heart
- With increased resistence to left ventricular filling, fluid backs up into the lungs
- Surface tension increases, the alveoli shrink, and the lungs become stiff, making breathing more difficult
- Hypoxemia and an altered $\dot{V}/\dot{Q}$ ratio develop
- Fluid moves into the larger airways, where it's coughed up as pink, frothy sputum

Acute pulmonary edema---

Signs and symptoms

- Symptoms include anxiety, tachycardia, acute respiratory distress, dyspnea at rest, change in level of consciousness, and an ascending fluid level within lungs
- Tachycardia and tachypnea may be accompanied by narrowed pulse pressures and hypotension; third and fourth heart sounds may be present; skin may be cold and clammy
- Dyspnea, increased respiratory rate, orthopnea, and pulmonary hypertension may occur
- Jugular veins may be distended, and PAWP may be elevated
- Coughing may produce blood-tinged or pink, frothy sputum
- Lung auscultation may reveal dependent crackles
- Other signs and symptoms may include confusion, decreased urine output, diaphoresis, drowsiness, lethargy, and restlessness

Acute pulmonary edema---

Diagnosis and treatment

- Chest X-ray, pulse oximetry, and ABG studies typically are prescribed
- A PA catheter is inserted to measure pressures
- A diuretic is administered to decrease edema
- Other drugs that may be administered include an inotropic drug to increase myocardial contractility, nitroglycerin to reduce preload and afterload, I.V. nitroprusside to reduce preload and afterload, and a vasopressor to maintain blood pressure
- Intubation and mechanical ventilation may be necessary to treat respiratory distress
- Morphine is administered to decrease preload, respiratory rate, and anxiety
- Patients who don't respond to drug therapy may be treated with an intra-aortic balloon pump, which temporarily assists the failed left ventricle, or with surgery (such as angioplasty, coronary artery bypass grafting, or valvular repair), depending on the underlying heart condition

Acute pulmonary edema---

Nursing interventions

- Administer oxygen to aid ventilation, improve Pao2, and reverse hypoxemia
- IV morphine (to decrease anxiety, respiratory distress, and decrease venous return)
- IV administration of rapid-acting loop diuretics, such as furosemide
- Place the patient in semi-Fowler's position to maximize oxygenation and increase comfort
- Carefully monitor fluid intake and output to assess the effectiveness of diuretic therapy and prevent sudden increases in venous return caused by oral and I.V. intake
- Frequently change the patient's position to prevent pressure ulcers and encourage lung expansion

Cor-pulmonale

- **Cor pulmonale** is a condition in which the right ventricle of the heart enlarges (with or without right-sided heart failure) as a result of diseases that affect the structure or function of the lung or its vasculature.
- Any disease affecting the lungs and accompanied by hypoxemia may result in cor pulmonale.
- The most frequent cause is severe COPD , in which changes in the airway and retained secretions reduce alveolar ventilation.
- Other causes are conditions that restrict or compromise ventilatory function, leading to hypoxemia or acidosis (deformities of the thoracic cage, massive obesity), or conditions that reduce the pulmonary vascular bed (primary idiopathic pulmonary arterial hypertension, pulmonary embolus).
- Certain disorders of the nervous system, respiratory muscles, chest wall, and pulmonary arterial tree also may be responsible for cor pulmonale

Cor-pulmonale---

- In cor pulmonale, hypertrophy and dilation of the right ventricle secondary to disease affect the structure or function of the lungs or their vasculature, resulting in right-sided heart failure
- The disorder can occur at the end stage of various chronic disorders of the lungs, pulmonary vessels, chest wall, and respiratory control center
- Pulmonary hypertension increases the heart's workload
- To compensate, the right ventricle hypertrophies to force blood through the lungs
- In response to hypoxia, the bone marrow produces more red blood cells, causing polycythemia; the resulting increased viscosity further aggravates pulmonary hypertension and increases right ventricular workload
- Causes include primary pulmonary hypertension, pulmonary embolism, asthma, connective tissue disorders, COPD (the cause in more than half of all cases), chronic severe tricuspid regurgitation, disorders affecting the pulmonary parenchyma, and neuromuscular disease
- Cor pulmonale accounts for approximately 6% to 8% of all types of heart disease in adults in the United States
- Patients with cor pulmonale are typically older than age 45; males are more likely to be affected than females

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Cor-Pulmonale---

Signs and symptoms

- Signs and symptoms include a history of dyspnea, chronic productive cough, fatigue, and weakness
- Other signs and symptoms include tachypnea, wheezing, chest wall retractions, hemoptysis, pitting edema in the extremities, distended jugular veins, an enlarged liver, and tachycardia with pansystolic murmur at the lower left sternal border

Cor-Pulmonale–-

Diagnosis

- ABG analysis reveals decreased Pao2 (usually less than 70 mm Hg and rarely more than 90 mm Hg), hypercapnia, and hypoxia
- Hematocrit is typically over 50%
- Serum liver enzyme levels may show an elevated level of aspartate aminotransferase
- Brain natriuretic peptide level may be elevated
- Chest x-ray, echocardiography, angiography, and magnetic resonance imaging (MRI) demonstrate right ventricular enlargement
- An ECG shows arrhythmias and may show atrial fibrillation and right bundle-branch block
- Pulmonary function studies reflect underlying pulmonary disease
- A hemodynamic profile shows increased pulmonary vascular resistance

Cor-Pulmonale---

Treatment

- The key to treatment is correcting the underlying problem
- Oxygen therapy improves oxygenation
- Phlebotomy is indicated for patients with COPD with a hematocrit of 55% or more
- Continuous positive airway pressure or biphasic positive air pressure is indicated for sleep apnea
- The patient should be on moderate sodium restriction and diuretics
- Patients should limit activity as tolerated; during the acute phase, patients should be on bed rest
- Beta selective agonists, such as epoprostenol , treprostinil , and iloprost , are used to treat primary pulmonary hypertension
- Bronchodilators administered by nebulizer include ipratropium , metaproterenol, and albuterol
- For patients with persistent disease, vasodilators include hydralazine, nifedipine , diltiazem , and prazosin
- The endothelin-1 receptor antagonists bosentan can help patients with pulmonary hypertension and severe symptoms to improve exertional tolerance and increase walking distance
- Antibiotics treat acute respiratory infections
- Anticoagulants help prevent thromboembolism

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Cor-Pulmonale---

Nursing interventions

- Monitor the patient's vital signs, and pay attention to his cardiac and respiratory status
- Reposition the patient often; elevate the head of the bed to increase thoracic expansion and ease the work of breathing
- Administer oxygen as ordered based on oxygen saturation levels obtained with pulse oximetry and ABG results
- Give prescribed drugs; if the patient will receive I.V. diuretics, ensure patent I.V. access
- Encourage the patient to take slow, deep breaths when using nebulized medications, as appropriate
- Provide frequent rest periods; cluster nursing activities to minimize oxygen and metabolic demands
- Teach the patient and family about the disorder, the patient's diagnosis, the underlying cause and its relationship to the patient's current condition, treatment, and follow-up care

VIII. Congenital Heart Diseases

	Relative Frequency of Lesions
• Ventricular septal defect	25-30
• Atrial septal defect (secundum)	6-8
• Patent ductus arteriosus	6-8
• Coarctation of aorta	5-7
• Tetralogy of Fallot	5-7
• Pulmonary valve stenosis	5-7
• Aortic valve stenosis	4-7
• Transposition of great arteries	3-5
• Hypoplastic left ventricle	1-3
• Hypoplastic right ventricle	1-3
• Truncus arteriosus	1-2
• Total anomalous pulm venous return	1-2
• Tricuspid atresia	1-2
• Double-outlet right ventricle	1-2
• Others	5-10

Classifications

Acyanotic (Noncyanotic CHD (L →P))

- Interrupted Aortic Arch
- Ventricular Septal Defect (VSD)
- Atrial Septal Defect (ASD)
- Patent ductus arteriosus (PDA)

Obstruction to blood flow:-

- Pulmonic stenosis (PS)
- Aortic stenosis (AS)
- Aortic coarctation

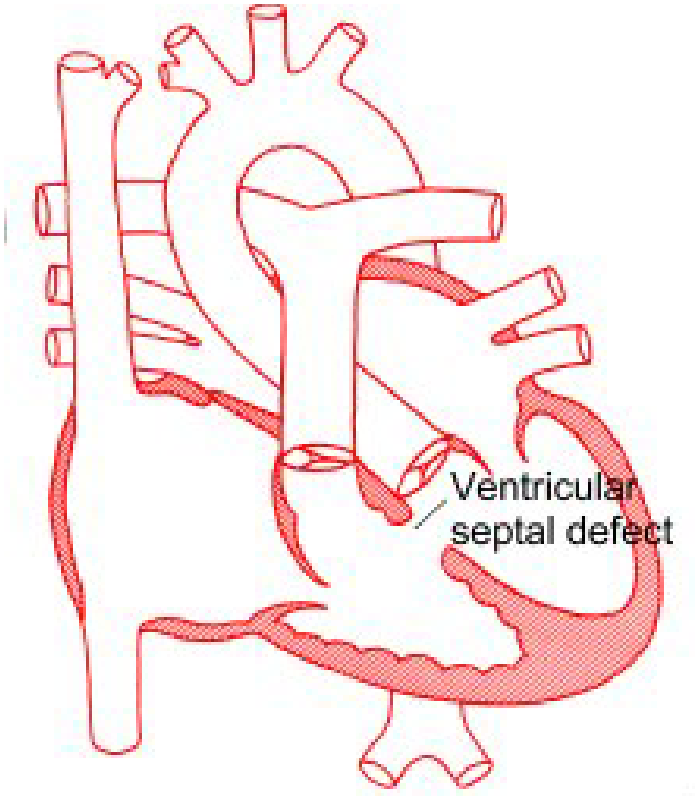
Cyanotic Defects (R-- L)

- **T**etralogy of Fallot (TOF)
- **T**ricuspid atresia (TA)
- **T**otal anomalous pulmonary venous return (TAPVR)
- **T**runcus arteriosus
- **T**ransposition of the great vessels
- **H**ypoplastic left heart syndrome (HLH)
- Pulmonary atresia (PA) / critical PS
- Double outlet right ventricle (DORV)

Ventricular Septal Defect (VSD)

- Developes between the 4th and 8th weeks of gestation
- Single ventricle is divided in two.
- Two portions of septum-membranous and muscular
- Single most common congenital heart malformation, accounting for almost 30% of all CHD
- Defects can occur in both the membranous portion of the septum (most common) and the muscular portion
- Left to right shunt
- LVto RV to pulmonary artery
- RA-normal in size
- RV dilates as does main PA, left atrium and left ventricle

Ventricular Septal Defect



Ventricular Septal Defects

Hemodynamics

- The left to right shunt occurs secondary to PVR being < SVR, not the higher pressure in the LV.
- This leads to elevated RV & pulmonary pressures & volume hypertrophy of the LA & LV.

4 Types

- *Perimembranous (or membranous)* – Most common
- *Infundibular (subpulmonary or supracristal VSD)* – involves the RV outflow tract
- *Muscular VSD* – can be single or multiple
- *AVSD* – inlet VSD, almost always involves AV valvular abnormalities.

Symptoms of VSD

- **Small VSD**, hemodynamically insignificant
  - Between 80% and 85% of all VSDs
  - < 3 mm in diameter
  - All close spontaneously
    - 50% by 2 years, 90% by 6 years
    - 10% during school years
  - Muscular close sooner than membranous
- **Moderate VSD**
  - 3-5 mm in diameter
  - Least common group of children (3-5%)
  - Without evidence of CHF or pulmonary hypertension, may be followed until spontaneous closure occurs
- **Large VSDs** with normal PVR
  - 6-10 mm in diameter
  - Usually requires surgery, otherwise…
  - Will develop CHF and FTT by age 3-6 months

Symptoms of VSD

- Rapid breathing
- Irritability
- Excessive Sweating
- Poor weight gain
- Congestive Heart Failure, usually within 6 to 8 weeks of life if defect is large
- Pulmonary Hypertension if defect is large

Ventricular Septal Defects(VSD)

P/E

- Grade II-IV/VI, medium- to high-pitched, harsh pansystolic murmur heard best at the left sternal border with radiation over the entire precordium
- II-III/VI harsh holosystolic murmur heard along the LSB, more prominent with small VSD, may be absent with a very Large VSD
- Prominent P2, Diastolic murmur.
- CHF, FTT, Respiratory infections, exercise intolerance hyperactive precordium
- Symptoms develop between 1 – 6 months

- Onset of systole produces holosystolic murmur
- Heard best at the 4th left ICS
- Widespread transmission even into pulmonary artery
- Loud!!!
- RV heave

Ventricular Septal Defect

Treatment

- Indicated for closure of a VSD associated with CHF *and* FTT or pulmonary hypertension
- Patients with cardiomegaly, poor growth, poor exercise tolerance, or other clinical abnormalities and a qP/qS > 2:1 typically undergo surgical repair at 3-6 mo

- Small VSD - no surgical intervention, no physical restrictions, just reassurance and periodic follow-up and endocarditis prophylaxis.
- Symptomatic VSD - Medical treatment initially with afterload reducers & diuretics.

Treatments for Ventricular Septal Defects

- Lasix and Aldactone to decrease symptoms of CHF
- Digoxin to increase effectiveness of myocardial function
- If surgery needed, patching or suturing the defect can be done
- Mortality from surgery is low

Indications for Surgical Closure

- Large VSD w/ medically uncontrolled symptomatology & continued FTT
- Ages 6-12 mo w/ large VSD & Pulm. HTN
- Age > 24 mo w/ Qp:Qs ratio > 2:1
- Supracristal VSD of any size, secondary to risk of developing AV insufficiency

ASD

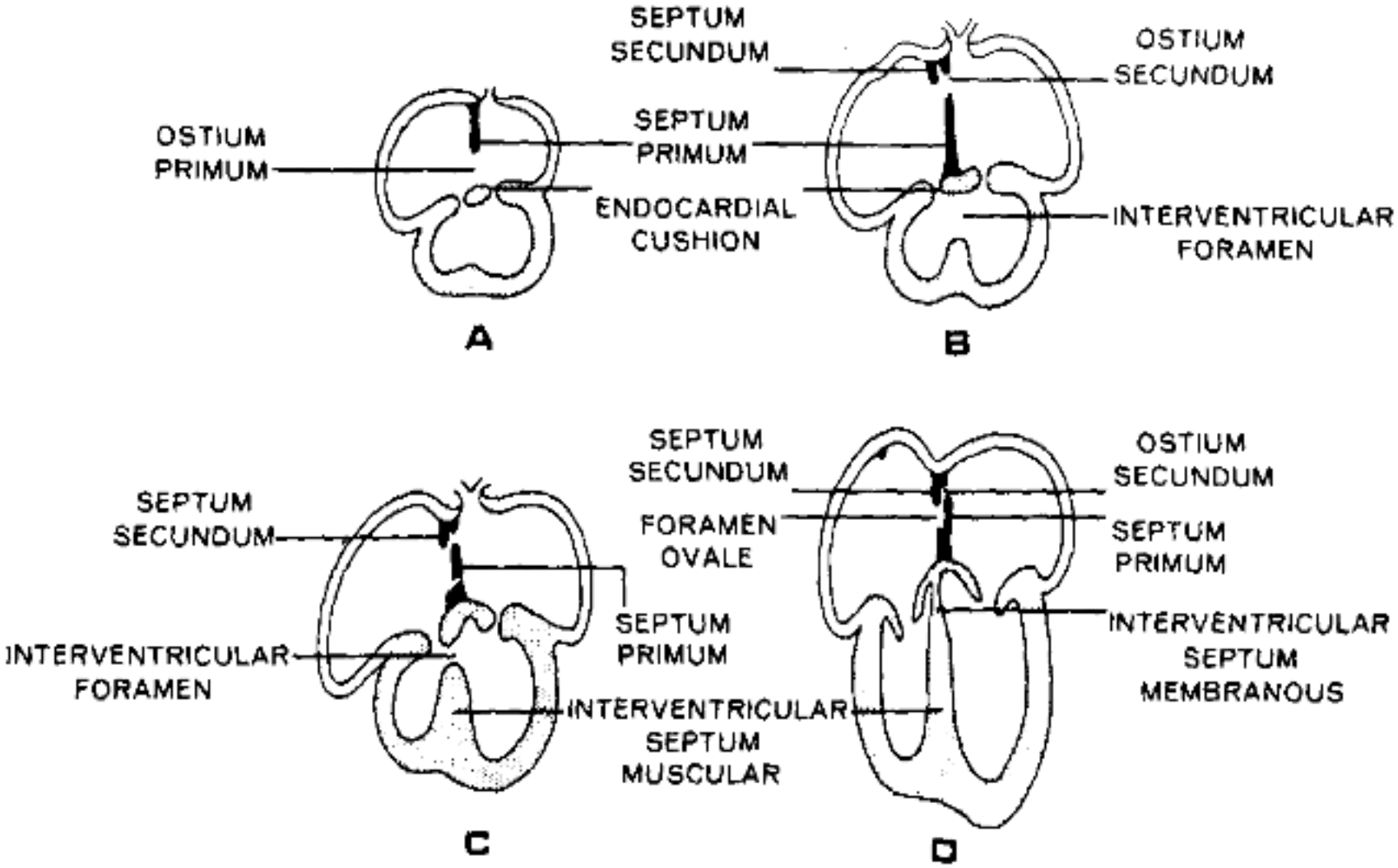
- Left to right shunt
- Increased right sided volume
- Results in dilitation of RA,RV and pulmonary vessels
- Left heart is unchanged!

Atrial Septal Defect(ASD)

- 1/1500 live births

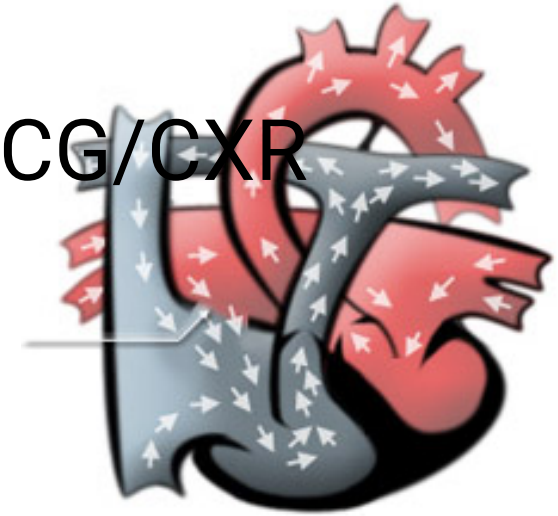
Major types

- **Secundum –– 75%**
  - Most common
  - In the middle of the septum in the region of the foramen ovale
- **Primum–– 15%**
  - associated with other endocardial cushion defects (cleft AV valves, inlet type VSD)
  - LAD
    - Low position
    - Form of AV septal defect
- **Sinus Venosus–– 10%**
  - large, associated with anomalous pulmonary venous drainage (usually R superior PV)
    - Least common
    - Positioned high in the atrial septum
    - Frequently associated with PAPVR
- **Coronary sinus** (rare)
  - associated with unroofed coronary sinus


SEPTUM SECUNDUM
OSTIUM SECUNDUM
OSTIUM PRIMUM
SEPTUM PRIMUM
ENDOCARDIAL CUSHION
INTERVENTRICULAR FORAMEN
A
B
SEPTUM SECUNDUM
FORAMEN OVALE
OSTIUM SECUNDUM
SEPTUM PRIMUM
INTERVENTRICULAR FORAMEN
SEPTUM PRIMUM
INTERVENTRICULAR SEPTUM MEMBRANOUS
INTERVENTRICULAR SEPTUM MUSCULAR
C
D

Atrial Septal Defect (ASD)

- Majority repaired in childhood, but may present in adolescence/adulthood
- **Asymptomatic**: Murmur, abnl ECG/CXR
- **Symptomatic**
  - Dyspnea/CHF, CVA/emboli
  - Atrial Fibrillation



_ Right ventricular heave

- S [redacted]
- Grade I-III/VI systolic murmur at the pulmonary area
- Widely radiating systolic murmur mimicking PPS in infancy

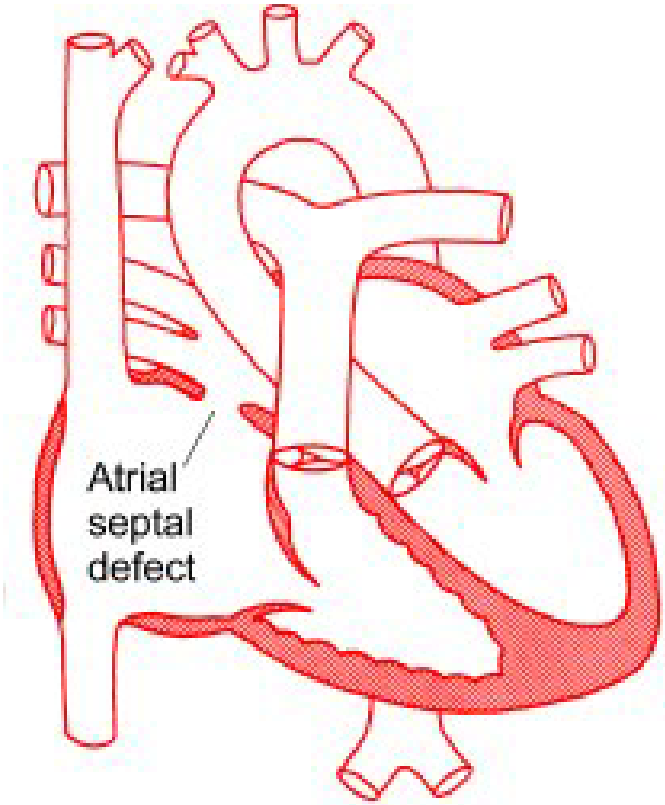
Sign & symptoms

- Rarely presents with signs of CHF or other cardiovascular symptoms.
- Most are asymptomatic but may have easy fatigability or mild growth failure.
- Cyanosis does not occur unless pulmonary HTN is present.
- Hyperactive precordium, RV heave, fixed widely split S2.
- II-III/VI systolic ejection murmur @ LSB.
- Mid-diastolic murmur heard over LLSB.
  - Systolic murmur is caused by increased flow across the pulmonary valve*NOT THE ASD*
  - Diastolic murmur is caused by increased flow across the tricupsid valve & this suggest high flow Qp:Qs is 2:1

Symptoms of Atrial Septal Defects

- Slender build
- Heart murmur resulting from increased blood flow through pulmonary valve
- Usually no significant exercise restriction unless defect is large.
- SOB or palpitations are possible.

Atrial Septal Defect



Auscultation

- Increased flow across the pulmonary valve produces a systolic ejection murmur and fixed splitting of the second heart sound.
- Fixed splitting of S2 may in part be due to delayed right bundle conduction.
- Increased flow across the TV produces a diastolic rumble at the mid to lower right sternal border.
- Older pt loses pulm ejection murmur as shunt becomes bidirectional signs of pulm HTN/ CHF may predominate

Wallo University

P/E - ASD

- Ejection murmur-2nd left intercostal space(LICS)
- Same volume of blood!!!
- Mid-diastolic filling with fixed volume consistently delays closure of pulmonic valve
- Fixed split second sound
- Ostium primum defect
- Same defects as secundum with addition of mitral regurgitation.
- Poor growth, -infant

Atrial Septal Defect

Treatment

- Closure generally recommended when ratio of pulmonary to systemic blood flow (qP/qS) is > 2:1
- Operation performed electively between ages 1 and 3 years

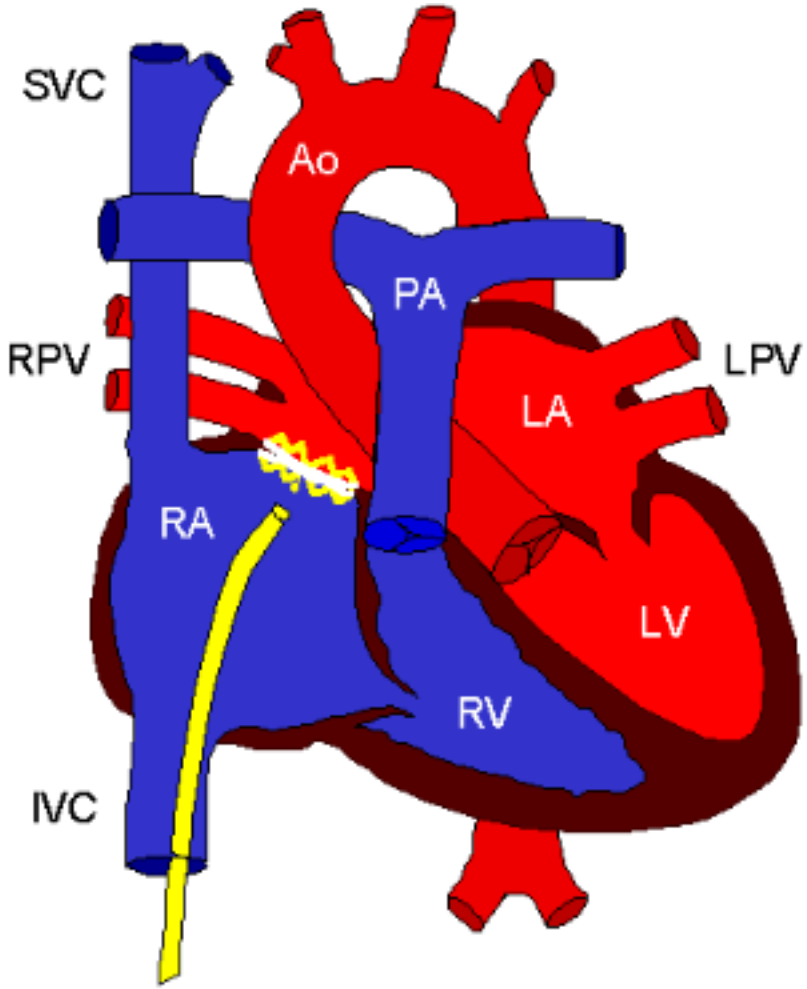
- Previously surgical; now often closed interventionally
- Percutaneous closure
  - Only for secundum (contra in others)
  - Adequate superior/inferior rim around ASD
  - No R-L shunting
- Surgical Closure
  - Good prognosis:
    - Closure age < 25, PA pressure < 40
    - If >25 or PA>40, decreased survival due to CHF, stroke, and afib

Atrial Septal Defect (ASD)

Treatment

- Surgical or catherization laboratory closure is generally recommended for secundum ASD w/ a Qp:Qs ratio >2:1
- Closure is performed electively between ages 2 & 5 yrs to avoid late complications.
- Surgical correction is done earlier in children w/ CHF or significant Pulm HTN
- Once pulmonary HTN w/ shunt reversal occurs this is considered too late.
- Mortality is < 1%.

Atrial Septal Defect



Patent Ductus Arteriosus (PDA)

PDA – Persistence of the normal fetal vessel that joins the PA to the Aorta.

- The ductus arteriosus connects the pulmonary artery to the descending aorta during fetal life.
- PDA results when the ductus fails to close after birth.
- Persistence of normal fetal vessel joining the pulmonary artery to the aorta
- Developes between the 5th and 7th weeks of gestation
- Aortic arch develops with proliferation from apex of truncus arteriosus.
- On the left, the distal portion maintains attachment to aorta and becomes ductus arteriosus
- In fetal life, ductus serves as a funtioning connection between the pulm artery and aorta.
- After birth, the partial pressure of o2 rises and the pulm arterioles dilate causing the ductus to close.

Weber University

Patent Ductus Arteriosus (PDA)

- Persistence of normal fetal vessel joining the pulmonary artery to the aorta
- Closes spontaneously in normal term infants at 3-5 days of age
- Closes spontaneously in normal term infants at 3-5 days of age
- Ultimately, the ductus fibroses and becomes the ligamentum arteriosum
- When it doesn’t close it is called a patent ductus arteriosus(redundant)
- High pressure aorta communicates with low pressure pulmonary artery
- Increases volume in lungs and subsequently into lv
- Similar to VSD

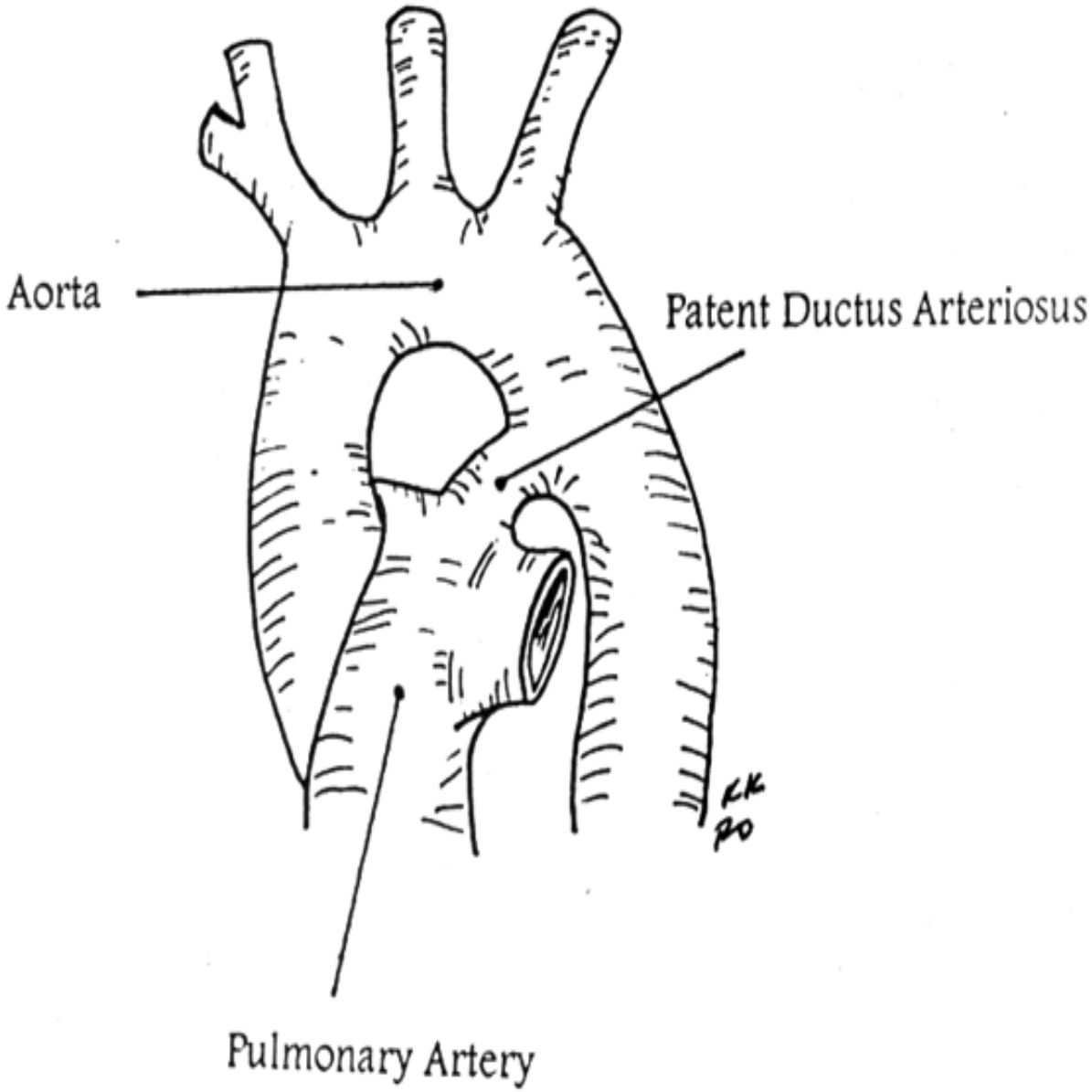
Patent Ductus Arteriosus (PDA)

- Normally closes in the 1st wk of life
- Accounts for 10% of all CHD, seen in 10% of other congenital hrt lesions and can often play a critical role in some lesions.
- Female : Male ratio of 2:1
- Often associated w/ coarctation & VSD
- As a result of higher aortic pressure, blood shunts L to R through the ductus from Aorta to PA
- Extent of the shunt depends on size of the ductus & PVR:SVR
- Small PDA, pressures in PA, RV, RA are normal
- Large PDA, PA pressures are equal to systemic pressures. In extreme cases 70% of CO is shunted through the ductus to pulmonary circulation.
- Leads to increased pulmonary vascular disease.
- Higher incidence of PDA in infants born at high altitudes (over 10,000 feet)
- More common in females

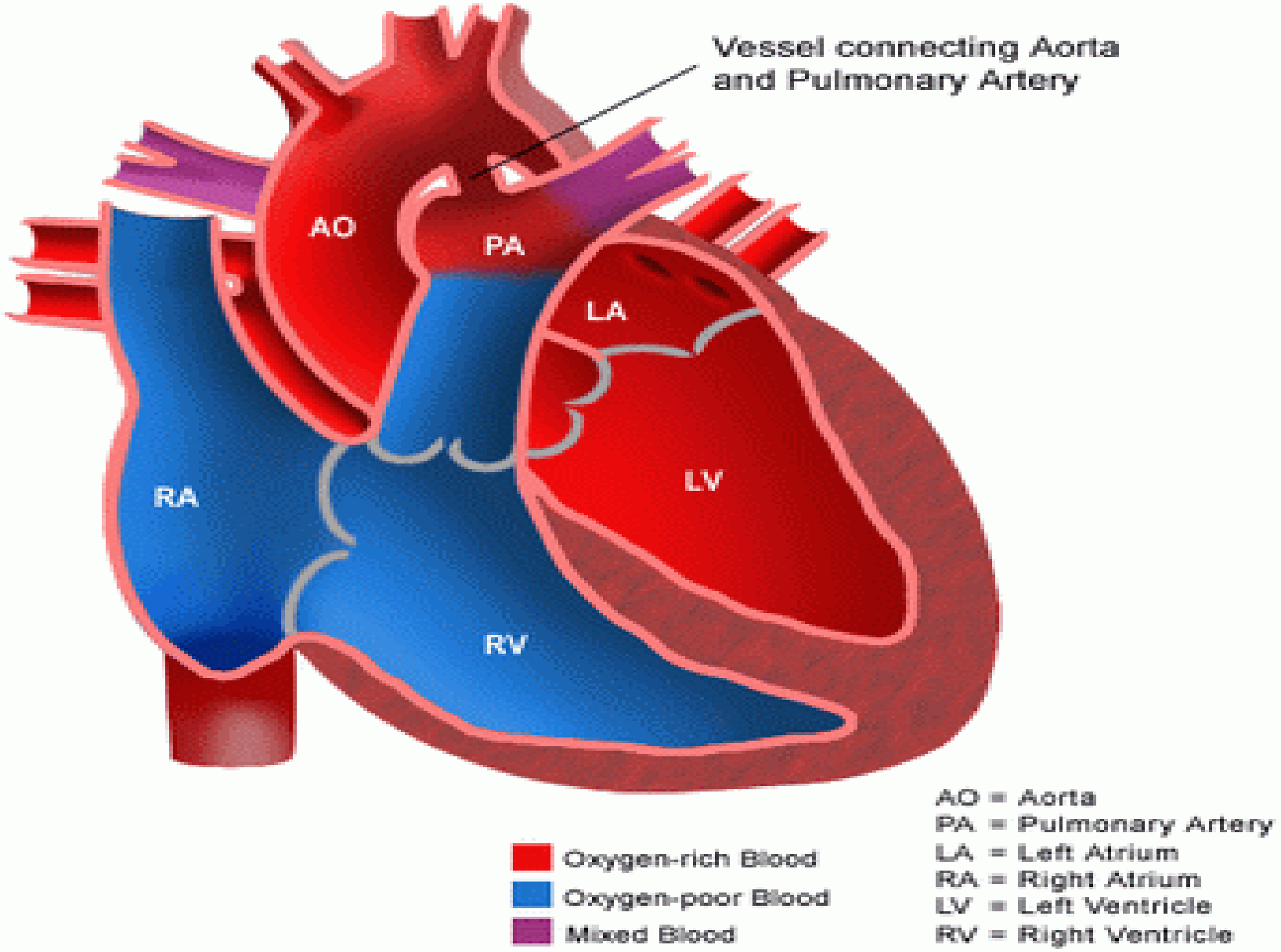
What TORCH infection is PDA associated with?

(Ans. Rubella)

Patent Ductus Arteriosus



Patent Ductus Arteriosus (PDA)



Pathophysiology

Blood flows from aorta to the pulmonary artery, creating a left to right shunt, resulting in left atrium and ventricle overload.

Increased pulmonary blood flow can result in pulmonary hypertension and reversal of the shunt, which is known as Eisenmenger's Syndrome. This results in flow of desaturated blood to the lower extremities.

- Symptoms
  - Children with small patent ductus are usually asymptomatic.
  - Large left to right shunts develop symptoms of congestive heart failure such as tachypnea, tachycardia, poor feeding and slow growth
- Physical exam
  - Continuous murmur heard best at the left sternal border.

Patent Ductus Arteriosus

Clinical Signs & Symptoms

- Occurs early in life
- Initially murmur is systolic, but as diastolic equilibration occurs, murmur becomes a classic to and fro or continuous murmur occurs.
- Small PDA's are usually asymptomatic
- Large PDA's can result in symptoms of CHF, growth restriction, FTT.
- Bounding arterial pulses
- Widened pulse pressure
- Enlarged heart, prominent apical impulse
- Classic continuous machinary systolic murmur
- Mid-diastolic murmur at the apex



S& Sxs

- Continuous murmur
- Left of sternum at 2nd or 3rd interspace
- Courses along sternum and along pulmonary artery
- Displace apex due to increased volume with a thrust
- Pulses are bounding and pulse pressure is widened
- Characteristically has a rough “machinery” murmur which peaks at S2 and becomes a decrescendo murmur and fades before the S1

PDA

- Lab Studies
  - CXR: enlarged cardiac silhouette secondary to left atrial and ventricular enlargement with prominent pulmonary vascular markings.
  - EKG: Left atrial enlargement, LVH
  - ECHO: Doppler flow through the ductus
- Treatment
  - Surgical division or ligation of the PDA



Patent Ductus Arteriosus
Treatment

- Treatment consists of surgical correction when the PDA is large except in patients with pulmonary vascular obstructive disease
- Transcatheter closure of small defects has become standard therapy
- In preterm infants indomethacin is used (80-90% success in infants > 1200 grams)
- Indomethacin, inhibitor of prostaglandin synthesis can be used in premature infants.
- Closure is required treatment heart failure & to prevent pulmonary vascular disease.
- Usually done by ligation & division or intra vascular coil.
- Mortality is < 1%

Pulmonary Stenosis (PS)



Pulmonary Stenosis is obstruction in the region of either the pulmonary valve or the subpulmonary ventricular outflow tract.

- Accounts for 7-10% of all CHD.
- Most cases are isolated lesions
- Maybe biscuspid or fusion of 2 or more leaflets.
- Can present w/or w/o an intact ventricular septum.

What syndrome is PS associated with?

Answer:

Noonan's Syndrome, secondary to valve dysplasia.

Pulmonary Stenosis

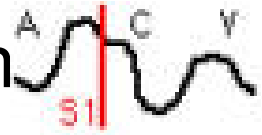
Hemodynamics

- RV pressure hypertrophy $\Rightarrow$ Pς failure
- RV pressures maybe > systemic pressure
- Post-stenotic dilation of main PA
- W/intact septum & severe stenosis $\Rightarrow$ P–Λ σηυντ τηρουγη ΠΦΟ $\Rightarrow$ cyanosis
- Cyanosis is indicative of Critical PS

Pulmonary Stenosis

Clinical Signs & Symptoms

- Depends on the severity of obstruction
- Asymptomatic w/ mild PS < 30mmHg
- Mod-severe: 30-60mmHg, > 60mmHg
- Prominent jugular a-wave, RV lift
- Split 2nd hrt sound w/ a delay
- Ejection click, followed by systolic murm



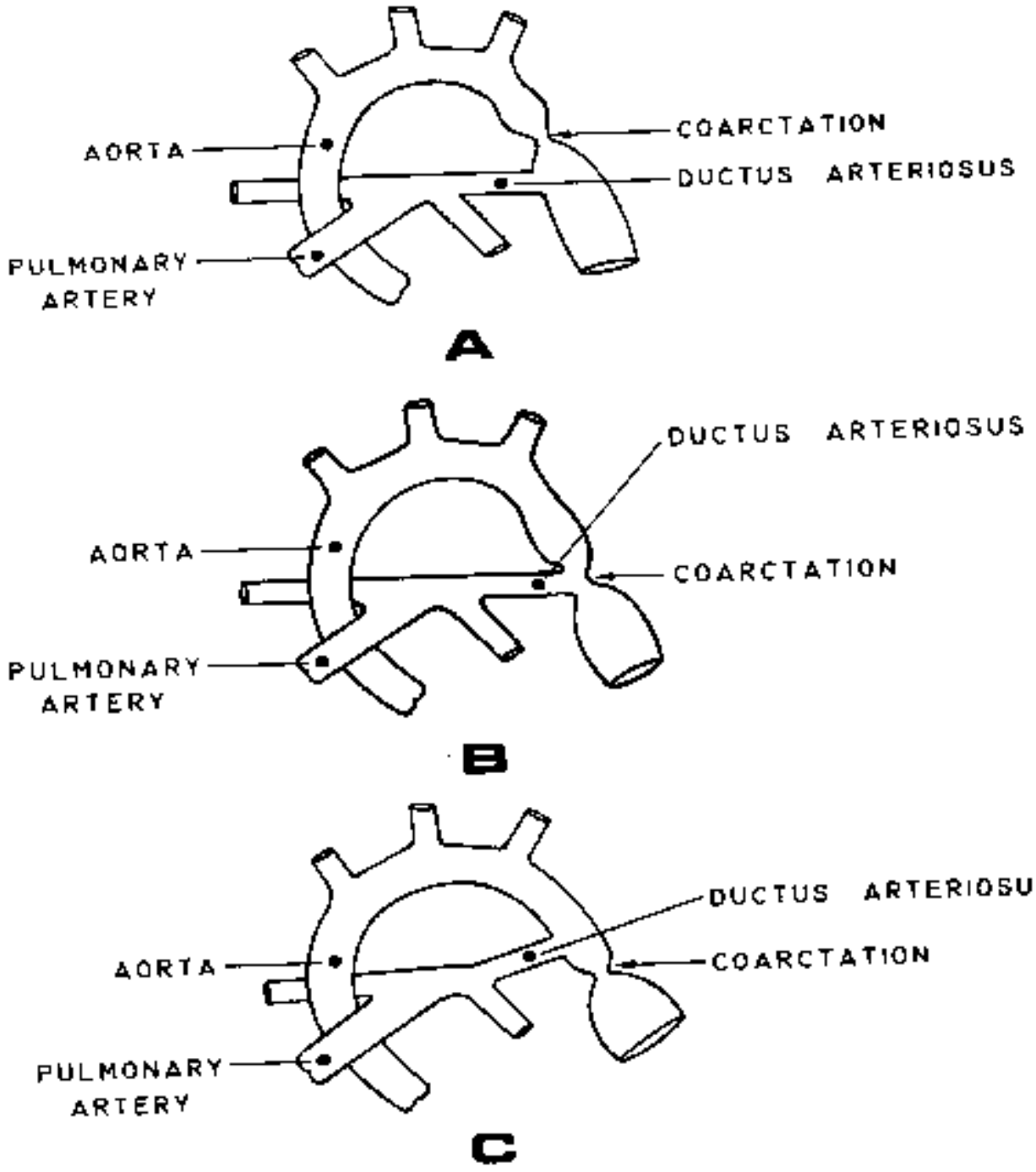
- Heart failure & cyanosis seen in severe cases.

Pulmonary Stenosis
Treatment

- Mild PS no intervention required, close follow-up.
- Mod-severe – require relieve of stenosis.
- Balloon valvuloplasty, treatment of choice.
- Surgical valvotomy is also a consideration.

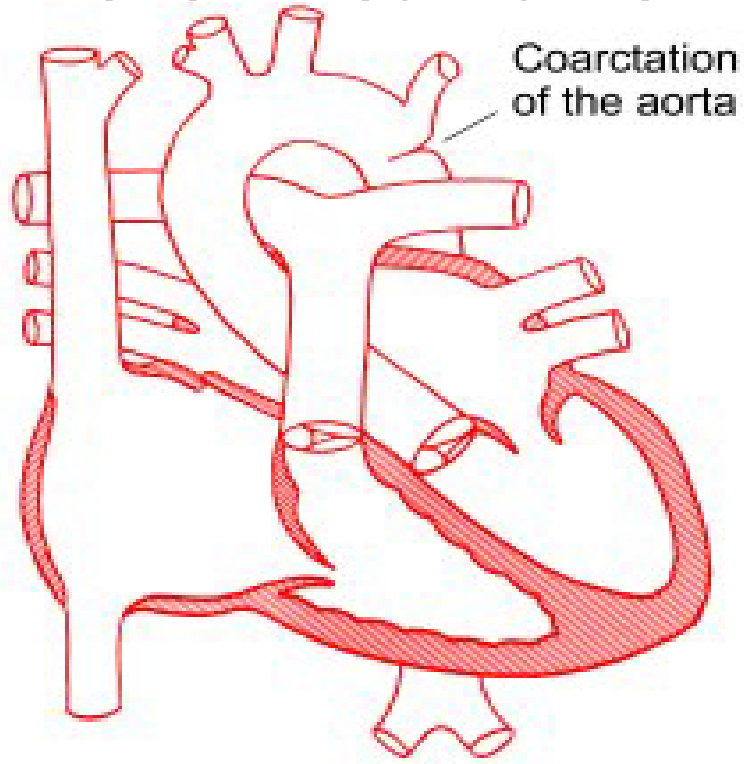
Coarctation of the Aorta

- **Coarctation**- is narrowing of the aorta at varying points anywhere from the transverse arch to the iliac bifurcation.
- 5th and 7th weeks of gestation, the aortic arch develops
- At area of patent ductus, aorta develops improperly, leaving a restrcted lumen.
- Location:proximal , at , or distal to insertin of ductus.
- 98% of coarctations are juxtaductal
- Rib notching occurs due to physical errosion of the undersurface of the ribs as a result of intercostal collateral circulation
- Associated with bicuspid aortic valve
- Male: Female ratio 3:1
- Accounts for 7 % of all CHD


AORTA
COARCTATION
DUCTUS ARTERIOSUS
PULMONARY ARTERY
A
DUCTUS ARTERIOSUS
AORTA
COARCTATION
PULMONARY ARTERY
B
DUCTUS ARTERIOSUS
AORTA
COARCTATION
PULMONARY ARTERY
C

Coarctation of the Aorta Hemodynamics

- Obstruction of left ventricular outflow ⇒ πρεσσυρε ηψπερτροπηψ οφ τηε Λς.



Coarctation of the Aorta

Clinical Signs & Symptoms

- Classic signs of coarctation are diminution or absence of femoral pulses.
- Higher BP in the upper extremities as compared to the lower extremities.
- 90% have systolic hypertension of the upper extremities.
- Pulse discrepancy between rt & lt arms.
- With severe coarc. LE hypoperfusion, acidosis, HF and shock.
- Differential cyanosis if ductus is still open
- II/VI systolic ejection murmur @ LSB.
- Cardiomegaly, rib notching on X-ray.

Coarctation of the Aorta

Treatment

- With severe coarctation maintaining the ductus with prostaglandin E is essential.
- Surgical intervention, to prevent LV dysfunction.
- Angioplasty is used by some centers.
- Re-coarctation can occur, balloon angioplasty is the procedure of choice.

Fallot's Tetralogy

- 3rd to 4th week, the common trunk divides into the pulmonary artery and the aorta
- 4th and 8th week, the ventricle divides into two

Defined by four findings

1) Infundibular stenosis
2) Ventricular septal defect
3) Right ventricular hypertrophy
4) Overriding of the aorta

Hemodynamics

- Diminished blood flow to the lungs and increased blood flow to the body
- Due to the stenosis of the infundibulum, pulmonary flow is diminished. The overriding aorta accepts most of the RV blood

Fallot's Tetralogy

Anatomic Defects

- Ventricular septal defect
- Overriding Aorta
- Pulmonary artery stenosis
- Right ventricular hypertrophy

Pathophysiology

Increased resistance by the pulmonary stenosis causes deoxygenated systemic venous return to be diverted from RV, through VSD to the overriding aorta and systemic circulation → systemic hypoxemia and cyanosis

S&Sxs

- Symptoms:
  - Dyspnea on exertion or when crying
  - Tet spells: irritability, cyanosis, hyperventilation and sometimes syncope or convulsions due to cerebral hypoxemia.
  - Patients learn to alleviate symptoms by squatting which increases systemic resistance and decreases the right-to-left shunt and directs more blood to the pulmonary circulation.
- This produces a right to left shunt and therefore produces cyanosis of periphery
- Children present with cyanotic hands and feet
- Children squat to enhance flow back to heart to oxygenate

Fallot's Tetralogy

- Physical exam
  - Clubbing of the fingers and toes
  - Systolic ejection murmur heard at the upper left sternal border created by turbulent blood flow through stenotic RV outflow tract
- Lab Studies
  - CXR: prominent RV
  - EKG: RVH, right axis deviation
  - ECHO: displays and quantifies extent of RV outflow tract obstruction

Tetralogy of Fallot

- Treatment:
  - Surgical closure of the VSD and enlargement of the pulmonary outflow tract
- Anesthetic Management:

-The goal is to control the magnitude of the right to left intracardiac shunt, which is increased by:

1) Decreased SVR
2) Increased PVR
3) Increased myocardial contractility

- Patient given beta blockers for prophylaxis against Tet spells
- Inhalation induction can be employed in an attempt to avoid Tet spells while placing an intravenous line.

-Physiologic monitoring includes standard ASA monitors and an arterial line. Echocardiography and EEG may also be employed.

Tetralogy of Fallot

- Concerns After Surgical Repair:
  - Endocarditis prophylaxis
  - Residual VSD secondary to incomplete closure
  - Residual RV outflow tract obstruction
  - Chronic pulmonary valve regurgitation results in a large volume load on the right ventricle that can lead to cardiomegaly and increased incidence of arrhythmias.
  - Right ventriculotomy during the repair leads to scarring which increases the risk of dysrhythmias and conduction abnormalities.
- Overall incidence of sudden death in TOF patients after surgical repair is about 0.3%.

Vascular disorders

A. Disorders of the arteries
B. Disorders of the veins
C. Disorders of the lymphatic system

A. Disorders of the arteries

Aneurysm

- An aneurysm is distension of an artery caused by structural weakening of the arterial wall
- Under hemodynamic pressure the weakened area enlarges, causing serious complications by compressing surrounding structures.
- It results from degeneration of the medial wall, which occurs as a normal part of the aging process as well as with hypertension, atherosclerosis, trauma or infection, immunologic conditions.

Disorders of the arteries..

- **Thoracoabdominal aneurysm**
  - *May originate in the ascending aorta & aortic arch (frequent site of dissection) or in the lower descending aorta & upper abdominal aorta.*
- ***Abdominal aneurysm***
  - *Originate in the abdominal aorta, typically b/n the renal arteries & iliac branches.*
  - *Many of these pts (mostly male) are asymptomatic.*

Disorders of the arteries..

- **Aneurysm–-**
  - Aneurysm my also occur in peripheral arteries (femoral, popliteal, renal, subclavian) or any major artery.
  - The distension may occur from one side or entire circumference, but the directing type is tear of the intima & separation of the medial layers causing hemorrhage or intramural hematoma.
- ***Intracranial aneurysm***
  - *Is a sacular dilation of a cerebral artery whose walls are congenitally weakened, resulting in hemorrhage (stroke), increased ICP, vasospasm, ischemia.*

Wollo University

Aneurysm––

Predisposing factors:-

- Local infections: Pyogenic or fungal
- Congenital weakness of vessels
- Arteriosclerosis ,trauma ,syphilis

Complications:-

- Fatal hemorrhage
- Paraplegia due to interruption of anterior spinal artery
- Ischemia of abdomen, myocardium, lower extremity, stroke, renal failure

Aneurysm---

Clinical manifestations

- Thoraco-abdominal aneurysm
  - Constant, boring pain or pressure in chest
  - Intermittent neurologic pain due to nerve compression
  - Dyspnea, cough, & hoarseness b/s of pressure against trachea & laryngeal nerve
  - Dysphagia
  - Dilated superficial veins of chest & cyanosis due to compression of chest vessels
  - Pulse or B/P variations b/n arms
- Abdominal aneurysm
  - Persistent or Intermittent abdominal pain, often localized to middle or lower Lt side of abdomen
  - Pulsating mass with bruits
  - B/P elevated in arm more than in thigh

Aneurysm---

Diagnostic evaluation:-

- Abdominal or chest X-ray
- CT scan or Ultrasonography
- Arteriography

Management

- Surgery to remove the aneurysm & restore vascular continuity
- Endovascular grafting
- Nursing interventions for complications & postoperative cases

Hypertension

- Is a systolic blood pressure greater than 140 mm Hg and a diastolic pressure greater than 90 mm Hg over a sustained period, based on the average of two or more blood pressure measurements taken in two or more contacts with the health care provider after an initial screening.
- Is the product of cardiac output multiplied by peripheral resistance
- Cardiac output is the product of the heart rate multiplied by the stroke volume.
- In normal circulation, pressure is exerted by the flow of blood through the heart and blood vessels

Hypertension…

- High blood pressure, known as hypertension, can result from
  - A change in cardiac output
  - A change in peripheral resistance, or
  - A change in both

  - The medications used for treating hypertension decrease peripheral resistance, blood volume, or the strength and rate of myocardial contraction

Hypertension–-

Pathophysiology

- Although the precise cause for most cases of hypertension cannot be identified, it is understood that hypertension is a multifactorial condition
- For hypertension to occur there must be a change in one or more factors affecting peripheral resistance or cardiac output
- In addition, there must also be a problem with the control systems that monitor or regulate pressure

Hypertension may be caused by one or more of the following:-

- Increased sympathetic nervous system activity related to dysfunction of the autonomic nervous system, leading to increased stress responses.

Hypertension–-

Pathophysiology

- Increased renal reabsorption of sodium, chloride, and water related to a genetic variation in the pathways by which the kidneys handle sodium
- Increased activity of the renin-angiotensin-aldosterone system, resulting in expansion of extracellular fluid volume and increased systemic vascular resistance, leads to vasoconstriction and retention of sodium and water The increase in blood volume leads to hypertension
- Decreased vasodilatation of the arterioles related to dysfunction of the vascular endothelium
- Resistance to insulin action, which may be a common factor linking hypertension, type 2 diabetes mellitus, hypertriglyceridemia, obesity, and glucose intolerance

HPN

Blood pressure is regulated by four bodily mechanisms:-

1/ **Arterial baroreceptors**

- Baroreceptors are located in the carotid sinus, aorta, and left ventricle
- They control blood pressure by altering the heart rate and/or causing vasoconstriction or vasodilatation

2/ **Regulation of body-fluid volume**

- Properly functioning kidneys either retain fluid when the client is hypotensive or excrete fluid when the client is hypertensive

3/ **Renin-angiotensin system**

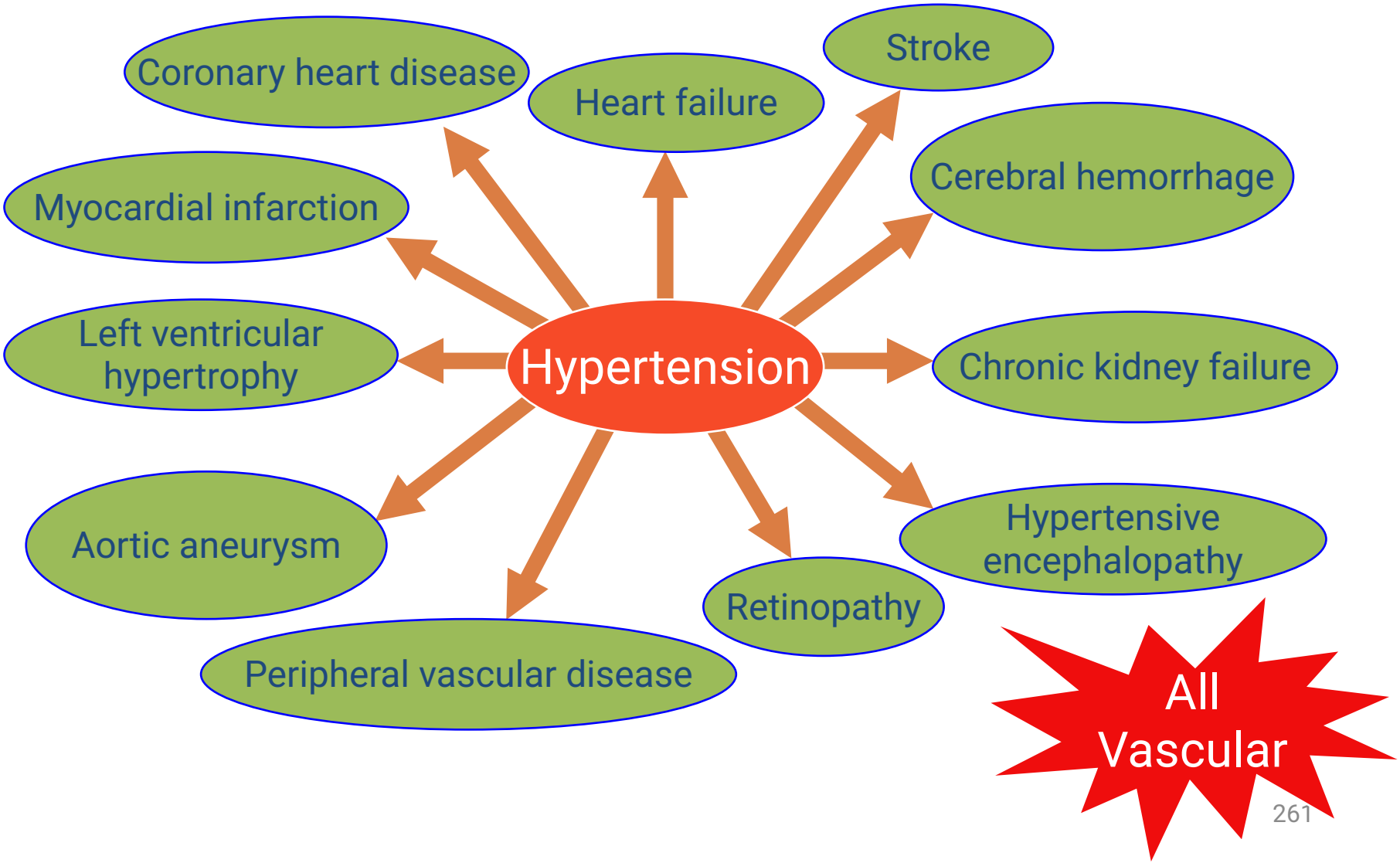
- Angiotensin II vasoconstricts and controls aldosterone release, which causes the kidneys to reabsorb sodium and inhibit fluid loss

4/ **Vascular autoregulation**

- This maintains consistent levels of tissue perfusion

Wollo University

Diseases Attributable to Hypertension



Target Organ Damage (TOD)

- Heart

Left ventricular hypertrophy (LVH)

Angina or prior myocardial infarction (CHD)

Prior Coronary revascularization PTCA or CABG

Heart failure (Systolic / Diastolic dysfunction)

- Brain

CVA Stroke or Transient Ischemic Attack (TIA)

- Kidney : Chronic kidney disease and CRF
- Vessels : Peripheral arterial disease PVD
- Eyes : Hypertensive Retinopathy

Category	Systolic BP in mmHg	Diastolic BP in mmHg	Follow up
Normal	<130	<85	Check in 2 years
High normal	130-139	85-89	Check in 1 years & discussed in life style modification
Hypertension			
Stage-I	140-159	90-99	Confirm with in 2 months
Stage-II	160-179	100-109	Evaluate or refer within 1 month
Stage-III	180-209	110-119	Evaluate or refer with in 1 wk
Stage-IV	>210	>120	Evaluate or refer immediately

Hypertension–-

Types of hypertension

There are two types of hypertension

- **Primary hypertension or essential hypertension**
  - The reason for the elevation in blood pressure cannot be identified.
  - Is more common in adolescents
  - Has multiple risk factors, including obesity and a family history of hypertension
- **Secondary hypertension**
  - Is the term used to signify high blood pressure from an identified cause.
  - Is more common in preadolescent children, with most cases caused by renal disease.

Hypertension––

1. **Non modifiable risk factors: Unable to be changed**
   - Age - The amount of collagen in arteries increases with age, causing the blood vessels to get stiff.
   - Family history of hypertension
   - Sex
   - Race and ethnicity

Hypertension—

2. Modifiable risk factors: Changeable

- **Tobacco smoking -** Smoking increases blood cholesterol, creating small plaques in arteries
- **Stress -** Stress reduction and relaxation can significantly improve your general health state as well as lower your blood pressure
- **Diet with high levels of saturated fat in it -** Fats will stimulate the process called atherosclerosis which makes a man more prone to all kinds of cardiovascular diseases
- **Alcohol abuse**
- **Obesity -** One of the most important risk factors. According to some researches done in the past, every extra kilogram generally will increase your blood pressure by two mmHg

NB. High blood pressure increases morbidity and mortality from: Cardio-vascular diseases, Stroke, Congestive heart failure and many renal diseases

Hypertension—

Clinical manifestations

- Hypertension is sometimes called —the silent killer because people who have it are often symptom free
- Rare: Headache, bloody nose, blurred vision, dizziness- late signs
- Physical examination may reveal no abnormalities other than high blood pressure
- Occasionally, retinal changes such as hemorrhages, exudates (fluid accumulation), arteriolar narrowing.
- In severe hypertension: Papilledema (swelling of the optic disc) may be seen, Target organ disease: Damage to blood vessels of heart, kidney, brain (cerebral hemorrhage & hypertensive encephalopathy)

Terms

- **Isolated systolic HPN** (S > 160 & D < 90), RX = CCB

Isolated systolic HPN **(**ISH) is universal after 65 yrs

- **Borderline isolated systolic HPN** (S= 140 - 160 & D = normal)
- **Labile HPN** (When a pt is hypertensive at one time & normal at another time)
- ***Malignant HPN*** (The presence of papilledema (retinal hemorrhage or exudates),Nephropathy & or encephalopathy (B/P > 240/120)
- **Accelerated HPN**(recent rise in B/P over previous hypertensive levels, which is associated with retinal damage without papilledema
- "**White coat hypertension** " – induced by stress at physical examination

Isolated Systolic Hypertension(ISH)

1. What is ISH ? –

   SBP 140+ , DBP < 90

2. What percentage of 65+ aged have ISH ?

   More than 90%

3. Which is more harmful – ↑ SBP or DBP ?

   Of course ↑ SBP

4. Why is ISH important ?

   Because of ↑↑ CVA and CHD mortality

Hypertension—

- **Assessment and diagnostic evaluation**
  - A thorough health history and physical examination are necessary
  - The retinas are examined, and laboratory studies are performed to assess possible target organ damage
  - Routine laboratory tests include:-
    - Urinalysis
    - Blood chemistry (ie, analysis of sodium, potassium, creatinine, fasting glucose, and total and high-density lipoprotein [HDL] cholesterol levels),
  - Electrocardiogram - Left ventricular hypertrophy can be assessed
  - Renal damage may be suggested by elevations in BUN and Creatinine levels

Hypertension---

Management of hypertension

A. General measures

- Treatment of cause if present
- Regular exercise (Brisk walking > 30min/d) 4-9mmhg
- Dietary salt (< 2.3 g/day of sodium, 2- 8 mmhg)
- Weight reduction (BMI = 18.5 - 24.9, 5-20mmhg/ 10kg wt loss)
- Avoidance of stress, emotions
- Smoking
- Limit alcohol(237ml = 8 oz wine, 710 ml =24 oz beer per day) 2- 4mmhg
- The DASH diet (8-14mmhg)

Lifestyle modifications

Modification	Approximate BP reduction (range)
Weight reduction	5–20 mm/10 kg wt loss
Adopt DASH eating plan	8–14 mmHg
Dietary sodium reduction	2–8 mmHg
Physical activity	4–9 mmHg
Abstinence from alcohol	2–4 mmHg

All put together reduce BP by 20 to 55 mmHg

Hypertension—

Benefits of treatment

- Reductions in **stroke** incidence, averaging 35–40 %
- Reductions in **MI**, averaging 20–25 %
- Reductions in **HF**, averaging >50 %

B. Specific drugs

Diuretics

- Thiazide (Ex. Hydrochlorothiazide)
  - Block reabsorption of Na^+ and Cl^- from distal tubulus
- Loop diuretics Ex. **Furosemide, Bumetanide,Torasemide**
  - Block active reabsorption of Na^+, Cl^-, K^+ from ascending limb of Henle´s loop
- Potassium-sparing diuretics

Ex. **Amiloride**: 5–20 mg/day, **Triamterene**: 150–250 mg/day,
Spironolactone: 12.5–50.0 mg/day for severe heart failure. Higher doses may be used for refractory oedema.
Eplerenone: 25–50 mg/day.

– To correct hypokalemia

- Aldestron antagonists (Sprinolactone)

Specific drugs–-

- **Sympatholytics**

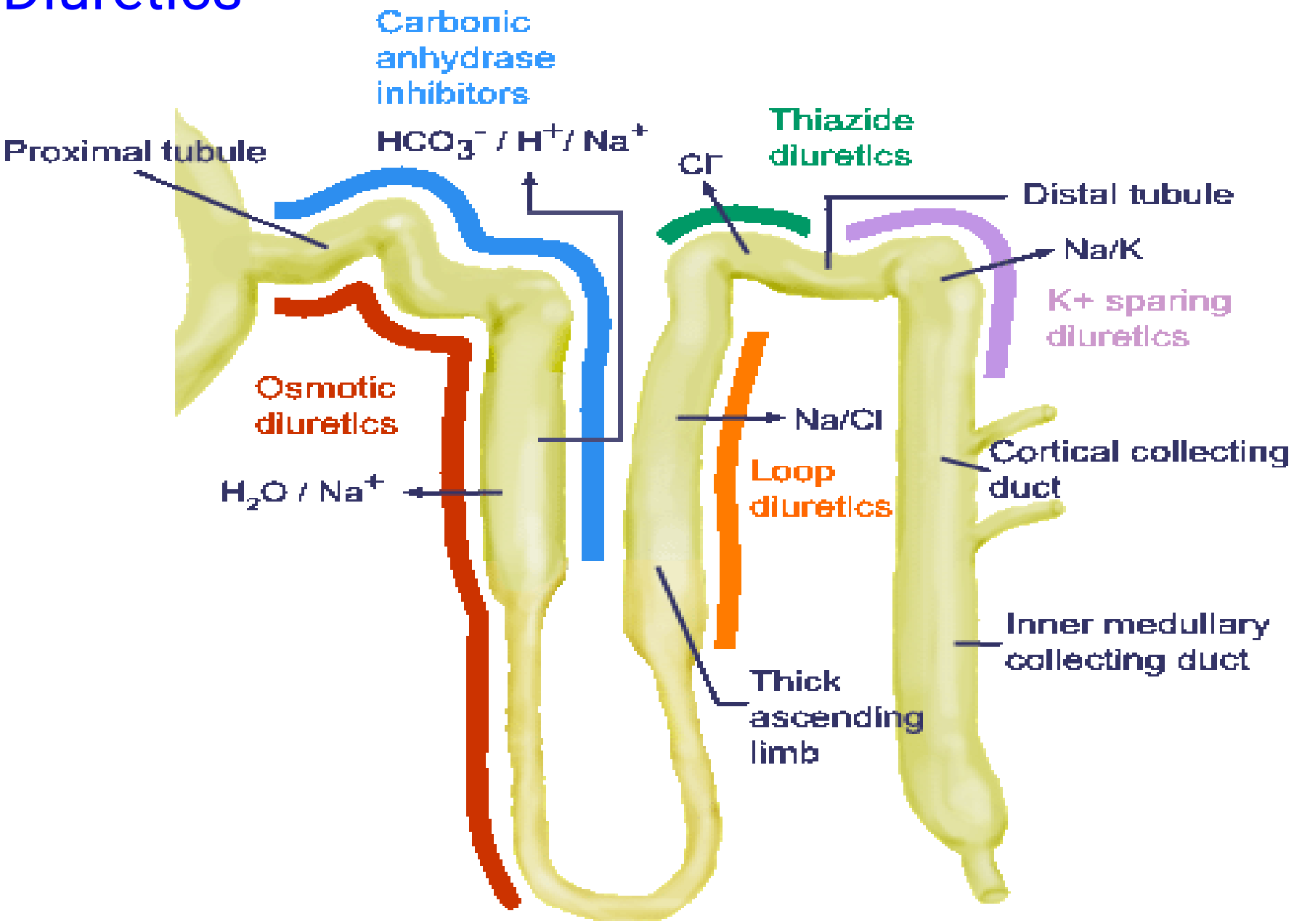
- Adrenergic neuron blockers e.g. Reserpine, Guannethedine)
- Alpha adrenergic blockers e.g. Prazosin
- Beta adrenergic blockers : e.g. Propanolol

- **Vasodilators:** e.g. Hydralizine, Sodium nitroprusside
- **Calcium channel blockers :** e.g. Verapamil, Nifedipine, Diltiazem

- Block influx of calcium to cell through slow L-type channels, lower its intracellular concentration what causes relaxation of smooth muscle in vessel wall, decrease of contractility, decrease of electrical irritability and conductivity

- **Angotensin converting enzyme inhibitors**: e.g. Captopril, Enalapril

Diuretics


Carbonic anhydrase inhibitors
$HCO_3^- / H^+ / Na^+$
Proximal tubule
Thiazide diuretics
Cl^-
Distal tubule
Na/K
K+ sparing diuretics
Osmotic diuretics
H_2O / Na^+
Na/Cl
Loop diuretics
Cortical collecting duct
Inner medullary collecting duct
Thick ascending limb

Hypertension–-

Complications of hypertension

- Atherosclerosis
- Coronary artery disease
- Myocardial infarction or failure
- Left ventricular hypertrophy
- Stroke
- Kidney/eye damage

Hypertensive crises

- Hypertensive **emergency** and hypertensive **urgency**
  - May occur in patients whose hypertension has been poorly controlled or in those who have abruptly discontinued their medications.
- **Hypertensive emergency**:
  - Is a situation in which blood pressure must be lowered immediately (not necessarily to less than 140/90 mm Hg) to halt or prevent damage to the target organs
  - Conditions associated with hypertensive emergency include: Acute myocardial infarction, intracranial hemorrhage.
  - Are acute, life threatening blood pressure elevations that require prompt treatment in an intensive care setting because of the serious target organ damage that may occur.

Hypertensive crises –-

- The medications of choice in hypertensive emergencies are those that have an immediate effect. Intravenous vasodilators, including:-
  - Sodium nitroprusside
  - Nicardipine hydrochloride
  - Fenoldopam mesylate , enalaprilat (I.V.), and
  - Nitroglycerin have an immediate action that is short lived (minutes to 4 hours), and they are therefore used as the initial treatment
- **Hypertensive urgency:**
  - Is a situation in which blood pressure must be lowered within a few hours
  - Hypertensive urgencies are managed with oral doses of fast-acting agents such as loop diuretics like furosemide [Lasix]), beta-blockers propranolol (Inderal).

Hypertension–-

Prevention

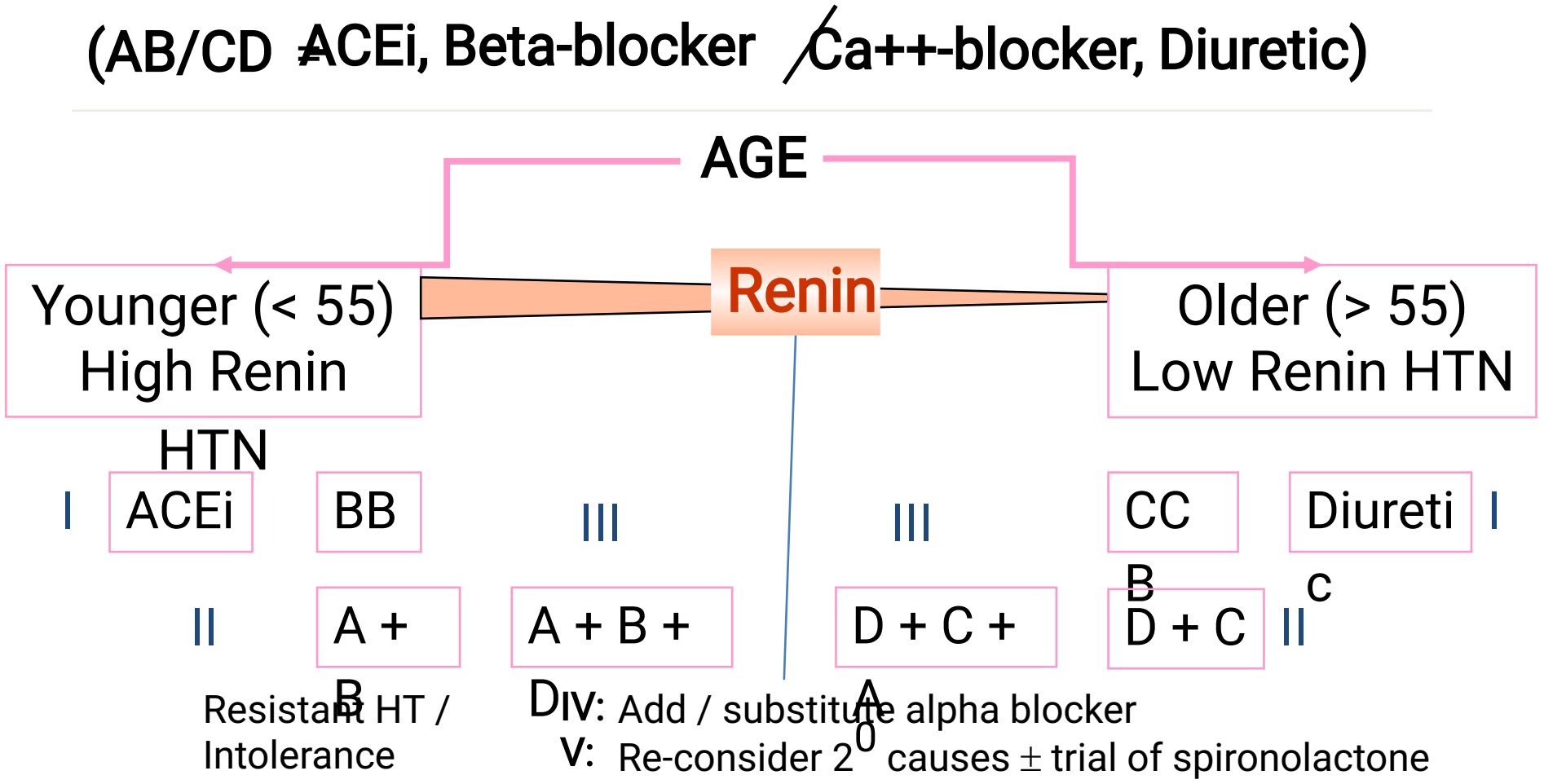
- The most important fact about this condition is that we are able to prevent it.
- The degree to which hypertension can be prevented depends on a number of features including:-
  - Current blood pressure level
  - Changes in end/target organs (retina, kidney, heart - among others)
  - Risk factors for cardiovascular diseases and
  - The age at presentation.
- Unless the presenting patient has very severe hypertension, there should be a relatively prolonged assessment period within which should be repeated measurements of blood pressure.

Hypertension–-

Prevention….

- Following this, lifestyle advice and non-pharmacological options should be offered to the patient, before any initiation of drug therapy.
- These measures include;
  - Weight reduction and regular aerobic exercise
  - Reducing sodium (salt) in the diet
  - Additional dietary changes: beneficial to reducing blood pressure includes the DASH diet (dietary approaches to **stop hypertension), which is rich in fruits and vegetables and low-fat or fat-free dairy foods.**
  - Discontinuing tobacco use and alcohol consumption.
  - Reducing stress

AB/CD Rule – HTN Treatment



Dickerson et al. Lancet 353:2008-11;1999

Hypertension – Rational Drug Combinations

ACEI and ARB = A

Beta Blockers = B

Calcium Channel (CCB) = C

Diuretics Drugs= D

Diuretics = D – Rank 1

ACEI and ARB = A – Rank 2

Beta Blockers = B – Rank 3

CCB = C – Rank 4

D and A combination is excellent -Ramace H, Losar H, Enace D

D and B combination next -Betaloc H, Atecard D, Tenoric

A and B combination Third -Losar A, Cardif Beta

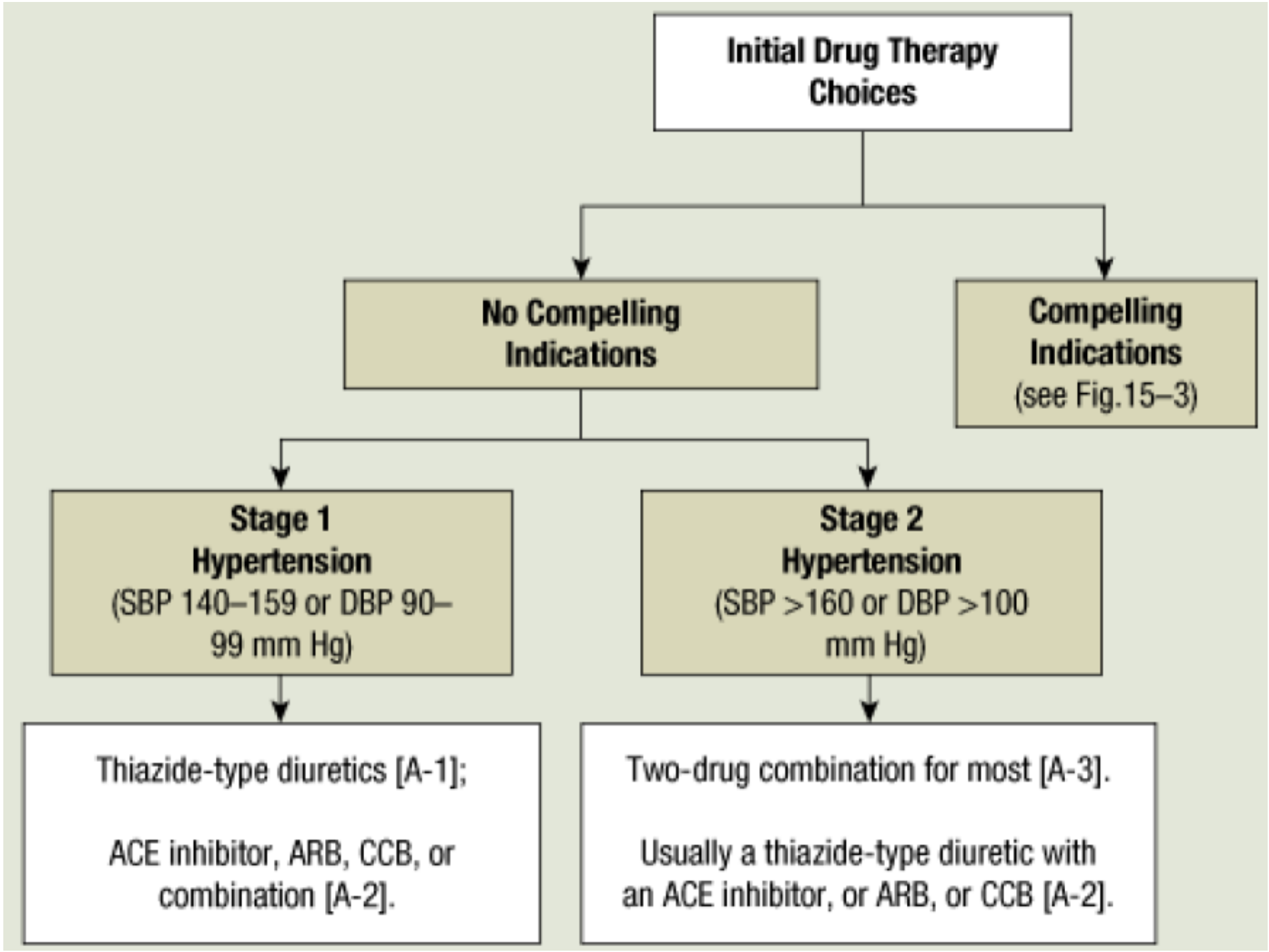
A and C combination fourth -Amlopres L, Hipril A, Amlo LS

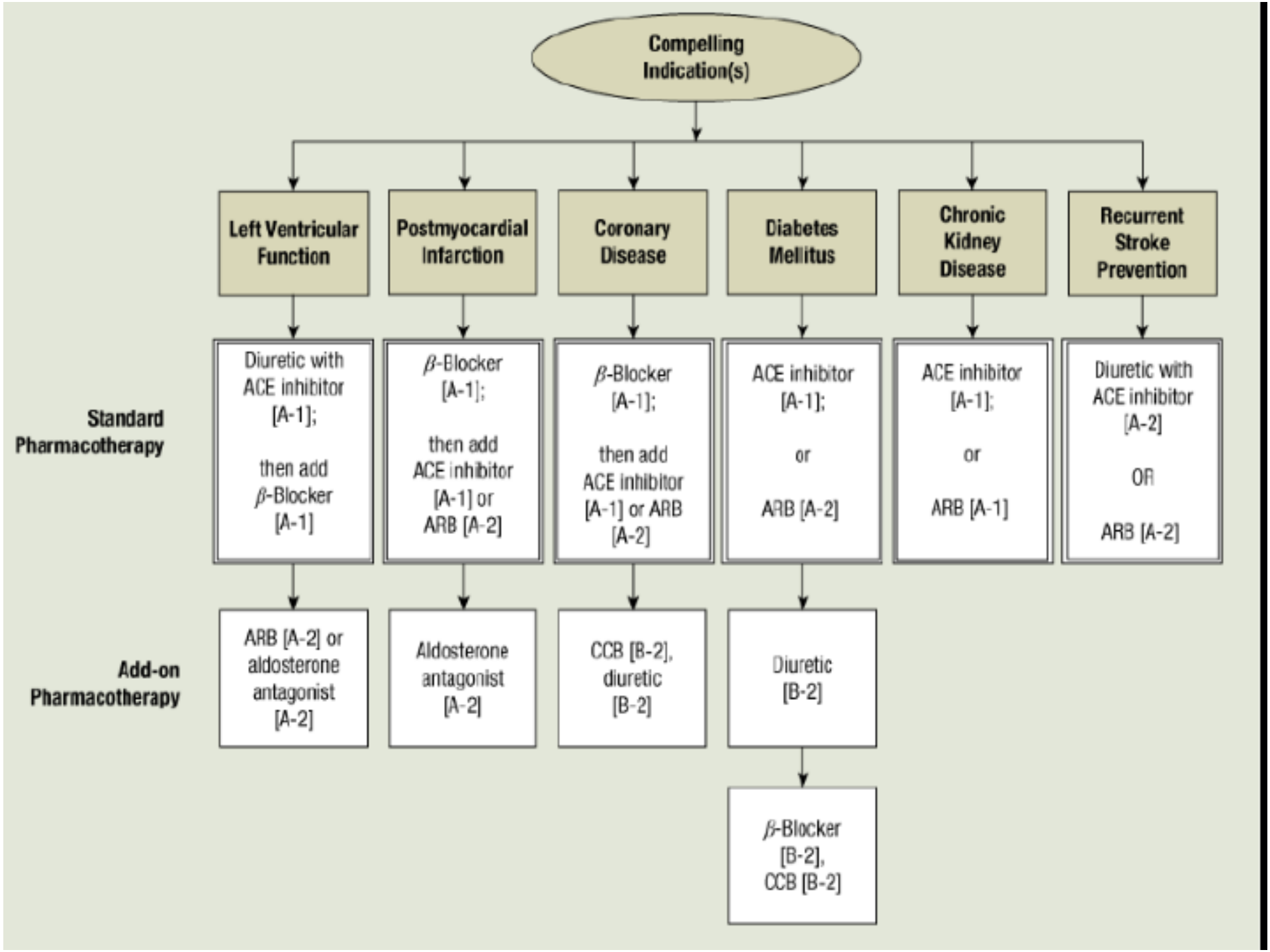
B and C combination fifth - Amlo AT, Amlobet, Beta Nicardia

D and C combination sixth -Amlogaurd H, Stamlo D

Compelling Indications

- Heart Failure
- Post Myocardial Infarction
- High Coronary Disease Risk
- Diabetes Mellitus
- Chronic Kidney Disease
- Recurrent Stroke Prevention


Initial Drug Therapy Choices
No Compelling Indications
Compelling Indications (see Fig.15–3)
Stage 1 Hypertension (SBP 140–159 or DBP 90–99 mm Hg)
Stage 2 Hypertension (SBP >160 or DBP >100 mm Hg)
Thiazide-type diuretics [A-1];
ACE inhibitor, ARB, CCB, or combination [A-2].
Two-drug combination for most [A-3].
Usually a thiazide-type diuretic with an ACE inhibitor, or ARB, or CCB [A-2].


Compelling Indication(s)
Left Ventricular Function
Postmyocardial Infarction
Coronary Disease
Diabetes Mellitus
Chronic Kidney Disease
Recurrent Stroke Prevention
Standard Pharmacotherapy
Diuretic with ACE inhibitor [A-1]; then add β-Blocker [A-1]
β-Blocker [A-1]; then add ACE inhibitor [A-1] or ARB [A-2]
β-Blocker [A-1]; then add ACE inhibitor [A-1] or ARB [A-2]
ACE inhibitor [A-1]; or ARB [A-2]
ACE inhibitor [A-1]; or ARB [A-1]
Diuretic with ACE inhibitor [A-2] OR ARB [A-2]
Add-on Pharmacotherapy
ARB [A-2] or aldosterone antagonist [A-2]
Aldosterone antagonist [A-2]
CCB [B-2], diuretic [B-2]
Diuretic [B-2]
β-Blocker [B-2], CCB [B-2]

Shock

Definitions:-

Shock 1:- inadequate organ perfusion and substrates to meet the tissue's oxygenation demand

Shock 2:- a condition in which system B/P is too low to deliver oxygen & nutrients to support vital organs & cellular function

Shock 3:- a life threatening medical condition that occurs due to inadequate substrate for aerobic cellular respiration

Shock 4:-acute circulatory failure leading to inadequate tissue perfusion & end organ injury

Shock

SUPPLY < DEMAND

- Failure to remove metabolic end-products
- Result of inadequate blood flow and/or oxygen delivery

Shock is not a synonym to hypotension!

Hypotension

o Is an indication of:-

1/ An abnormality of HR,SV, or PR

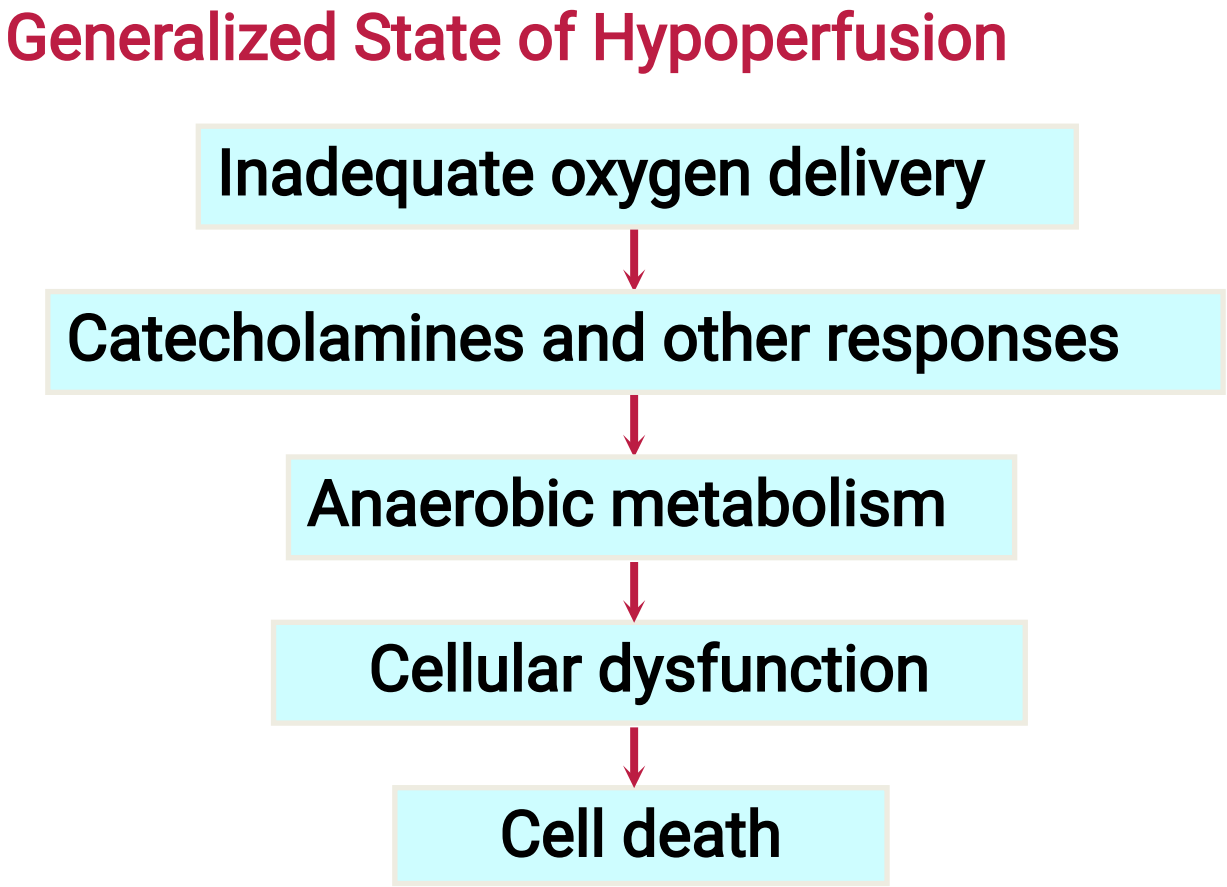
2/ Failure of the others to compensate

o In adults:-

- Systolic BP < 90 mm Hg & Diastole < 60 mmhg), specially in the arteries of the systemic circulation + S& Sxs
- Mean arterial pressure (MAP) ↓ 60 mm Hg
- Reduction of systolic BP > 40 mm Hg from the patient's baseline pressure

$$MAP = \frac{SBP + 2DBP}{3} = DBP + 1/3 \text{ pulse pressure}$$

What is shock?



Pathophysiology of shock syndrome–– 1

Cells switch from aerobic to anaerobic metabolism

↓

lactic acid production

↓

Cell function ceases & swells

↓

membrane becomes more permeable

↓

electrolytes & fluids seep in & out of cell

↓

Na+/K+ pump impaired

mitochondria damage

↓

cell death

Shock—Classifications

1/ **Hypovolumic**(the fluid) shock
- Loss of blood, plasma, electrolytes
  - Burn pts, dehydration(vomiting, diarrhea ,diaphoresis, DKA, ascites)

2/ **Cardiogenic**(the pump) shock
- Assoc w/ cardiac impairment(MI, stenosis, regurgitation, VSD, arrhythmia, cardiomyopathy, HF)

3/ **Distributive** (the tubing)shock
- Occur with profound **peripheral vasodilatation**
- CO & total body fluid volume are normal, but organ perfusion pressure are markedly compromized
  - Anaphylactic shock, septic shock /Toxic Shock Syndrome(TSS)
  - Neurogenic shock
  - Endocrine Ex. Hypothyroidism, Thyrotoxicosis , Addison's, etc

4/ **Obstructive** shock
- Obstruction of the blood flow outside of the heart but CO is normal

Ex. Cardiac Tamponade, Pneumothorax, pericarditis, pulm HPN

Effects of shock

- Decreased oxygen delivery
- Fall in body temperature
- Secretion of adrenalin
- Metabolic disorder
- Impairment of liver function
- Shock kidney
- Shock lung
- Reduced coronary perfusion
- Hypoxic brain damage

Stages of shock

- Pre-shock
- Shock
- End-organ dysfunction

Stages: Pre-shock

- Warm or compensated shock
- Regulatory mechanisms are able to compensate for diminished perfusion
- Low-preload:-
  - Tachycardia
  - Peripheral vasoconstriction
  - Decrease in blood pressure
- Low-after load:-
  - Peripheral vasodilation
  - Hyperdynamic state

Stages: Shock

- Compensatory mechanisms become overwhelmed, resulting in:-
  - Tachycardia
  - Tachypnea
  - Metabolic acidosis
  - Oilguria
  - Cool, clammy skin
- Usually occur with loss of 20-25% of effective blood volume

Shock—Pathophysiology

- Bp decreases when blood volume, heart contraction, or peripheral resistance fails
- Low CO, microcirculation: decreased oxygen, nutrients for cells
- **Compensation mechanism**
  - SNS, adrenal medulla stimulated
  - Renin secreted
  - Increased secretion of ADH
  - Secretion of glucocorticoids
  - Acidosis stimulates respiration
- **Complications of decompensation of shock**
  - Acute renal failure
  - Adult respiratory distress syndrome (ARDS)
  - Hepatic failures
  - Hemorrhagic ulcers
  - Infection of septicemia
  - Decreased cardiac function

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Pathophysiology––

- Shock affects mitochondria first
- Without oxygen mitochondria convert fuels to lactate
  → lactic acid
- Failure of the Krebs cycle: Oxygen is the final electron accepter to form water

ATP + H 

- Acidosis results from the accumulation of acid when during anaerobic metabolism the creation of ATP from ADP is slowed.

  H^+ shift extracellularly and a metabolic acidosis develops

Pathophysiology–-

- ATP production fails, the Na^+/K^+ pump fails resulting in the inability to correct the cell electronic potential.
- Cell swelling occurs leading to rupture and death.
- Oxidative Phosphorylation stops & anaerobic metabolism begins leading to lactic acid production.

Shock—signs and symptoms

- **1st signs**
  - **Hypotension**, thirst, agitation, restlessness
  - Often missed
- **2nd signs**
  - Cold, moist, pale skin; diaphoretic skin, tachycardia; oliguria
  - Tachypnea, shallow respirations
  - Compensation
  - Vasoconstriction
- Direct effects
  - Decrease bp and blood flow
  - Acidosis

- **Prolonged**
  - Alteration in level of consciousness, anxiety
  - Compensated metabolic acidosis progresses to decompensated
  - Acute renal failure
  - Monitoring(decreased urinary output)

	Hypovolemic	Cardiogenic	Septic	Neurogenic	Anaphylactic
Heart rate	Increased	Increased	Increased	Decreased	Increased
Blood pressure	Decreased	Decreased	Decreased	Decreased	Decreased
Preload	Decreased	Increased	Decreased	Decreased	Decreased
Afterload	Increased	Increased	Decreased	Decreased	Decreased

Empiric criteria for shock

4 out of 6 criteria have to be met

I. ILL appearance or altered mental status
II. Heart rate >100 bpm
III. Respiratory rate > 22 bm (or PaCO 
IV. Urine output < 0.5 ml/kg/hr
V. Arterial hypotension > 20 minutes duration
VI. Lactate > 4mmol/l (Pyruvate Dehydrogenase lactate)

OR

- CVP < 8 (N = 8 -12 mmHg)
- MAP < 65 mmHg (N = 65 - 95)

Shock—treatment

- Start treatment as soon as you suspect Pre-shock or Shock
- DO NOT WAIT for hypotension
- Primary problem must be treated
- Position the pt
- Prevent metabolic disturbances
- Prevent/treat hypothermia
- Treat pain if it is there b
- Maximize oxygen supply
- Epinephrine reinforces heart action and vasoconstriction
- Dopamine, dubutamine increase heart function
- Good prognosis in early stages
- Mortality increases as irreversible shock develops

✓ Hypovolemic shock
  – Whole blood, plasma, electrolytes, bicarbonate required

✓ Anaphylactic shock
  – Antihistamines, corticosteroids

✓ Septic
  – Antimicrobials, glucocorticoids

Shock—treatment —

- Mortality increases as irreversible shock develops

✓ Hypovolemic shock
  – Whole blood, plasma, electrolytes, bicarbonate required

✓ Anaphylactic shock
  – Antihistamines, corticosteroids

✓ Septic
  – Antimicrobials, glucocorticoids

- Intubation
- Warm the Pt
- Manage pain and anxiety
- Fluids - 20 ml/kg bolus x 3 (1-2 lit over 10 min for adults)
  – Normal saline, Ringer's lactate
  (Until MAP 60 mmHg or adequate peripheral pulses)
  – Use packed red blood cells if no improvement to fluids

Monitor the pts progress

Hypovolemic shock

- *Causes*
  - Hemorrhage
  - Vomiting
  - Diarrhea
  - Dehydration
  - Third-space loss
  - Burns
- *Signs*
  - ⇩ ♍♋❒♎♓♋♍ □◆♦□◆♦
  - ⇩ ⚐✌⚐⚐
  - ⇧ ♦✞☼

Hypovolemic shock –– S & Sxs

- Tachycardia and tachypnea
- Weak, tready pulses
- Hypotension
- Skin cool & clammy
- Mental status changes
- Decreased urine output: dark & concentrated

Classes of Hypovolemic Shock

	Class I	Class II	Class III	Class IV
Blood Loss	< 750	750-1500	1500-2000	> 2000
% Blood Vol.	< 15%	15 – 30%	30 – 40%	> 40%
Pulse	< 100	> 100	> 120	> 140
Blood Pressure	Normal	Normal	Decreased	Decreased
Pulse Pressure	Normal	Decreased	Decreased	Decreased
Resp. Rate	14 – 20	20 – 30	30 – 40	> 40
UOP	> 30	20 – 30	5 – 15	negligible
Mental Status	sl. Anxious	mildly anx	confused	lethargic
Fluid	crystalloid	crystalloid	blood	blood

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Treatment – Hypovolemic shock

- Reverse hypovolemia vs. hemorrhage control
- Transport times < 15 minutes showed pre-hospital fluids were ineffective, however, if transport time > 100 minutes fluid was beneficial.
- Pressors?-not indicated

Fluid administration

- Crystalloids are cheaper, 2-3 litres in 20-30 minutes
- Blood must supplement
- One big vein, or smaller veins
- Use larger Vigo/canula
- Warming fluid may be needed

Effects of acute blood loss

- Reduced oxygen transfer from the lungs to the red cells- due to decreased venous return, and CO
- Reduced oxygen storage by the red cells- reduced total amount of hg
- Reduced oxygen transport and delivery to the tissues –mismatching of the pulmonary blood flow and ventilation

Compensatory response to acute blood loss

- Restoration of plasma volume
- Restoration of CO- activation of sympathetic NS,
- Increase the rate and force of contraction
- Circulatory compensation- vasoconstriction
- Stimulation of ventilation
- Changes in the oxygen dissociation curve
- Hormonal changes

Blood volume

- Neonate- 85-90ml/kg
- Children and adolescents-70-80ml/kg
- Adult- 60-70ml/kg
- Compensatory vasoconstriction may hide the signs of acute blood loss until at least 10% of BV is lost
- Healthy pts may lose up to 20% of their BV before signs of hypovolemia occur
- Blood loss > 30% may exceed the ability of crystalloids to replace blood volume without jeopardizing the O



Blood loss estimation

- Clinical findings- history, V/S, capillary refill, mental state, urine out put, response to IV fluids, level of Hct …
- Amount and speed of blood loss- socked towels and swabs, contents of suction machine, inspection of the surgical area

Interventions

- Fluid resuscitation
  - Vascular access?
  - Type?
  - Volume?
- Monitor response
- Prevent hypothermia



Steps in shock management

1. Stabilization
2. Monitoring
3. Etiologic diagnoses
4. Specific management

Interventions

- **Fluid resuscitation**
  - **Vascular access?**
  - **Type?**
  - **Volume?**
- **Monitor response**

o Skin: tem, capillary refill

o Renal: increased urinary output

o Vital signs: PR BP

o CNS: improved level of consciousness

o Respiration

❖ Rapid responders

❖ Transient responders

❖ Non-responders

- **Prevent hypothermia!**



Class I Hemorrhage

750 mL BVL (15%)

- Slightly anxious
- Normal blood pressure
- Heart rate < 100 / min
- Respirations 14-20 / min
- Urinary output 30 mL / hour

Crystalloid

Class II Hemorrhage

750-1500 mL BVL (15-30%)

- Anxious
- Normal blood pressure
- Heart rate > 100 / min
- Decreased pulse pressure
- Respirations 20-30 / min
- Urinary output 20-30 mL / hour

Crystalloid,
? blood

Class III Hemorrhage

1500-2000 mL BVL (30-40%)

- Confused, anxious
- Decreased blood pressure
- Heart rate > 120 / min
- Decreased pulse pressure
- Respirations 30-40 / min
- Urinary output 5-15 mL / hour

Crystalloid, blood components, operation

Class IV Hemorrhage

>2000 mL BVL (>40%)

- Confused, lethargic
- Hypotension
- Heart rate > 140 / min
- Decreased pulse pressure
- Respirations >35 / min
- Urinary output negligible

Definitive control, blood components

Cardiogenic shock

- ***Cause***

Results from pump failure and decreased cardiac output

- ***Signs***
  - ⇩ ☃♋❒♎♓♋♍ □◆♦□◆♦
- Main categories:
  - Myopathies
  - Arrythimia
  - Mechanical
  - Extracardiac/obstructive

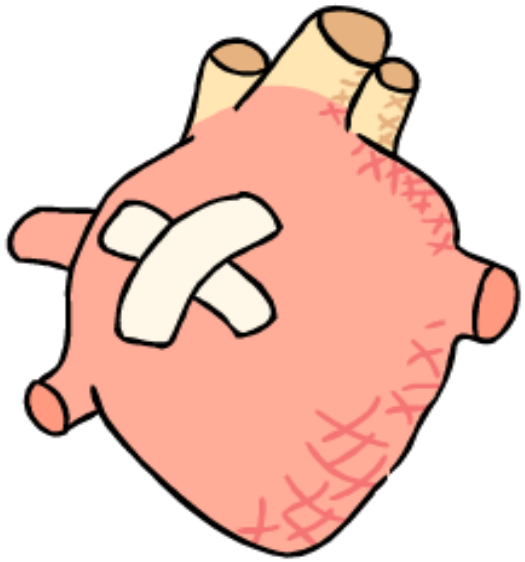
The failing heart

- Small amount of Fluids first, then cautious pressers

Cardiogenic shock---

- **Preload augmentation** - Consider Fluids

  Morphine as needed (Decreases preload, anxiety)
- **Contractility**
  - Dopamine
  - Dobutamine
  - Phosphodiesterase inhibitor
- **Afterload reduction**
  - Nitroglycerin
  - Dobutamine



Management cardiogenic shock

OPTIMIZING PUMP FUNCTION:

- **Pulmonary artery monitoring is a necessity !!**
- Aggressive airway management: Mechanical Ventilation
- Judicious fluid management
- Vasoactive agents
  - **Dobutamine**
  - **Dopamine**
- Morphine as needed (Decreases preload, anxiety)
- Cautious use of diuretics in CHF
- Vasodilators as needed for afterload reduction
- Short acting beta blocker, esmolol, for refractory tachycardia

Hemodynamic goals of cardiogenic shock

Optimized Cardiac function involves cautious use of combined fluids, diuretics, inotropes, vasopressors, and vasodilators to :

- Maintain adequate filling pressures (LVEDP 14 to 18 mmHg)
- Decrease Afterload (SVR 800-1400)
- Increase contractility
- Optimize CO/CI
- **Preload augmentation** - Consider fluids
- **Contractility**
  - Dopamine, dobutamine
  - Phosphodiesterase inhibitor
- **Afterload reduction**
  - Nitroglycerin, Dobutamine

Distributive shock

- ***Types***
  - Sepsis
  - Anaphylactic
  - Acute adrenal insufficiency
  - Neurogenic
- ***Signs***
  - ± Cardiac output , ⇩ PAOP, ⇩♦✞☼

SIRS - Distributive Shock

- Prompt volume replacement - fill the tank
- Early antibiotic administration
- Inotropes - first try Dopamine
- If MAP < 60
  - Dopamine = 2 - 3 μγ/κγ/μιν
  - Nor epinephrine = titrate (1-100 μg/min)

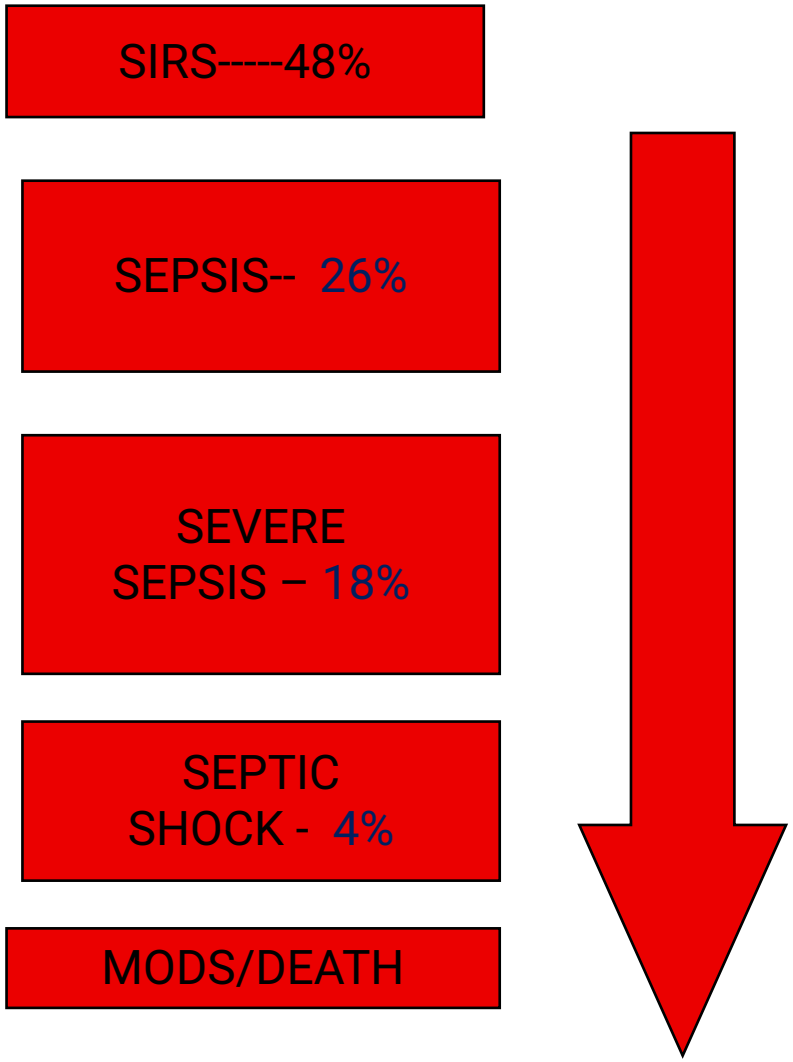
Wello University

Sepsis

- **Fluids- Initial resuscitation**
- CVP: 8- 12 mm Hg
- MAP ≥ 65 μμ Hγ
- UOP 0.5 ml/kg/hr
- Mixed venous oxygen sat 70%
- **Consider:**
  - Transfusion to Hgb ≥ 10
  - Dobutamine up to 20 g/kg/min
- **Correct the cause**
- Antibiotics , Debridement
- **Vasopressors**
  - Phenylephrine _ Levophed

Stages of Sepsis

48% of patients with SIRS developed part of sepsis continuum



Stages of Sepsis

SIRS = Systemic Inflammatory Response Syndrome

- Defined by the presence of ≥ 2 of the following:-
  - *Body temp < 36 °C or > 38 °C*
  - *Heart Rate > 90 bpm*
  - *RR > 20 bpm*
  - *WBC < 6,000 cells/mm³ or > 12,000 cells/mm³ , or greater than 10% band*

Management

- Prompt volume replacement - fill the tank
- Early antibiotic administration
- Inotropes - first try Dopamine
- If MAP < 60
  - Dopamine = 2 - 3 μγ/κγ/μιν
  - Nor epinephrine = titrate (1-100 μg/min)

Stages of Sepsis –-

- SIRS
- **Sepsis**- SIRS in response to a confirmed infectious process
- **Severe sepsis** – sepsis with one or more signs of organ dysfunction or hypoperfusion
- **Septic shock-** sepsis with refractory arterial hypotension or hypoperfusion abnormalities in spite of adequate fluid resuscitation

Septic shock **=** Sepsis + Refractory hypotension

- Unresponsive to initial fluids 20- 40cc/kg – Vasopressor dependant

- **MODS** – multiple organ dysfunction syndrome

2 or more organs

Ghana Emergency Medicine Collaborative
Advanced Emergency Trauma Course

Sepsis

- **Fluids**
- **Correct the cause**
  - Antibiotics
  - Debridement
- **Vasopressors**
  - Phenylephrine
  - Levophed

Initial resuscitation

- CVP: 8- 12 mm Hg
- MAP ≥ 65 μμ Hγ
- UOP 0.5 ml/kg/hr
- Mixed venous Oxygen Sat 70%
- Consider:
  - Transfusion to Hgb ≥ 10
  - Dobutamine up to 20 g/kg/min

Vasopressors

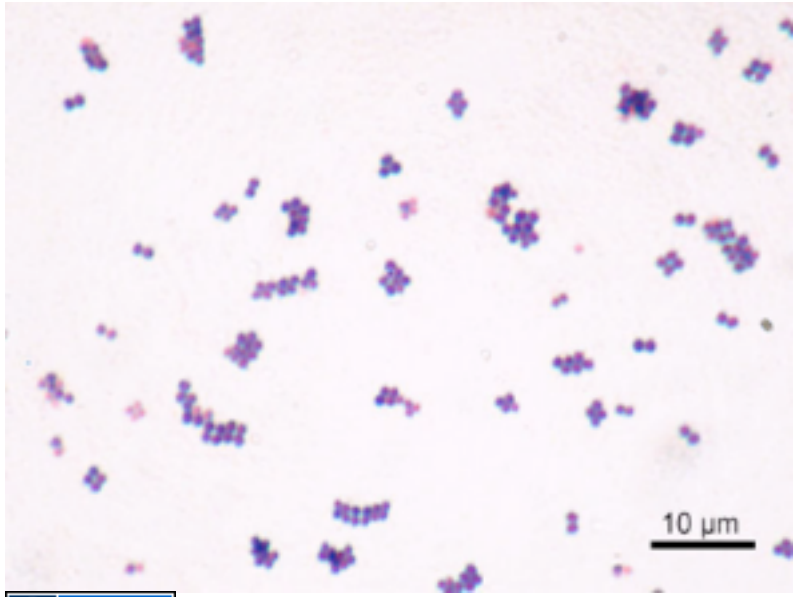
- Assure adequate fluid volume
- Administer via CVL
- Do not use dopamine for renal protection
- Requires arterial line placement
- **Vasopressin:**
  - Refractory shock
  - Infusion rate 0.01 – 0.04 Units/min

Steroid use in sepsis

- Refractory shock 200-300 mg/day of hydrocortisone in divided doses for 7 days
- ACTH test
- Once septic shock resolves, taper dose
- Add fludrocortisone 50 μγ po q day

Septic shock

- A blood borne infection widely disseminated to many areas of the body
- Common features are high fever, vasodilatation (especially in affected tissues)
- Sludging of the blood, and RBC agglutination resulting in DIC



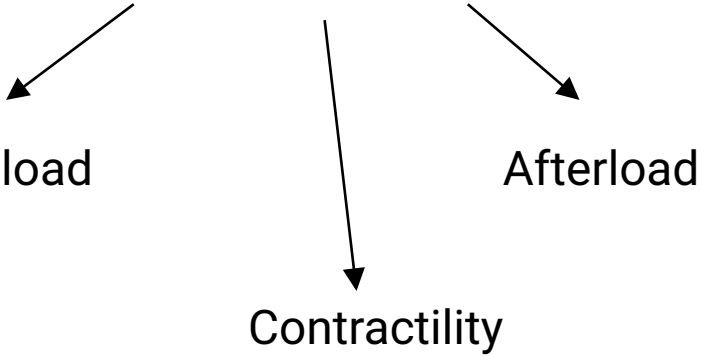
PD-INEL

http://en.wikipedia.org/wiki/File:
Staphylococcus_aureus_Gram.jpg

Equation

MAP = CO x SVR

(HR x Stroke volume)



Resuscitation in Sepsis: EGDT

- The theory is to normalize…
- **Preload - 1.**
- **Afterload - 2.**
- **Contractility - 3.**

Preload - 1.

- Dependent on intravascular volume
  - If depleted intravascular volume (due to increased endothelial permeability) - PRELOAD DECREASES
- Can use the CVP as measurement of preload
  - Normal = 8-12 mm Hg
- How do you correct decreased preload (or intravascular volume)
  - Give fluids
  - Rivers showed an average of **5 L** in first 6 hours
- What is the end point?

Resuscitation in Sepsis: EGDT

Afterload - 2

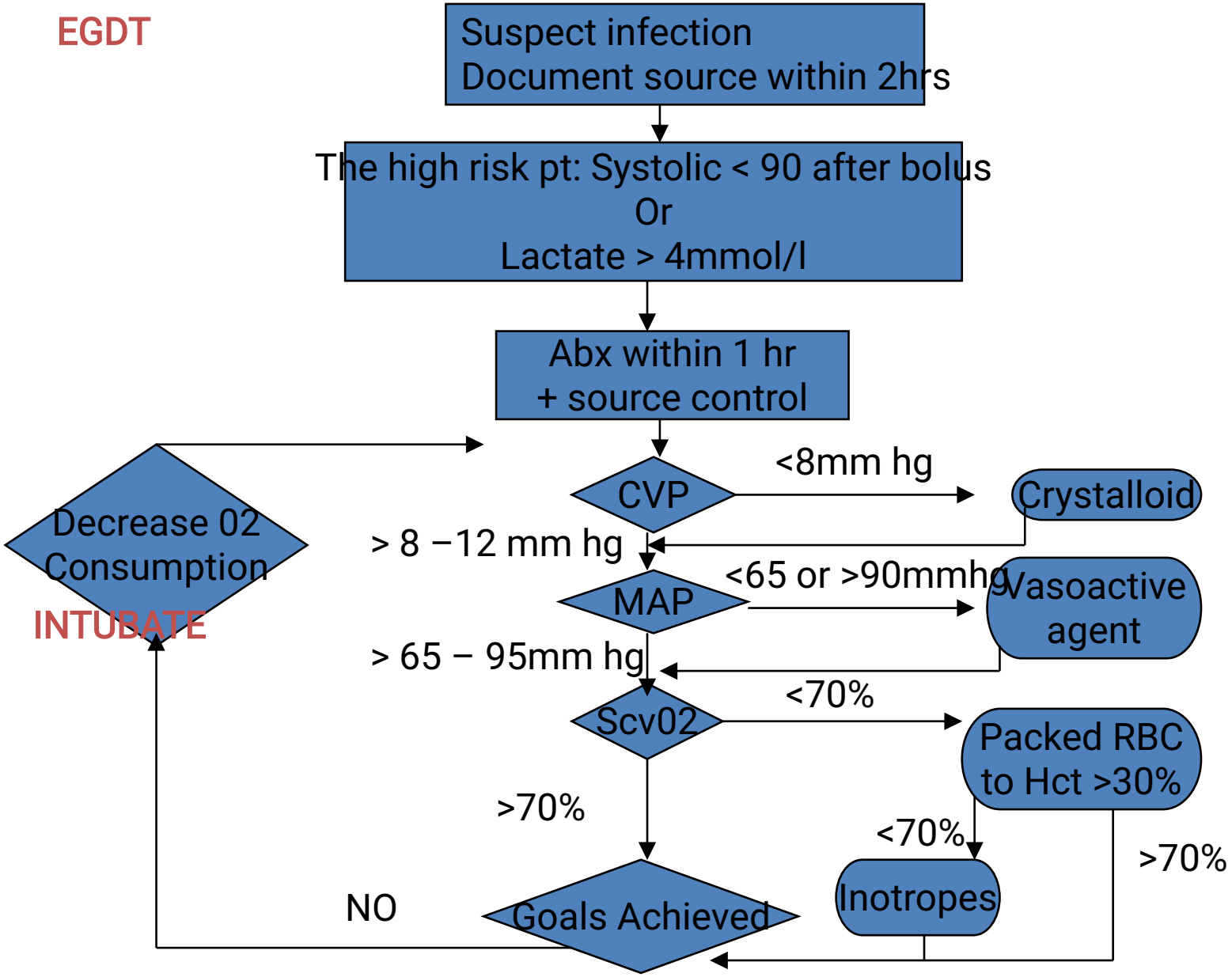
- Afterload determines tissue perfusion
  - Using the MAP as a surrogate measure - Keep between 60-90 mm Hg
  - In sepsis afterload is decreased d/t loss of vessel tone
- How do you correct decreased afterload?
- Use vasopressor agent
  - Norepinephrine
  - Alternative Dopamine or Phenylpehrine

Contractility - 3rd

- Use the central venous oxygen saturation (ScvO2) as a surrogate measure
- Shown to a be a surrogate for cardiac index
- Keep > 70%

  How to improve ScvO2 > 70%?
- Optimize arterial O2 with non-rebreather
- Ensure a hematocrit > 30 (Transfuse to reach a hematocrit of > 30)
- Use Inotrope - Dobutamine 2.5ug/kg per minute and titrated (max 20ug/kg)
- Respiratory Support - Intubation (Don’t forget to sedate and paralyze)

EGDT


Suspect infection
Document source within 2hrs
The high risk pt: Systolic < 90 after bolus
Or
Lactate > 4mmol/l
Abx within 1 hr
+ source control
CVP
<8mm hg
Crystalloid
> 8 –12 mm hg
MAP
<65 or >90mmhg
Vasoactive agent
> 65 – 95mm hg
Scv02
<70%
Packed RBC to Hct >30%
>70%
<70%
>70%
Inotropes
Goals Achieved
NO
Decrease 02 Consumption
INTUBATE

Toxic shock syndrome(TSS)

Staphyloccocal
&
Streptococcal Origins

- Rare, potentially life threatening disorder.
- Occurs when toxins made by certain types of bacteria (S.aureus) are released into the bloodstream.

TSS- Signs and Symptoms



- Fever
- Confusion
- Diarrhea
- Dizziness or fainting
- Headaches
- Pelvic Pain
- Sore throat

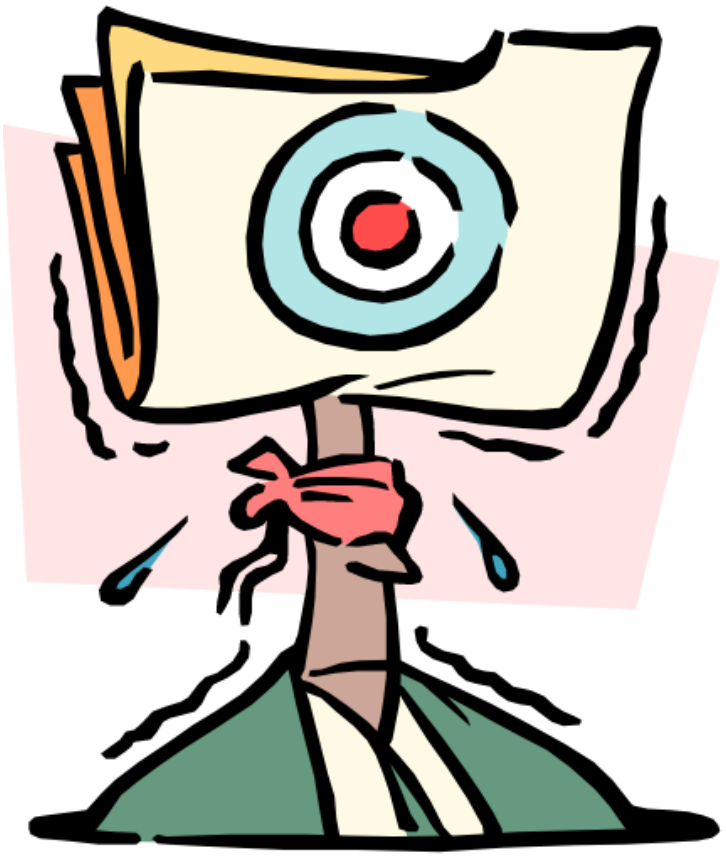
More Signs and Symptoms



- Sunburn like rash anywhere on the body, but usually on the soles of the hands and feet.
- Vomiting
- Photophobia-discomfort looking @ light.
- Myalgia-aching muscles
- Low blood pressure

Causes and Risks

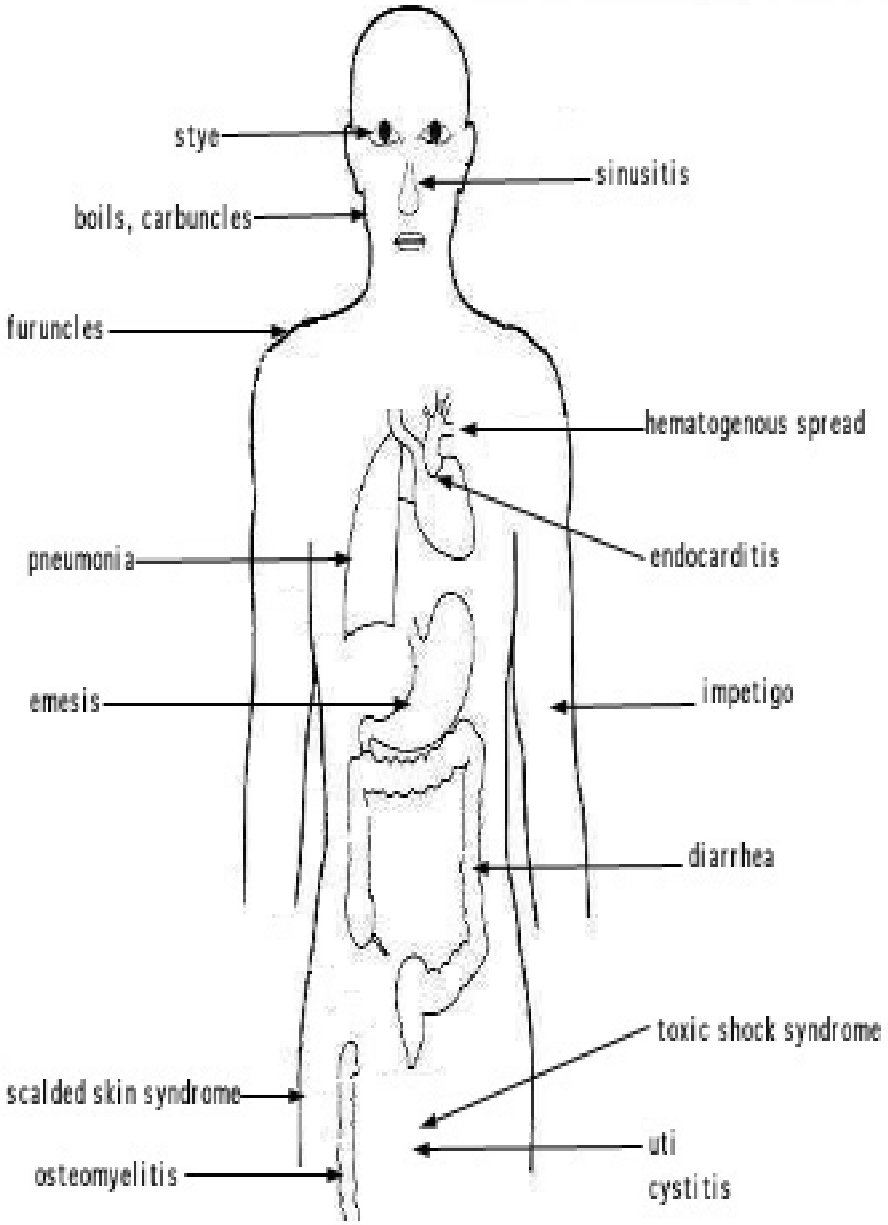
- Having recently delivered a baby.
- Recently had surgery
- Leaving a diaphragm or cervical cap in for 36 hrs. Or longer.
- Wearing the same tampon for longer than 8 hours.
- Also been reported in those following nasal surgery or use of nasal packing for a nose bleed. (Rare)



Prevention

- Alternate tampons with pads every other day during the heaviest flow.
- Avoid or minimize use of superabsorbant tampons.
- Change tampons every 6-8 hrs
- Use pads while sleeping.
- Wash hands regularly
- Change dressings to wounds at least every day
- Keep area clean and dry
- Practice aseptic technique

How is it diagnosed?



- History and Physical exam
- Blood C/X (cultures)
- CBC (complete blood count)
- Kidney Function tests
- Liver Function tests
- Spinal Tap
- Throat cultures
- U/A (urinalysis)
- Vaginal C/X (cultures) for S. aureus

TSS- Treatment

- Aggressive Therapy
- IV Antibiotics
- Blood Transfusions
- Corticosteroids
- Electrolyte replacements
- B/P meds
- Ventilator if lungs are damaged



Neurogenic shock

- A type of distributive shock that results from the loss or suppression of sympathetic tone
- Causes massive vasodilatation in the venous vasculature, ⊠ venous return to heart, ⊠ cardiac output.
- Most common etiology: Spinal cord injury above T6
- Neurogenic is the rarest form of shock!

Pathophysiology of Neurogenic Shock

Disruption of sympathetic nervous system

Loss of sympathetic tone

Venous and arterial vasodilatation

Decreased venous return

Decreased stroke volume

Decreased cardiac output

Decreased cellular oxygen supply

Impaired tissue perfusion

Impaired cellular metabolism

Assessment, diagnosis and management of neurogenic shock

PATIENT ASSESSMENT

- Hypotension
- Bradycardia
- Hypothermia
- Warm, dry skin
- RAP ↓
- PAWP ↓
- CO ↓
- Flaccid paralysis below level of the spinal lesion

MEDICAL MANAGEMENT

- Goals of therapy are to treat or remove the cause & prevent cardiovascular instability, & promote optimal tissue perfusion

Management of neurogenic shock

Hypovolemia- Rx with careful fluid replacement for
BP< 90mmHg, Urine O <30cc/hr

Changes in LOC

Observe closely for fluid overload

Vasopressors may be needed

Hypothermia- warming txs

-Avoid large swings in pts body temperature

Treat Hypoxia

Maintain ventilatory support

Management of - neurogenic shock

- Observe for Bradycardia-major dysrhythmia
- Observe for DVT- venous pooling in extremities make patients high-risk>>P.E.
- Use prevention modalities [TEDS, ROM,Sequential stockings, anticoagulation]
  - Alpha agonist to augment tone if perfusion still inadequate
    - **dopamine at alpha doses (> 10 mcg/kg per min)**
    - **ephedrine (12.5-25 mg IV every 3-4 hour)**
  - Treat bradycardia with atropine 0.5-1 mg doses to maximum 3 mg
    - **may need transcutaneous or transvenous pacing temporarily**

Adrenal Crisis- distributive shock

- ***Causes***
  - Autoimmune adrenalitis
  - Adrenal apoplexy = hemorrhage or infarct
  - Heparin may predispose bleeding.
- Steroids may be lifesaving in the patient who is unresponsive to fluids, inotropic, and vasopressor support. Which one?

Obstructive shock

- ***Causes***
  - Cardiac Tamponade
  - Tension Pneumothorax
  - Massive Pulmonary Embolus
- ***Signs***
  - ⇩ ♍♋❒♎♓♋♍ □◆♦□◆♦
  - ⇧ 🏳✌🏳🏳
  - ⇧ 🩸✞☼

Vasopressor agents?

- Augments contractility, after preload established, thus improving cardiac output.
- Risk tachycardia and increased myocardial oxygen consumption if used too soon

Vasopressors & Inotropic agents

A. Dopamine

- Low dose (0.5 - 2 μγ/κγ/μιν) = dopaminergic
- Moderate dose (3-10 g/kg/min) = -effects
- High dose (> 10 g/kg/min) = -effects

Side effects

- Tachycardia
- > 20 g/kg/min to Norepinephrine

B. Dobutamine

- β–αγονιστ 5 – 20 μγ/κγ/μιν
- Potent inotrope, variable chronotrope
- Caution in hypotension (inadequate volume) may precipitate tachycardia or worsen hypotension

C. Norepinephrine

- Potent α–αδρενεργιχ vasopressor
- Some β-adrenergic, inotropic, chronotropic
- Dose 1 - 100 μg/min
- Unproven effect with low-dose dopamine to protect renal and mesenteric flow.

D. Epinephrine

- α– ανδ β–αδρενεργιχ εφφεχτσ
- Potent inotrope and chronotrope
- Dose 1 - 10 μg/min
- Increases myocardial oxygen consumption particularly in coronary heart disease

B. Disorders of the veins

Phlebitis

- Phlebitis is defined as inflammation of a vein related to a chemical or mechanical irritation, or both
- It is characterized by
  - a reddened, warm area around the insertion site or along the path of the vein,
  - Pain or tenderness at the site or along the vein, and swelling
- The incidence of phlebitis increases with
  - The length of time the intravenous line is in place,
  - The composition of the fluid or medication infused
  - The size and site of the cannula inserted
  - Improper anchoring of the line, and
  - The introduction of micro-organisms at the time of insertion.

Phlebitis–-

- Treatment consists of
  - Discontinuing the IV and restarting it in another site, and
  - Applying a warm moist compress to the affected site.
- Phlebitis can be prevented by using
  - Aseptic technique during insertion
  - The appropriate size catheter and needle size for the vein,
  - Considering the composition of fluids and medications when selecting a site,
  - Observing the site for any complications very hour, and
  - Anchoring the catheter or needle well.

Thrombophlebitis

- **Thrombophlebitis**
  - Refers to the presence of a clot plus inflammation in the vein
  - Is marked by inflammation of the venous wall and thrombus formation of the deep or superficial veins
  - It is evidenced by
    - Localized pain, redness, warmth, and swelling around the insertion site or along the path of the vein, immobility of the extremity because of discomfort and swelling, sluggish flow rate, fever, malaise, and leukocytosis.
  - Deep vein thrombophlebitis may lead to occlusion of the vessels or systemic embolization such as pulmonary embolism

Thrombophlebitis…

- Several conditions may lead to thrombophlebitis, including
  - hypercoagulability (such as from cancer, blood dyscrasias, or oral contraceptives);
  - injury to the venous wall (such as from I.V. injections, fractures, antibiotics, or infection); and
  - venous stasis (such as from varicose veins, pregnancy, heart failure, or prolonged bed rest)
- Treatment includes
  - discontinuing the IV infusion,
  - applying a cold compress first to decrease the flow of blood and
  - increase platelet aggregation followed by a warm compress,
  - elevating the extremity, and
  - restarting the line in the opposite extremity.

Thrombophlebitis…

- If the patient has signs and symptoms of thrombophlebitis, the IV line should not be flushed (although flushing may be indicated in the absence of phlebitis to ensure cannula patency and to prevent mixing incompatible medications and solutions).
- Thrombophlebitis can be prevented by avoiding trauma to the vein at the time the IV is inserted, observing the site every hour, and checking medication additives for compatibility

Thrombophlebitis–-

Signs and symptoms

- Deep vein thrombophlebitis will sometimes cause no clinical symptoms or physicial findings; when they do occur, they may include
  - Cramping pain, edema, tenderness to touch, fever, chills, and malaise
- Superficial thrombophlebitis produces visible and palpable signs, such as
  - heat, pain, swelling, tenderness, and induration along the affected vein’s length

Thrombophlebitis….

Diagnosis and treatment

- Diagnostic tests may include photoplethysmography, Doppler ultrasonography, and venography; laboratory tests include a CBC
- Superficial thrombophlebitis may require no specific therapy other than treatment for symptoms
- An anticoagulant (initially I.V. heparin or low-molecular-weight heparin followed by oral warfarin is administered to prolong clotting time
- Thrombolytic therapy (such as streptokinase [Streptase]) is indicated for acute, extensive deep vein thrombophlebitis
- Embolectomy, venous ligation, or insertion of a vena caval umbrella or filter may also be indicated

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Thrombophlebitis–-

Nursing interventions

- If the patient is receiving a thrombolytic, heparin, or warfarin, monitor him for signs and symptoms of bleeding
- Assess the patient for signs and symptoms of pulmonary embolism, such as crackles, dyspnea, tachypnea, hemoptysis, tachycardia, and chest pain
- Make sure the patient maintains bed rest and elevates the affected extremity
- Apply moist, warm compresses to improve circulation to the affected area and relieve pain
- Tell the patient to avoid prolonged sitting and standing to help prevent recurrence
- Teach the patient how to properly apply and use antiembolism stockings

Deep Vein Thrombosis (DVT)

- **Deep veins –**
  - Obstruction of the deep veins of the legs produces edema & swelling of the extremity because the out flow of venous blood is inhibited
  - The mount of swelling can be determined by measuring the circumference of the leg at various levels with a tape measure.
  - One leg is compared with the other at the some level to determine size may be difficult to detect
  - The affected leg, may feel warmer than the un affected leg, & the superficial veins may appear to be more prominent.
  - Tenderness, which usually occurs later, is produced by inflammation of the vein wall and can be detected by gently palpating the leg

Deep Vein Thrombosis (DVT)…

- "Homans' sign (pain in the calf after the foot is sharply dorsiflexed) is not specific for deep venous thrombosis because it can be elicited in any painful condition of the calf.
- In same cases, signs of a pulmonary embolus are the first indication of a DVT.

Interventions

- Encourage rest
- Facilitate bed rest and elevation of the extremity above the level of the heart
- Administer intermittent or continuous warm moist compresses as ordered (to prevent thrombus from dislodging and becoming an embolus, do NOT massage the affected limb)

Deep Vein Thrombosis (DVT)––

Interventions…..

- Administer medications as prescribed. Ex. Anticoagulants
- Thrombolytic therapy is effective in dissolving thrombi quickly and completely. It must be initiated within 5 days after the onset of symptoms to be most effective.
- The advantage is the prevention of valvular damage and consequential venous insufficiency, or “postphlebitis syndrome.”
- Analgesics

Varicose veins

- **Varicose veins (varicosities)**
  - Are abnormally dilated, tortuous, superficial veins caused by in competent venous values.
  - Most commonly, this condition occurs in the lower extremities, the saphenous veins, or the lower trunk; however, it can occur else where in the , such as esophageal varices
  - The condition is most common in women and in people whose occupations require prolonged standing, such as salespeople, hair stylists, teachers, nurses, ancillary medical personnel, and construction workers

Varicose veins …

- A hereditary weakness of the vein wall may contribute to the development of varicosities, and it is not uncommon to see this condition occur in several members of the same family
- Varicose veins are rare before puberty.
- Pregnancy may cause varicosities.
- The leg veins dilate during pregnancy because of hormonal effects related to distensibility, increased pressure by the gravid uterus, and increased blood volume which all contribute to the development of varicose veins

Varicose veins---

Pathophysiology

- Varicose veins may be considered primary (without involvement of deep veins) or secondary (resulting from obstruction of deep veins)
- A reflux of venous blood in the veins results in venous stasis.
- If only the superficial veins are affected, the person may have no symptoms but may be troubled by the appearance of the dilated veins

Varicose veins---

Pathophysiology and manifestations

- Varicose veins may be considered primary (with out involvement of deep veins) or secondary (resulting from obstruction of deep veins)
- A reflex of venous blood in the veins results in venous stasis. If only the superficial veins are affected, the person may have no symptoms but may be troubled by the cosmetic appearance of the dilated veins. Symptoms, if present, may take the form of dull aches, muscle cramps, and increased muscle fatigue in the lower legs. Ankle edema and a feeling of heaviness of the legs may occur. Nocturnal cramps are common.
- When deep venous obstruction results in varicose veins, patients may demonstrate the signs and symptoms of chronic venous insufficiency and edema, pain, pigmentation, and ulcerations, susceptibility to injury and infection is increased.

Varicose veins–-

Clinical manifestations

- Symptoms, if present, may take the form of dull aches, muscle cramps, and increased muscle fatigue in the lower legs
- Ankle edema and a feeling of heaviness of the legs may occur.
- Nocturnal cramps are common. When deep venous obstruction results in varicose veins, patients may develop the signs and symptoms of chronic venous insufficiency: edema, pain, pigmentation, and ulcerations
- Susceptibility to injury and infection is increased

Varicose veins---

Prevention

- The patient should avoid activities that cause venous stasis, such as
  - Wearing tight socks or a constricting panty girdle,
  - Avoid crossing the legs at the thighs, and
  - Avoid sitting or standing for long periods.
  - Changing position frequently

Varicose veins—

- **Prevention…**
  - Elevating the legs when they are tired, and getting up to walk for several minutes of every hour promote circulation.
  - The patient should be encouraged to walk 1 or 2 miles each day if there are no contraindications.
  - Walking up the stairs rather than using the elevator or escalator is helpful in promoting circulation.
  - Swimming is also good exercise for the legs.
  - The overweight patient should be encouraged to begin a weight-reduction plan.

Varicose veins---

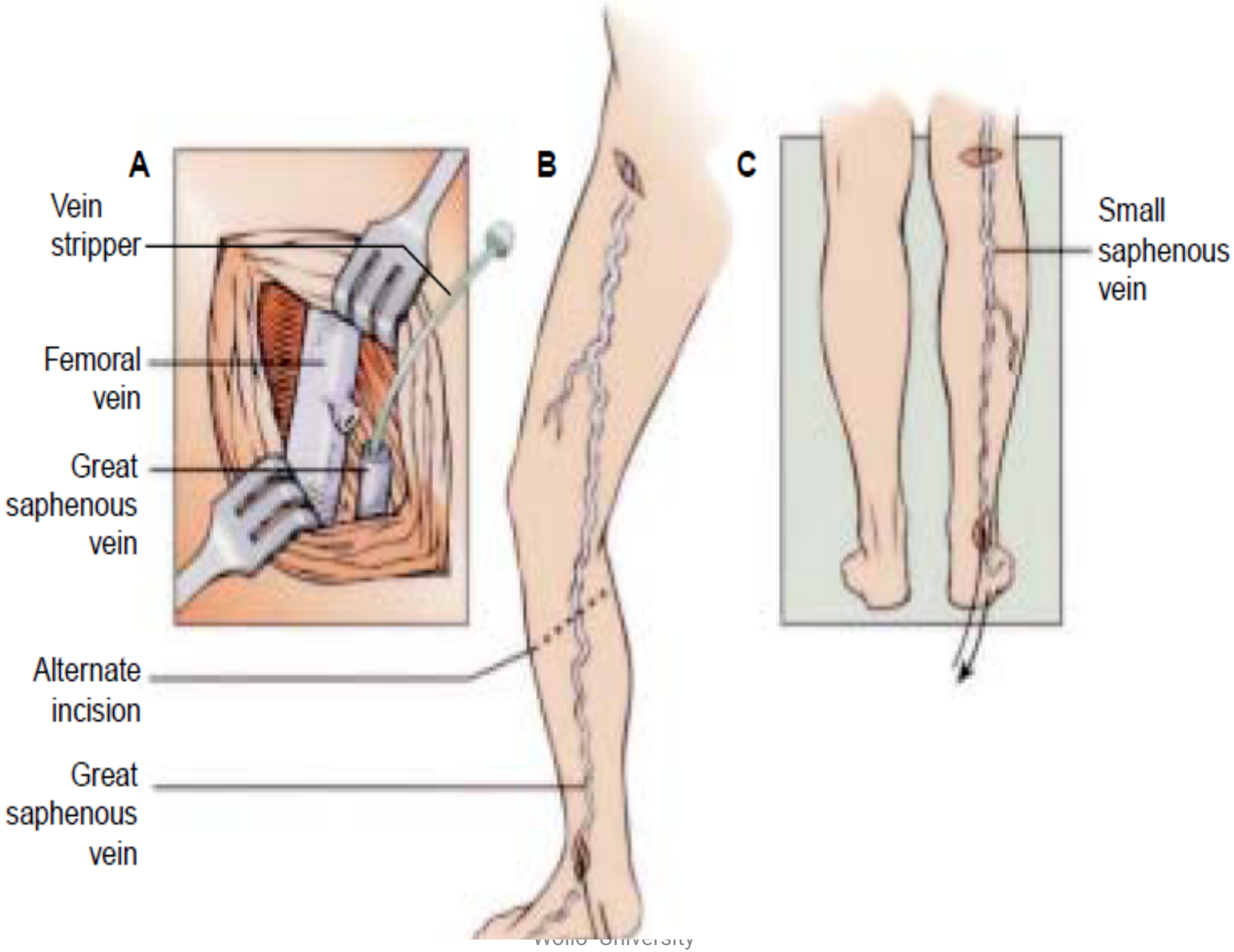
Surgical managemet

- Surgery for varicose veins requires that the deep veins be patent and functional.

- The patient is placed under general anesthesia, and the saphenous vein is ligated and divided.
- The vein is ligated high in the groin where the saphenous vein meets the femoral vein.

Post Operative Nsg management

- Bed rest is maintained for 24 hours after which the patient begins every 2 hours for 5 to 10 minutes.
- Elastic compression of the leg.
- Exercise and movements of the legs and elevation of the foot of the bed.
- Analgesics


A
B
C
Vein stripper
Femoral vein
Great saphenous vein
Alternate incision
Great saphenous vein
Small saphenous vein

C. **Disorders of lymphatic system**

A. Non-hodgkin's/Hodgkin's/ lymphoma B. Lymphadenitis C. Lympangitis D. Multiple myeloma

- The lymphomas are neoplasms of cells of lymphoid origin
- These tumors usually start in lymph nodes but can involve lymphoid tissue in the spleen, the gastrointestinal tract (eg, the wall of the stomach), the liver, or the bone marrow
- They are often classified according to the degree of cell differentiation and the origin of the predominant malignant cell. Lymphomas can be broadly classified into two categories: Hodgkin’s disease and non-Hodgkin’s lymphoma (NHL)

Lymphomas

Hodgkin's lymphoma

- Most cases involve adults
- Possible causes include viral infections and exposure to chemical agents
- The presence of Reed-Sternberg cells (B lymphocytes that have become cancerous) is diagnostic for Hodgkin's disease

Non-Hodgkin's lymphoma

- More common in clients older than 50 years
- Possible causes :- Gene damage, viral infections, autoimmune disease, and exposure to radiation or to toxic chemicals

Lymphomas––

The lymphomas are neoplasm's of the cells of lymphoid origin

These tumors usually start in lymph nodes, but can involve lymphoid tissue in the spleen, the gastro intestinal tract (for example, the wall of the stomach) the liver or the bone marrow they often spread to all of these areas & to extra lymphatic tissues (lungs, kidneys, skin) by the time of death

The cause of these tumors is unknown

Hodgkin's diseases

- Hodgkin's diseases like other lymphomas is a malignant disease of unknown origin that originates in the lymphatic system & involves the lymph nodes
- It is more common in men & tends to peak in the early 20s & after age 50. The "Reed-sternberg cell, agigantic atypical tumor cell, is the pathologic hall mark & essential diagnostic criterion for Hodgkin's disease
- The cause is unknown but a viral etiology is suspected

Hodgkin's lymphoma –-

Clinical manifestations

1. Usually begins as a painless enlargement of one or more lymph nodes on one side of the neck

a. Individual nodes are rubbery & **painless** (firm but not hard)

b. Lymph nodes of other regions (the other side of the neck) also enlarge in the same manner.

2. Mediastinal & retroperitoneal lymph nodes may also enlarge, causing severe pressure symptoms

a. Dyspnea from pressure against the trachea

b. Dysphagia from pressure against the esophagus.

c. Laryngeal paralysis & brachial, number, or sacral neuralgias from pressure on nerves.

d. Edema of one or both extremities & effusion in to the pleura from the pressure on veins.

e. Obstructive jaundice from pressure on the bile duct

Hodgkin's lymphoma –––

Clinical manifestations–––

3. Later, spleen may become palpable & liver may enlarge
4. In some patients the first nodes to enlarge are in the axillae or groin, occasionally mediastinal or peritoneal nodes, or enlargement of spleen may be the only sign.
5. Eventually progressive anemia develops
6. Fever varies with pathologic involvement.
7. Un treated this is a progressive disease. Causing weight loss, cachexia, infection, anemia, anasarca & hypo tension, soaking night sweats, generalized unexplained itching ,pain in lymph node after alcohol
8. Death is likely in 1 to 3 years with out treatment

Hodgkin's lymphoma –-

Clinical manifestations

- All organs are vulnerable to invasion by Hodgkin's disease. The symptoms result from compression of organs by the tumor, such as cough and pulmonary effusion (from pulmonary infiltrates), jaundice (from hepatic involvement or bile duct obstruction), abdominal pain (from splenomegaly or retroperitoneal adenopathy), or bone pain (from skeletal involvement). Herpes zoster infections are common.
- A cluster of constitutional symptoms has important prognostic implications. Referred to as "B symptoms," they include fever (without chills), drenching sweats (particularly at night), and unintentional weight loss of more than 10%. "B symptoms" are found in 40% of patients and are more common in advanced disease.
- A mild anemia is the most common hematologic finding. The WBC count may be elevated or decreased. The platelet count is typically normal, unless the tumor has invaded the bone marrow, suppressing hematopoiesis.
- The erythrocyte sedimentation rate (ESR) and the serum copper level are used by some clinicians to assess disease activity. Patients with Hodgkin's disease have impaired cellular immunity, as evidenced by an absent or decreased reaction to skin sensitivity tests (eg, *Candida, mumps).*

Hodgkin's Lymphoma –-

Diagnostic evaluation

- Lymph node biopsy and the finding of the **Reed-Sternberg cell**

1. Laboratory studies, complete blood studies, liver & renal function studies
2. Bone marrow biopsy & liver & spleen scans
3. Chest X-rays & extensive bone scans

Management

- Treatment is determined by the stage of the disease instead of the histologic type.
- Hodgkin's disease is potentially curable by radiotherapy if it has not extended beyond the lymph node chains, spleen, & oronaso pharynx
- Patients who do not have extension of the disease should have the benefit of "curative" radiotherapy.
- Patients who have any sign of spread beyond treatable areas should receive a combination of chemotherapy & palliative radiotherapy.

Staging lymphomas

Hodgkin's disease and non-Hodgkin's lymphoma are classified into stages so that treatment protocols can be established and outcomes predicted. The Ann Arbor classification system is used.

Stages

Stage I

- Involvement of a single lymph node region

or

- Localized involvement of a single extranodal organ or site

Stage II

- Involvement of two or more lymph node regions on the same side of the diaphragm

or

- Localized involvement to a single associated extranodal organ or site and its regional lymph nodes with or without other lymph node regions on the same side of the diaphragm

Stage III

- Involvement of lymph node regions on both sides of the diaphragm that may also be accompanied by localized involvement of an extranodal organ or site, by involvement of the spleen, or by both

Stage IV

- Disseminated involvement of one or more extralymphatic sites with or without associated lymph node involvement

or

- Isolated extralymphatic organ involvement with distant (nonregional) nodal involvement

Extralymphatic sites include the liver, lungs, bone marrow, pleurae, bone, and skin as well as tissues separate from but near to major lymphatic clusters. The spleen is considered a lymphatic site.

Letter designation

The stages are also accompanied by a letter designation that refers to patient symptoms as follows:

- **A** means absence of the specific symptoms listed in **B** but may include other common symptoms such as pruritus.
- **B** includes the presence of at least one of the following three symptoms:

– temperature greater than 100.4° F (38° C)
– unexplained loss of more than 10% of body weight in the preceding 6 months
– drenching night sweats

Non-Hodgkin's lymphoma(NHLs)

- Non-Hodgkin's lymphoma describes a group of malignant neoplasms arising from abnormal lymphocytes that affect the immune system; this disease is usually disseminated when diagnosed and produces systemic symptoms
- Most NHLs involve malignant B lymphocytes; only 5% involve T lymphocytes. In contrast to Hodgkin's disease, the lymphoid tissues involved are largely infiltrated with malignant cells
- It may be associated with viral infection, congenital immunodefi ciency, or immuno-suppression after organ transplantation
- The prognosis is poor but depends on the histologic type and progression of the disease

Signs and symptoms

- Non-Hodgkin's lymphoma can affect any organ or tissue and produces wide range of symptoms
- Signs and symptoms resemble those of Hodgkin's disease
- Fever, painless lymphadenopathy, profuse sweating (especially at night), severe pruritus, abdominal pain or swelling, and weight loss are common signs and symptoms

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Non-Hodgkin's lymphoma–-

Diagnosis and treatment

- Diagnostic procedures are similar to those for Hodgkin's disease; however, no Reed-Sternberg cells are present
- Staging is similar to Hodgkin's disease
- Radiation therapy, chemotherapy, and stem cell transplantation are the principal treatments
- Biological response modifiers, such as interferon and monoclonal antibodies, also may be used to change the tumor-host relationship
- Radio-immunotherapy may also be used
- Surgery

Nursing interventions

- Educate the patient and his family about transfusions, chemotherapy, and stem cell transplantation
- Initiate interventions similar to those for leukemia
- Prevent infection
- Support nutritional status

Lymphadenitis & Lymphangitis

Lymphadenitis: - is an inflammation of a lymph node

Lymphangitis: - is an inflammation of lymphatic vessel or vessels

- Lymphadenitis: - is an acute inflammation of the lymphatic channels. It arises most commonly from a focus of infection in an extremity
- Usually, the infectious organisms are the hemolytic streptococcus
- The characteristic red streaks that extend up the arm or the leg from an infected wound out line the course of the lymphatic vessels as they drain
- The lymph nodes located along the course of the lymphatic channels also become enlarged, red & tender (acute lymphadenitis) and can become neurotic & from an abscess (suppurative lympahdenitis)

Lymphadenitis & Lymphangitis–-

- The nodes involved most often are those in the groin the axilla, or the cervical region
- Because these infections are nearly always causes by organisms that are sensitive to antibiotic, it is unusual to see abscess formation
- Recurrent episodes of lymphangitis are often associated with progressive lymph edema
- After acute attacks an elastic stocking or sleeve should be worn on the affected extremity for several months to prevent long-term edema

Lymphadenitis

1. Lymph-node tuberculosis (tuberculosis lymphadenitis)

- One of the commonest presentations of extra pulmonary TB (being documented in more than 25% of cases) lymph node disease is particularly frequent among HIV infected patients.
- Lymph nodes tuberculosis presents as painless, swelling of the lymph nodes, most commonly at cervical & supra clavicular sites
- Systemic symptoms are usually limited to HIV infected patients, and concomitant lung disease may or may not be present. The diagnosis is established by fine needle aspiration or surgical biopsy
- AFB are seen in up to 50 percent of cases, cultures are positive in 70 to 80 present & histologic. Examination shows, granvlomas usually are not seen. Differential diagnosis includes a variety of infectious conditions as well as neo plastic diseases such as lymphoma or metastatic carcinomas.

Lymphadenitis & Lymphangitis

2. Lymphadenitis

- Non-tuberculosis mycobacterium (NTM) " NTM" are among the causes of localized lymphadenitis. The disease occurs mostly in children b/n the age of 1 & 5 Years
- Painless swelling of one nod e or a group of nodes usually affects the anterior cervical chain
- Nodes may rapidly increase in size, with the formation of fistulas to the skin
- M. Scrofulaceum or “MAC" organisms most commonly cause NTM lymphadenitis, although many other species may be involved. Once tuberculosis has been excluded, the treatment of choice is excision with out chemotherapy. When excision is dangerous because of proximity to the facial nerve, aspiration combined with chemotherapy may be effective

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Burkittis lymphoma

- B-cell lymphoma
- EBV-virus, chromosomal translocation (8;14 or 8;22), malaria
- American type
- Jaw involvement: Max:Mand.=2:1
- Histology: “Starry sky” pattern
- Treatment: cyclophosphamide

Hematologic system

Includes:-

- **Blood cells** (RBCS, WBCs, Platelets)
- **Plasma** (55 % of blood) fluids, electrolytes, nutrients, waste products, plasma proteins
- **Plasma proteins** (Albumin, globulins, fibrinogen, clotting factors)
- **Bone marrow**
- **Reticuloendothelial system** (RES) or mononuclear phagocytic system includes: spleen, lymph nodes, liver ,etc

Hematologic system–-

Albumin: maintain fluid balance in vascular system.

Globulins

- Alpha globulins: for transportation
- Beta globulins: for transportation

- They carry various substances in bound forms

Ex. TBG carries thyroxin, Transferrin carries iron

- Gamma globulins : refers to immunoglobulins /antibodies

Hematologic system---

RES are special tissue macrophages involved in the removal of old & defective RBCs and other defective cells from the circulation

RES plays role in phagocytosis, cleaning of foreign bodies particularly bacteria from blood,lymph,& interstitial space.

Major organs involved in RES are the spleen, lymph nodes & the liver

Ex. Kupffer cells, Alveolar macrophages,peritonial macrophages

Spleen

Functions

- **Hematopoiesis**

- Blood cell production during fetal life,& resume hematopoiesis later in adulthood when bone marrow fails)

- **Site of cellular maturation**

- Spleen sequesters newly released reticulocytesfrom the bone marrow removing their nuclear fragment before the return to the circulation as fully matured RBCs

- **Site of filtration**

- It removes old and defective RBCSfrom the circulation. It returns iron component of the Hgb to the bone marrow for reuse of iron

- **Immunologic function**

- Spleen has reach supply of lymphocytes

- **Site of storage**

- It stores nearly 30 % of platelets mass & nearly 100-300 ml blood i.e 4% of total blood volume

Lymph nodes

- Are parts of the lymphatic system
- Superficial lymph nodes (Ex. cervical, axillary, inguinal, retro auricular lymph nodes)
- Deep lymph nodes (Intra abdominal lymph nodes)

Function

- Filtration of bacteria and other foreign particles carried by lymph vessels

Liver

Functions in hematologic system

- **Site of filtration**

- It filters out old, defective, dead cells, bacteria,& other foreign particles

- **Site of production**

- For prothrombin & most of clotting factors. Bile production of liver is critical to the formation of vit. K in the intestines.

- **Hematopoiesis**

- During fetal life & in adults with bone marrow failure ,liver takes place the function of production of blood cells, extramedulary hematopoiesis

- **Site of storage**

- Large quantities of whole blood & blood cells

- **Hgb metabolism**

- Liver converts bilirubin in to bile & stores extra iron with a storage protein called **ferritin**

Blood

- Composition:-
  - Approx **45**% by Vol. Solid Components
    - » Red Blood Cells (45%)
    - » White Cells (1%)
    - » Platelets
  - Approx **55** % Liquid (plasma)

    91 % = water

    7% = plasma proteins

    - Albumin 70%
    - Coagulation protein 10%
    - Complement protein 10%
    - Immunity 7%

    2 % other solutes (Nutrients, electrolytes)

Blood functions

✓Maintain homeostasis

- Transportation

- Carries O [redacted]

- Carries nutrients (glucose, AA, lipids, vitamins, medications) to cells

- Carries wastes away from cells(CO [redacted]

- Carries antibodies & WBCs

- Carries clotting factors for hemostasis

- Regulate

PH, body temperature, water content, electrolytes

- Provide intracellular communications
  (Endocrine, pancreas, cytokines)
- Protection & defense
- Self repair mechanism (Clotting cascade)

X. Hematologic disorders

RBC disorders

1. Anemia
2. Polycythemia

WBC disorders

1. Leucopenia
2. Leukocytosis
3. Myelofibrosis
4. **Multiple myeloma (Plasma cell dyscrasias)**
5. **Leukemia**
6. **Lymphomas**
   - Hodgkin's Lymphoma (HL)
   - Non Hodgkin's Lymphoma (NHL)
7. Burkittis lymphoma

Neoplastic disorders

Bleeding (Clotting) disorders

1. Thrombocytopenia
2. Thrombocytosis
3. ITP (Idiopathic (Autoimmune)Thrombocytopenic Purpura)
4. TTP (Thrombotic Thrombocytopenic Purpura)
5. HUS (Hemolytic Uremic Syndrome)
6. VWD (Von Willebrand's Disease)
7. Hemophilia A & B
8. DIC (Disseminated Intravascular Coagulation)
9. Vitamin K deficiencies

A/ RBC disorders

Anemia is defined by reduction in Hgb concentration, Hct concentration or RBC count

- As a result the amount of oxygen delivered to body tissues is also diminished
- It is not a specific disease state by itself but a sign of an underlying disorder
- It is by far the most common hematologic condition.
- **Anemia is generally defined as a hematocrit < 40% (hemoglobin < 13.0 g/dL) in men or < 37% (hemoglobin <12.0 g/dL) in women. (WHO definition)**
  - Revised WHO criteria for patient's with malignancy Hgb < 14 in M and Hgb < 12 in F

Classifications of anemia

1. **Hypoproliferative anemia** (resulting from defective RBC Production): -
   - Iron deficiency anemia
   - Megaloblastic anemia: Vitamin B12 deficiency, & folate deficiency
   - Decreased erythropoietin production (e.g. from Renal dysfunction)
   - Cancer/inflammation
2. **Bleeding (resulting from RBC loss):-**
   - Bleeding from GI tract, menorrhagia, epistaxis, trauma
3. **Hemolytic anemia: -**
   - Altered erythropoiesis: sickle cell anemia, thalassemia
   - Hypersplenism (hemolysis)
   - Drug-induced anemia
   - Autoimmune anemia
   - Mechanical heart valve related anemia

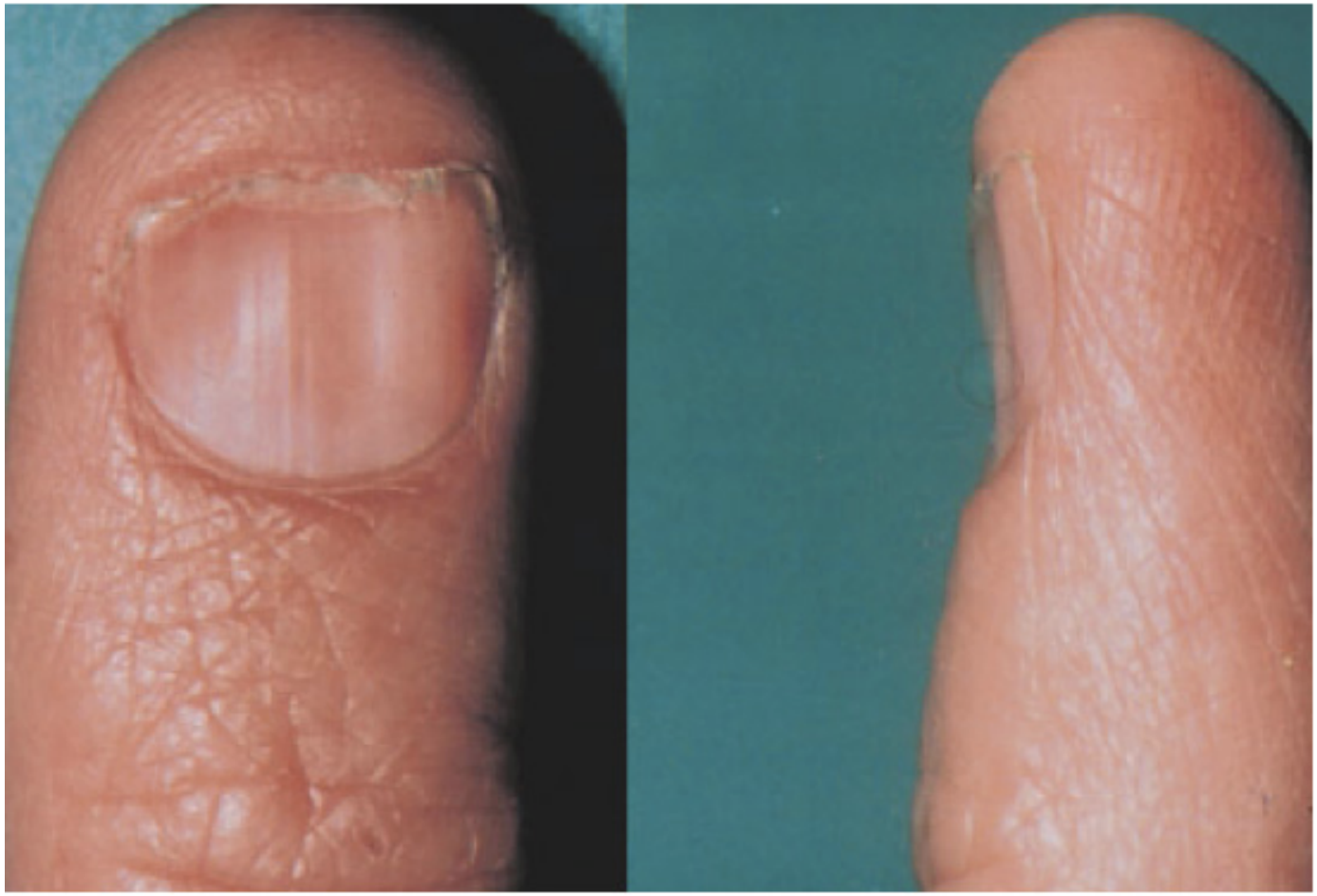
Kinetic approach

- **Decreased RBC production**
  - Lack of nutrients (B [redacted]
  - Bone marrow disorder
  - Bone marrow suppression
- **Increased RBC destruction**
  - Inherited and acquired hemolytic anemias
- **Blood loss**

Morphological approach

- Microcytic (MCV < 80 fl, femtoliters (10^{-15})
  - Reduced iron stores cause small RBCs
  - Reduced heme synthesis
  - Reduced globin production
- Normocytic (80 < MCV < 100)
- Macrocytic (MCV > 100)
  - Liver disease, B12, folate

Physical manifestation : “Spoon nails” in Iron deficiency



Labs

- ✓ History taking and a physical exam (pallor, jaundice, etc)
- ✓ CBC with Diff
- Leukopenia with anemia may suggest aplastic anemia
- Increased neutrophils may suggest infection
- Increased monocytes may suggest Myelodysplasia
- Thrombocytopenia may suggest hyperspleenism, marrow involvement with malignancy, autoimmune destruction, folate deficiency
- ✓ Reticulocyte count
- ✓ Peripheral smear (biconcave disk,size,color)

Labs

- ✓ **Mean corpuscular hemoglobin (MCH)**
  - To determine the amount of hemoglobin per RBC
  - Normochromic – normal amount of Hgb per cell
  - Hypochromic – decreased Hgb per cell
- ✓ **Hemoglobin electrophoresis**
  - Separates normal hemoglobin from abnormal.
  - It is used to detect sickle cell disease.
- ✓ **Sickle cell test**
  - Evaluates the sickling of red blood cells in the presence of decreased oxygen tension
- ✓ **Schilling test**
  - Measures vitamin B [redacted]
  - It is used to differentiate between malabsorption and pernicious anemia
- ✓ **Bone marrow examination**
  - Diagnoses aplastic anemia (failure of bone marrow to produce RBCs as well as platelets and WBCs)

D/D

RBC indices	Classification	Possible causes
Normal MCV, MCH, MCHC	Normocytic, normochromic anemia	• Acute blood loss • Sickle cell disease
Decreased MCV, MCH, MCHC	Microcytic, hypochromic anemia	• Iron deficiency anemia • Anemia of chronic illness • Chronic blood loss
Increased MCV	Macrocytic anemia	• Vitamin B_{12} deficiency • Folic acid deficiency

Iron deficiency anemia

- It is a condition in which total body iron content is inadequate for optimal development of erythrocytes.
- It is the most common type of anemia in all age groups & resulted from:-
  - Chronic blood loss
  - Iron malabsorption, as in small bowel disease
  - Increased iron requirement, as in pregnancy or periods of rapid growth
  - Insufficient intake caused by inadequate diet or Wt-loss

Complications:-

- Growth retardation in children; Heart failure
- Ischemic organ disease such as MI or stroke

Wolfe University

Iron deficiency anemia….

- High RDW
- High TIBC(TIBC reflects an indirect measurement of serum transference)
- Low serum iron level
  - Low ferritin (serum ferritin is an indicator of total body iron stores)
  - Low Hgb
  - Low reticulocyte count

Gold standard

- **Bone marrow aspiration**
  - Prussian blue staining shows lack of iron in erythroid precursors and macrophages
  - However, it is invasive and costly

What causes Iron deficiency?

- Blood Loss (occult or overt): PUD, Diverticulosis, Colon Cancer
- Decreased Iron absorption: atrophic gastritis, celiac disease
- Foods and medications: Phytate, calcium, soy protein, polyphenols decrease iron absorption
- Uncommon causes: intravascular hemolysis, pulmonary hemosiderosis, EPO, gastric bypass
- Decreased intake (rare)

❖ Iron deficiency anemia is the most common anemia

Iron deficiency…..

Pharmacologic & nursing interventions:

- Iron therapy – Oral FeSo4 , Ferrous gluconate, for 6-12 months
- Iron dextran IV or IM, but prior to parenteral administration of a full dose, a small test dose should be administered to avoid the risk of anaphylaxis.
- Iron rich diets: organ meat (beef or calf's, chicken liver), other meats, beans, leafy green vegetables, fruits
- Iron rich diets with a source of Vit-C
- Take iron on empty stomach 1hr before or 2hrs after meal
- Take high fiber diets to minimize constipation
- Inform the pt that stools will become dark in color
- Do not take within 2 hr of milk or antacids

Aplastic anemia

- Bone marrow hypoplasia resulting in pancytopenia (insufficient numbers of RBCs, WBCs, Platelets)
- May be idiopathic or caused by exposure to chemical toxins, radiation, viral infections, certain drugs (CAF), or may be congenital

Treatment: -

- Removal of causative agent or toxin
- Allogeneic bone marrow transplantation in severe cases
- Immunosuppressants, bone marrow stimulating factors
- Platelet & RBC transfusions, Antibiotics

- **Pernicious anemia**
  - Vitamin B 
    - Small bowel disease
    - Gastric resection
    - Absence of intrinsic factor or genetic cause
    - Strict vegetarians who consume no meat or dairy products
  - **Features:** GI symptoms & neuropathy
  - **Rx:** Monthly parenteral replacement with Cyanocobalamin
- ***Folate deficiency***
  - Folic acid, a vitamin that is necessary for normal RBC production, is stored as compounds referred to as folates.
  - Dietary deficiency, alcoholism, malabsorption disease of the small bowel, pregnancy, & some medications
  - **Rx**: Oral folic acid replacement (Fefol) & diet, Rx of underlying cause.
  - Folate is found in green vegetables and liver

Wollo University

- **Polycythemia**
  - Refers to an increased volume of RBCs.
  - It is a term used when the hematocrit is elevated (to more than 55% in males, more than 50% in females).
  - Is classified as either primary or secondary
- **Polycythemia vera**
  - Polycythemia vera, or primary polycythemia,
  - Is a proliferative disorder in which the myeloid stem cells seem to have escaped normal control mechanisms.
  - The bone marrow is hypercellular, and the RBC, WBC, and platelet counts in the peripheral blood are elevated.
  - However, the RBC elevation is predominant; the hematocrit can exceed 60%
  - Over time, the bone marrow may become fibrotic, with a resultant inability to produce as many cells

Polycythemia vera––

Clinical manifestations

- Patients typically have spleenomegaly
- The symptoms result from the increased blood volume
  - headache, dizziness, tinnitus, fatigue, paresthesias, and blurred vision
- The symptoms result from increased blood viscosity
  - angina, claudication, dyspnea, and thrombophlebitis), particularly if the patient has atherosclerotic blood vessels.
- Another common and bothersome problem is generalized pruritus, which may be caused by histamine release due to the increased number of basophils.
- Erythromelalgia, a burning sensation in the fingers and toes, may be reported and is only partially relieved by cooling.

Polycythemia vera–-

Assessment and Diagnostic Findings

- Diagnosis is made by finding an elevated RBC mass (a nuclear medicine procedure), a normal oxygen saturation level, and an enlarged spleen.
- Other factors useful in establishing the diagnosis include elevated WBC and platelet counts.
- The erythropoietin level is not as low as would be expected with an elevated hematocrit; it is normal or only slightly low.

Polycythemia vera–-

Complications

- Patients with polycythemia vera are at increased risk for thromboses resulting in a CVA (brain attack, stroke) or heart attack (MI); thrombotic complications are the most frequent cause of death.
- Bleeding is also a complication, possibly due to the fact that the platelets (often very large) are somewhat dysfunctional.
- The bleeding can be significant and can occur in the form of nosebleeds, ulcers, and frank gastrointestinal bleeding.

Polycythemia vera–-

Medical management

- The objective of management is
  - To reduce the high blood cell mass.
- Phlebotomy is an important part of therapy and can be performed repeatedly to keep the hematocrit within normal range.
  - This is achieved by removing enough blood (initially 500 ml once or twice weekly) to deplete the patient's iron stores, thereby rendering the patient iron deficient and consequently unable to continue to manufacture RBCs excessively.
- Patients need to be instructed to avoid iron supplements, including those within multivitamin supplements.
- If the patient has an elevated uric acid concentration, allopurinol (Zyloprim) is used to prevent gouty attacks.

Polycythemia vera–-

- If the patient develops ischemic symptoms, dipyridamole (eg, Persantine) is sometimes used.
- Radioactive phosphorus (32P) or chemotherapeutic agents (eg, hydroxyurea [Hydrea]) can be used to suppress marrow function, but they may increase the risk for leukemia.
- Patients receiving hydroxyurea appear to have a lower incidence of thrombotic complications; this may result from a more controlled platelet count.
- Anagrelide (Agrylin) inhibits platelet aggregation and can also be useful in controlling the thrombocytosis associated with polycythemia vera.

Polycythemia vera––

Nursing management

- The nurse's role is primarily that of educator.
  - Risk factors for thrombotic complications should be assessed, and patients should be instructed regarding the signs and symptoms of thrombosis.
- Patients with a history of bleeding are usually advised to avoid aspirin and aspirin-containing medications, because these medications alter platelet function.
- Minimizing alcohol intake should also be emphasized to further diminish any risk for bleeding.
- For Pruritus, the nurse may recommend bathing in tepid or cool water, along with applications of cocoa butter–based lotions and bath products

B/ WBC disorders

Normal results

- Neutrophils (40 - 70%) = 3000 – 7000 cells/ml

- Segments (40-70 %) = Mature neutrophilis , Right shift

- Bands/stabs (0 – 3 %) = Immature neutrophilis , Left shift

- Lymphocytes (20 - 40%) = 1500 - 3000 cells/ml
- Monocytes (2- 8%)
- Eosinophils (1- 4%)
- Basophils (0.5 -1 %)

1. Leukopenia
   - Neutropenia, Lymphopenia
2. Leukocytosis ***(proliferative disorder)***
   - Neutrophilia , Eosiniphilia, Basophilia, Lymphocitosis, Monocitosis
3. Leukemia
4. Multiple myeloma
5. Myelofibrosis

Leukopenia

- TWBC lower than the reference range for the age is defined as leucopenia
- Leucopenia may affect one or more lineages and it is possible to be severely neutropenic or lymphopenic without a reduction in total white cell count.

Granulocytosis

Increase in the count of all or one of the granulocytic component Ex. Neutrophils, Eosinophils & Basophils

Agranulocytosis

Decrease in the count of all or one granulocytic component

Leukopenia

- Total white blood count lower than the reference range for age is defined as leucopenia (< 4,000/cmm for an adult)
- The major contribution to a leucopenia usually comprises a reduction of PB neutrophils .

Causes

- Aplastic anemia
- Chemotherapy
- Influenza or other viral infection
- Widespread bacterial infection
- Radiation therapy or exposure

Causes of leukopenia

- Infections :

  -Viral as infectious hepatitis,influenza, rubella and others .

  -Bacterial as typhoid fever,brucellosis, miliary TB .

  -Rickesttial and protozoal infections .

- Drugs :

Selective neutropenia,Agranulocytosis, Aplastic anaemia

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- Chemotherapy
- Influenza or other viral infection
- Widespread bacterial infection
- Radiation therapy or exposure

Causes of leukopenia–-

- Megaloblastic anaemia
- Hypersplenism
- Leucoerythroblastic anaemia(Multiple Myeloma,metastatic carcinoma….)
- Acute leukaemia
- Myelodysplasia
- Aplastic anaemia

Causes of leukopenia–-

- Cyclical neutropenia
- Chronic idiopathic neutropenias
- Paroxysmal nocturnal hemoglobinuria
- Ionizing radiation & cytotoxic drugs
- Miscellaneous conditions :
  myxoedema,anaphylactoid shock,
  hypopituitarism & SLE

Neutropenia

A decreased percentage of neutrophils may be due to:

Aplastic anemia
Chemotherapy
Influenza or other viral infection
Widespread bacterial infection
Radiation therapy or exposure
Leukemia & lymphoma
Spleen destruction

- Neutropenia refers specifically to a decrease in neutrophils. It commonly is defined as a circulating neutrophil count of less than 1500 cells/μL.
- Agranulocytosis, which denotes a severe neutropenia, is characterized by a circulating neutrophil count of less than 200 cells/μL.
- Neutropenia can be:
  - Acquired
  - Congenital
    - *Kostmann's syndrome*
      - It occurs sporadically as an autosomal recessive disorder, causes severe neutropenia while preserving the erythroid and megakaryocyte cell lineages that result in red blood cell and platelet production.
      - The total white blood cell count may be within normal limits, but the neutrophil count is less than 200/μL. Monocyte and eosinophil levels may be elevated (compensatory).

Acquired neutropenia

- Accelerated removal - removal of neutrophils from the circulation exceeds production
  - Inflammation
  - Infection, viral or bacterial
- Increased destruction:
  - Drug-induced granulocytopenia
    - Treatment of cancer – chemotherapy (*e.g.,* alkylating agents, antimetabolites)
  - Irradiation
  - Autoimmune disorders or drug reactions
    - May cause increased and premature destruction of neutrophils
  - Splenomegaly
    - Neutrophils may be trapped in the spleen along with other blood cells
  - Felty's syndrome
    - A variant of rheumatoid arthritis, there is increased destruction of neutrophils in the spleen
  - Neoplasms involving bone marrow (e.g., leukemias and lymphomas, myeloma)

Acquired neutropenia

- Alcoholism
- Carentiale states:
  - Folic acid
  - Vitamin B12
  - Iron
  - Cooper
- Aplastic anemia
  - All of the myeloid stem cells are affected, resulting in anemia, thrombocytopenia, and agranulocytosis;
- Idiopathic neutropenia that occurs in the absence of other disease or provoking influence.

RX of neutropenia

- **Antibiotics**
- **Bone marrow biopsyolony**
- **Granulocyte colony stimulating factor**

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- **Granulocyte colony stimulating factor**

Causes neutropenia

- Racial
- Congenital
- Cyclical neutropenia
- Marrow aplasia
- Marrow infiltration
- Megaloblastic anemia
- Acute infections
  Typhoid, MiliaryTB, viral hepatitis
- Drugs
- Irradiation exposure
- Immune disorders
  HIV,SLE, Felty syndrome, Neonatal isoimmune and autoimmune neutropenia
- Hyperslplenism

Management of neutropenia

- Remove the cause if possible
- Treat any infection aggressively
- Role of

  - Growth factors

  - Splenectomy

Cyclical neutropenia

- Regular recurring episodes of severe neutropenia (<0.2 x 10^9/L) usually lasting for 3 3-6 days
- Can be familial & inherited with maturation arrest

Three suggested mechanisms for cyclical neutropenia

- Stem cell defect & altered response to growth factors
- Defect in humoral or cellular stem cell control
- Periodic accumulation of an inhibitor

Leukocytosis

Causes

- Neutrophilia (commonest cause)
- Eosinophilia
- Basophilia
- Lymphocytosis
- Monocytosis

Causes of Leukocytosis

Neutrophilic leukocytosis	Acute bacterial infections, especially those caused by pyogenic organisms; sterile inflammation(myocardial infarction, burns)
Eosinophilic leukocytosis (eosinophilia)	Allergic disorders such as asthma, allergic skin diseases; parasitic infestations; drug reactions; certain malignancies (e.g., Hodgkin disease and some non-Hodgkin lymphomas); collagen vascular disorders and some vasculitides; atheroembolic disease

Basophilic leukocytosis (basophilia)	Rare, often indicative of a myeloproliferative disease (e.g., chronic myelogenous leukemia)
Monocytosis	Chronic infections (e.g., tuberculosis), bacterial endocarditis and malaria; collagen vascular diseases (e.g., systemic lupus erythematosus) and inflammatory bowel diseases (e.g., ulcerative colitis)
Lymphocytosis	Accompanies monocytosis in many disorders associated with chronic immunologic stimulation (e.g., tuberculosis); viral infections (e.g., hepatitis)

Causes of neutrophilia

- Infections (pyogenic bacteria)
- Inflammations produced by :
  Toxins, infectious agents, neoplasms or burns .
- Following haemorrhage.

Reactive changes :

Left shift , toxic granulation , high LAP score

- Chronic granulocytic leukaemia
- Other myeloproliferative disorders

Diagnosis :

Low LAP score .

Philadelphia chromosome .

Causes of neutrophilia–-

- Acute infection
- Acute stress
- Eclampsia
- Gout
- Myelocytic leukemia
- Rheumatoid arthritis
- Rheumatic fever
- Thyroiditis
- Trauma
- *Neutrophils can increase in response to bacterial infection or inflammatory disease. Severe elevations in neutrophils may be caused by various bone marrow disorders, such as chronic myelogenous leukemia.*

Causes neutrophilia

- Infection
  - Bacterial
- Inflammatory conditions
- Autoimmune disorders
- Gout
- Neoplasia
- Metabolic conditions
- Uraemia
- Acidosis
- Haemorhage
- Corticosteroids
- Marrow infiltration
- Myeloproliferative

Neutrophilia

Increase in the number of neutophils and/or its precursors > 7,000/cmm

- Physiological: -
  - In newborns
  - Pregnancy
  - In labor
  - Post-partum
  - After exercise
- Drugs or toxics:-
  - Administration of corticosteroids - may increase the release of neutrophils from the bone marrow and reduce their migration into tissues;
  - Acute poisoning with Pb, Hg, some venoms;
- Reactive neutrophilia - the result of increased release of neutrophils from MB to compensate their high affinity for tissues. It is frequently accompanied by deviation to the left of the leukocyte formula (leukemoid reaction);
- Metabolic and endocrine diseases:
  - Diabetic ketoacidosis
  - Acute renal failure
  - Acute gout crise
- In some malignant hematologic diseases: CGL, MPCD (PV, ET, CMML)

Eosinophilia

- Allergic diseases: asthma, allergic rhinitis, eczema, atopic dermatitis
- Parasitic infections
- Fungal and other infections
- Tuberculosis
- Hematologic malignancies (CGL, AL with Eo, LAM2 and 4, rarely in MDS) and nonhematologic (lung, vaginal, skin, stomach carcinoma, malignant melanoma)
- Idiopathic - is diagnosis of exclusion
- Drugs: aspirin, beta blockers, penicillin, cephalosporins, NSAIDs, etc.
  - Bacterial infections usually does not cause eosinophilia, but eosinopenia.

Causes

- Allergic reaction ,autoimune
- Blood cancer
- Collagen vascular disease
- Parasitic infection
- Hypothyroidism

Causes of eosinophilia

- Allergy
  Atopic ,drug sensitivity & pulmonary eosinophilia .
- Infections
  Parasites ,recovery from infection
- Malignancy
  Hodgkin’s disease , NHL & MPD
- Skin disorder
- Drugs
- GIT disorders
- Hypereosinophilic syndrome

- Decreased levels of eosinophils can occur as a result of infection.

Eosinophilia

- Hypereosinophilic syndrome

Criteria of diagnosis

- Peripheral blood eosinophil >1.5 x 10^9/L
- Persistence of counts more than 6 months
- End organ damage (Heart, Lung, Skin,Neurological)
- Absence of any obvious cause for eosinophilia

Basophilia

- Blood cancer
- Hypothyroidism
- Surgical, procedure complication
  - Splenectomy
- Infectious disorders (specific agent)
  - Herpes Zoster
  - Influenza
  - Chickenpox/herpes zoster virus
  - Smallpox (variola)
- Infected organ, abscesses
- Inflammatory disorders
- Neoplastic disorders:
  - Acute myeloid leukemia
  - Hodgkin's disease
  - Myelodysplastic syndrome
  - Chronic myeloid leukemia
  - Polycythemia vera
  - Lymphoma/malignant, non-Hodgkins
- Allergic, collagen autoimmune disorders
- Congenital, developmental disorders
  - congenital hemolytic anemia
- Hereditary, familial, genetic disorders
  - Spherocytosis
- Vegetative, autonomic, endocrine disorders
  - Hypothyroidism (myxedema)
  - Leukemoid reaction
- Drugs
  - Foreign protein injection

A decreased percentage of basophils may be due to:

- Acute allergic reaction

Lymphocytosis

- Causes of absolute lymphocytosis include:
  - Acute viral infections, such as infectious mononucleosis (glandular fever), hepatitis and cytomegalo virus infection
- Other acute infections such as pertussis,Healing TB, typhoid fever
  - Protozoal infections, such as toxoplasmosis
  - Chronic intracellular bacterial infections such as tuberculosis or brucellosis
  - Chronic lymphocytic leukemia
- Causes of relative lymphocytosis include:
  - Age less than 2 years
  - Acute viral infections
  - Connective tissue diseases
  - Thyrotoxicosis
  - Splenomegaly with splenic sequestration of granulocytes
  - Exercise
  - Stress

Causes

- Chronic bacterial infection , Protozoal linfections Ex. Toxoplasmosi
- Infectious hepatitis , Infectious mononucleosis
- Lymphocytic leukemia , Multiple myeloma
- **Viral infections** (such as infectious mononucleosis, CMV, Rubella, hepatitis, adenoviruses, chicken pox,dengue mumps, measles)
- Allergic drug reactions,Hyperthyroidism, Splenectomy, Serum sickness

Causes of lymphocytosis

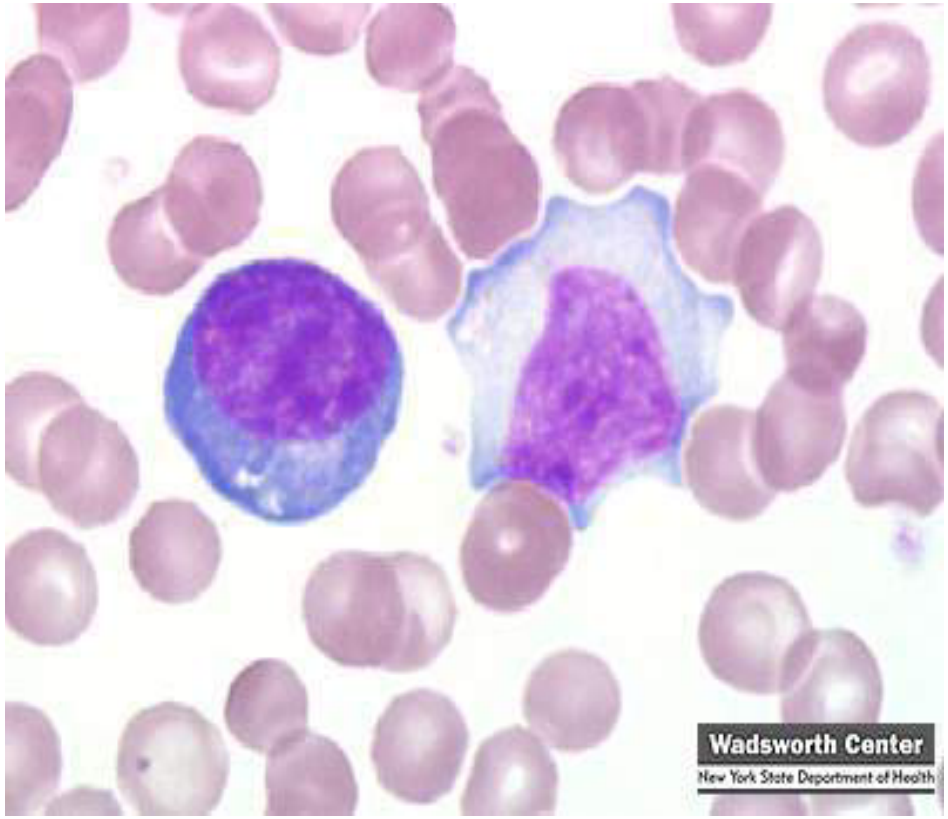
- Acute infections: Infectious mononucleosis, acute infectious lymphocytosis, mumps, rubella, pertussis
- Chronic infections: tuberculosis,syphilis, brucellosis,infectious hepatitis .
- Thyrotoxicosis (usually only relative)
- Chronic lymphocytic leukaemia

Lymphopenia

Causes

- Chemotherapy
- HIV infection
- Leukemia
- Radiation therapy or exposure
- Sepsis
- Decreased lymphocyte levels can indicate diseases that affect the immune system, such as lupus, and the later stages of HIV infection.

Infectious mononucleosis



- Infectious mononucleosis is caused by the Epstein-Barr virus, a DNA herpes-type virus that infects B lymphocytes.
- Patients present with mild to severe adenopathy, hepatosplenomegaly, fever, malaise, pharyngitis, and a characteristic peripheral blood smear demonstrating reactive lymphocytes.

Infectious mononucleosis–-

- Caused by Epstein-Barr Virus (EBV) herpes virus
- Infects B -cells
- Bood picture shows leucocytosis with absolute lymphocytosis and increased number of atypical lymphocytes
- **Transmission** – person to person

  Sneezing, coughing, saliva

IP = 4-6 wks

Diagnosis :-

- Monospot test
- EBV serology

Monocytosis

- Monocytosis often occurs during chronic inflammation. Diseases that produce this state:
  - Infections: tuberculosis, brucellosis, listeriosis, subacute bacterial endocarditis, syphilis, and other viral infections and many protozoal and rickettsial infections;
  - Blood and immune causes: chronic neutropenia and myeloproliferative disorders;
  - Autoimmune diseases and vasculitis: systemic lupus erythematosus, rheumatoid arthritis and inflammatory bowel disease;
  - Malignancies: Hodgkin's disease and certain leukaemias, such as chronic myelomonocytic leukaemia (CMML) and monocytic leukemia;
  - Recovery phase of neutropenia or an acute infection;

Causes of monocytosis

- Chronic bacterial infections :- tuberculosis,bacterial endocarditis, brucellosis
- Other infections :malaria,Kala-azar, trypanosomiasis ,typhus……..
- Hodgkin's disease
- Monocytic & myelomonocytic leukaemia

Causes

- Chronic inflammatory disease
- Parasitic infection
- Tuberculosis
- Viral infection (for example, infectious mononucleosis, mumps, measles)

Causes of monocytosis

- Infections
  - Chronic infection (TB, typhoid fever, infective endocarditis)
  - Recovery from acute infection
- Malignant disease
  MDS, AML, HD, NHL
- Connective tissue disorders
- Ulcerative colitis, Sarcoidosis , Crohn’s disease
- Post splenectomy

Monocytopenia

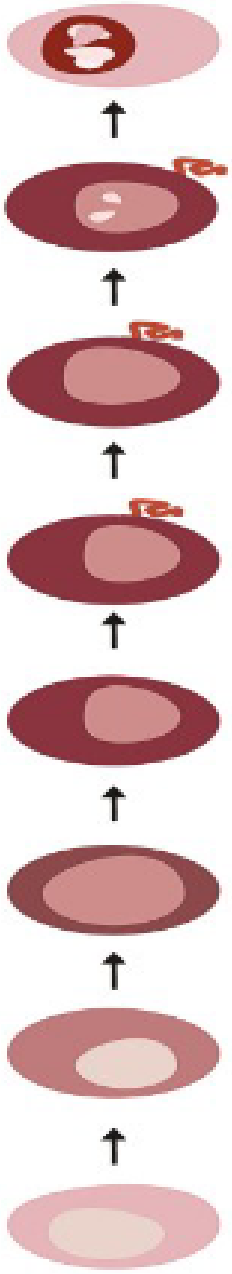
- The causes of monocytopenia include:
  - Acute infections
  - Stress
  - Aplastic anemia
  - Hairy cell leukemia
  - Acute myeloid leukemia
  - Treatment with myelotoxic drugs
  - Treatment with glucocorticoids

  Decreased monocyte levels can indicate bone marrow injury or failure and some forms of leukemia.

The neoplastic disorders

- The neoplastic disorders of hematopoietic and lymphoid origin represent the most important of the white blood cell disorders.
- They include three somewhat overlapping categories:
  - The lymphomas (Hodgkin’s disease and non-Hodgkin’s lymphoma)
  - The leukemias
  - Plasma cell dyscrasias (multiple myeloma)

The Maturation Stage of the Cell of Origin determines the Type of Lymphoma



Plasma Cell

Activated B Cell

B Cell after antigen exposure

B Cell before antigen exposure

Immature B-cell

Pre-B Cell

Lymphoid Stem Cell

Pluripotent Stem Cell



Waldenström's Macroglobulinemia

Diffuse large B-cell lymphoma

Follicular Lymphoma

Mantle Cell

CLL/SLL

Small Non-cleaved Lymphoma (Burkitt's)

Precursor B-Cell Acute Lymphblastic Lymphoma/Leukemias

Chronic Myeloid Leukaemia (CML)

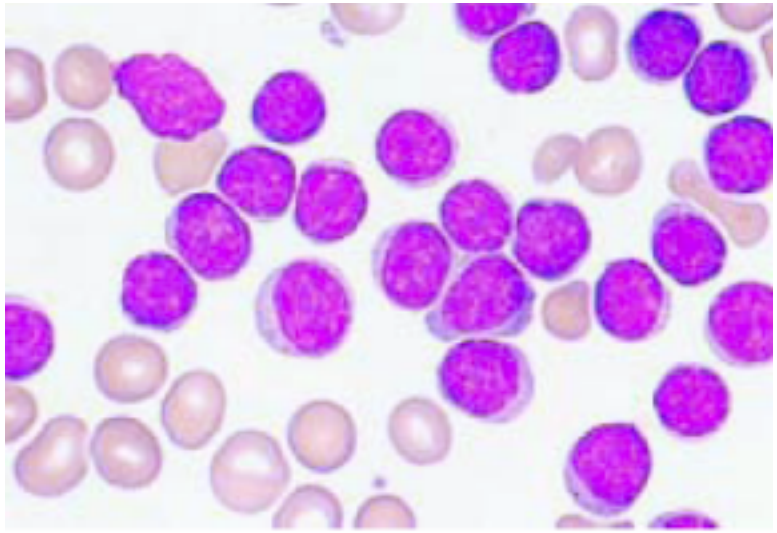
B-cells originate in Bone Marrow ...

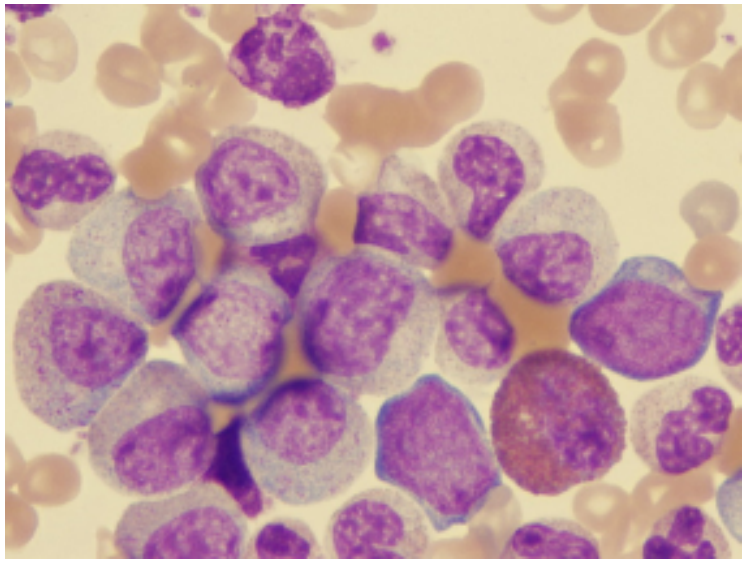
Leukemias

- Leukemias are malignant neoplasms of the hematopoietic stem cells with diffuse replacement of bone marrow.
- Leukemias are classified according to cell type
  - Lymphocytic
  - Myelogenous
- and whether the disease is *acute or chronic* .
- The lymphocytic leukemias involve immature lymphocytes and their progenitors that originate in the bone marrow but infiltrate the spleen, lymph nodes, CNS, and other tissues.
- The myelogenous leukemias involve the pluripotent myeloid stem cells in bone marrow and interfere with the maturation of all blood cells, including the granulocytes, erythrocytes, and thrombocytes.
- The acute leukemias (*i.e.*, ALL, which primarily affects children, and AML, which primarily affects adults) have a sudden and stormy onset, with symptoms of depressed bone marrow function:
    - Anemia
    - Fatigue
    - Bleeding
    - Infections
  - Bone pain
  - Lymphadenopathy
  - Splenomegaly
  - Hepatomegaly

Leukemias

- The chronic leukemias, which largely affect adults, have a more insidious onset.
- CLL often has the most favorable clinical course, with many persons living long enough to die of unrelated causes.
- The course of CML is slow and progressive, with transformation to a course resembling that of AML.





Leukemia

- Leukemia is a group of malignant disorders of the hematopoietic system that involves the bone marrow and lymph nodes and is characterized by the uncontrolled proliferation of immature WBCs
- Cause unknown, but it's associated with down's syndrome and other chromosomal abnormalities, chronic exposure to chemicals such as benzene, use of drugs that cause a plastic anemia, radiation exposure, and chemotherapy
- In leukemia, WBC (leukocyte) proliferation interferes with the production of other cells, leading to thrombocytopenia and anemia; the immature leukocytes decrease immuno-competence and increase susceptibility to infection
- In leukemia, the white blood cells are not functional. They invade and destroy bone marrow, and they can metastasize to the liver, spleen, lymph nodes, testes, and brain

Leukemia–––

Factors involved in the development of leukemia include:-

- Ionizing radiation (radiation therapy, environmental)
- Exposure to certain chemicals and drugs (chemotherapy agents and drugs that suppress bone marrow)
- Genetic factors (for example, hereditary conditions)
- Immunologic factors (for example, immuno-suppression)

The leukemias are commonly classified according to the stem cell line involved, either lymphoid or myeloid:-

o Acute lymphocytic leukemia(ALL) & myelogenous leukemia (AML)

o Chronic lymphocytic leukemia(CLL) and myelogenous leukemia (CML)

Incidence and cure rates

- **ALL (common)**: various factors influence the prognosis for children (age at diagnosis, gender, cell type involved); less than 50% of adults can be cured
- **CLL**: Most cases involve people older than 60. This disease does not occur in children. Immature lymphocytes that infiltrate spleen,lymph node & CNS
- **AML**: Most common leukemia among adults; prognosis is poor
- **CML**: Most cases involve adults. The disease is uncommon in children. Immature granulocytes, RBCs, & thrombocytes

Types of leukemia

The following chart compares the incidence and signs and symptoms for various types of leukemia.

Leukemia	Incidence	Signs and symptoms
Acute leukemias		
Acute lymphocytic leukemia (ALL)	• Primarily affects children ages 2 to 4. Incidence declines dramatically after age 10. Survival rates are higher in children than adults, and the prognosis for ALL is better than that of acute myeloblastic leukemia.	• Anemia with fatigue and pallor • Bleeding, such as ecchymosis, gingival or rectal bleeding, and petechiae • Decreased hematocrit, hemoglobin level, and platelet count
Acute myelogenous leukemia (AML), also known as acute nonlymphocytic leukemia	• Primarily affects those over 40; the incidence increases with age. Survival rates for AML have improved with the use of autologous and allogenic bone marrow transplants.	• Infections, particularly those of the mucous membranes, respiratory tract, and skin • Normal, decreased, or increased white blood cell count • Bone or joint pain, fever without infection, hepatosplenomegaly, lymphadenopathy, and weight loss
Chronic leukemias		
Chronic lymphocytic leukemia (CLL)	• Primarily affects men, with 72 being the average age at diagnosis. CLL runs a long course (4 to 10 years) and rarely progresses to acute leukemia.	• Enlarged spleen • Lymphadenopathy
Chronic myelogenous leukemia, or chronic granulocytic leukemia	• Primarily affects those over age 20; the incidence increases with age.	• Hepatomegaly

Leukemia---

Signs and symptoms of acute leukemia

- Bone pain
- Joint swelling
- Enlarged liver and spleen
- Weight loss
- Fever
- Poor wound healing (for example, infected lesions)
- Signs of anemia (fatigue, pallor, tachycardia, dyspnea on exertion)
- Signs of bleeding (ecchymoses, hematuria, bleeding gums)

Diagnosis and treatment (leukemia)––

- Diagnosis is usually based on bone marrow examination that reveals leukemic blast cells and on a CBC
- Chemotherapy, with or without stem cell transplantation, is standard treatment for leukemia
- Patients with low-risk chronic lymphocytic leukemia may not receive treatment if they’re asymptomatic
- Stem cell transplantation is used to treat leukemia
- White Blood Cell (WBC) Count: Often elevated (20,000 to 100,000/mm3) with leukemia prior to treatment; decreased with treatment
- Hemoglobin, hematocrit, and platelets: Decreased
- Bleeding times: Increased
- Bone marrow aspiration and biopsy: Identification of prolific quantities of immature leukemic blast cells and protein markers indicating the specific type of leukemia – lymphoid or myeloid

After bone marrow aspiration:-

- Apply pressure to the site for 5 to 10 min & assess vital signs frequently
- Apply pressure dressing
- Monitor for signs of bleeding and infection for 24 hr

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Leukemia–-

Surgical management of leukemia

- Bone marrow transplantation
- Used as a treatment for leukemia.
- Closely matched donor stem cells are used.
- Autologous cells are the client's own cells that are collected before chemotherapy.
- Syngeneic cells are donated from the client's identical twin.
- Allogenic: donor stem cells are transplanted into the patient , such as a relative or from umbilical cord blood
- Before transplantation, the patient undergoes chemotherapy and total body radiation to eliminate the leukemic cells; these procedures destroy all the patient's bone marrow in preparation for grafting from the donor
- Transfusion therapy may be necessary to treat severe thrombocytopenia, leukopenia, and anemia resulting from the disease process or from treatment
- Most RBC transfusions involve 250 to 300 mL/unit of packed RBCs; whole blood is seldom transfused to treat leukemia
- Following transplantation, the client is at high risk for infection and bleeding until the transfused stem cells grow

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Nursing interventions (leukemia)––

- Follow infection-control procedures if the WBC count is low—for example, place a severely neutropenic patient in a private room, with no flowers, plants, or fresh fruits; avoid unnecessary invasive procedures; limit patient contact with infected personnel or visitors; and teach hand hygiene techniques to the patient, family, and visitors
- If the patient has a low platelet count, monitor blood counts and take precautions to prevent bleeding—for example, avoid parenteral injections, limit the number of venipunctures, and advise the patient to use an electric razor for shaving
- Prevent or manage stomatitis by inspecting the oral cavity daily and encouraging oral care with peroxide or saline solution on a regular basis
- Teach the patient how to care for an indwelling vascular access device and how to detect signs of infection
- Educate patients undergoing chemotherapy about the effects, adverse effects, and length of treatment
- Provide supplemental feedings as prescribed
- Conserve the patient's energy, but promote his independence
- Provide emotional support to the patient and his family, and encourage the patient and his family to verbalize feelings

Multiple Myeloma

- Multiple myeloma is a malignancy of plasma cells that can invade the bone marrow, lymph nodes, liver, spleen, and kidneys and leads to bone destruction throughout the body
- The malignant plasma cells produce an increased amount of a specific nonfunctional immunoglobulin
- Multiple myeloma is a plasma cell cancer of the osseous tissue and accounts for 10% to 15% of all hematologic malignancies.
- It is characterized by the uncontrolled proliferation of an abnormal clone of plasma cells, which secrete primarily IgG or IgA.
- There is an atypical proliferation of one of the immunoglobulins, called the M protein, a monoclonal antibody.
- Levels of normal immunoglobulins are usually depressed. This contributes a general susceptibility to bacterial infections.
- In some forms of multiple myeloma, the plasma cells produce only Bence Jones proteins, abnormal proteins that consist of the light chains of the immunoglobulin molecule.
  - Because of their low molecular weight, Bence Jones proteins are partially excreted in the urine.
  - Many of these abnormal proteins are directly toxic to renal tubular structures, which may lead to tubular destruction and, eventually, to renal failure.

Pathogenesis

- The cause of multiple myeloma is unknown.
  - It does not appear to be caused by previous exposure to toxic agents (*e.g.,* solvents such as benzene, paints, pesticides).
  - Interestingly, an association with human herpesvirus 8 has been described, but the role of this virus in the pathogenesis of the disease remains to be established.
- Cytokines are important in the pathogenesis of the disorder.
  - The multiple myeloma plasma cell has a surface-membrane receptor for interleukin-6, which is known to be a growth factor for the disorder.
  - Another important growth factor for the myeloma cell is interleukin-1, which has important osteoclastic activity.

Multiple myeloma

- It is a malignant disease of the most mature form of B lymphocyte, the plasma cell. It is not classified as a lymphoma
- Plasma cells secrete immunoglobulins, proteins necessary for antibody production to fight infection.
- Multiple myeloma is a malignancy of plasma cells that can invade the bone marrow, lymph nodes, liver, spleen, and kidneys and leads to bone destruction throughout the body
- The malignant plasma cells produce an increased amount of a specific nonfunctional immunoglobulin
- The disorder usually occurs in black men between ages 50 and 70; men are 50% more likely to develop the disease than women
- Other risks include radiation exposure, family history, obesity, and occupational exposure in petroleum-related industries

Signs and symptoms

- Decrease bone density, bone pain, hypercalcemia, pathological fractures, renal failure
- Confusion, weakness, dizziness, weight loss,, skeletal deformities, and constant, severe bone pain may occur
- Anemia, leukopenia, and thrombocytopenia—with resulting bleeding, infection, shortness of breath, weakness, and protein in blood and urine—may also occur

Multiple myeloma–-

Diagnosis and treatment

- Serum and urine protein electrophoresis showing immunoglobulin or beta2-microglobulin are indicative of multiple myeloma; urine studies may show Bence Jones proteins and hypercalciuria
- Bone X-rays or MRI may show bone destruction
- Bone marrow aspiration and biopsy may show an increased number of immature plasma cells
- **Chemotherapy** suppresses plasma cell growth, and radiation therapy may be used to treat bone lesions and relieve pain
- **Radiation therapy** may be used to damage myeloma cells and stop growth
- Patients with multiple myeloma may also receive interferon to slow the growth of the myeloma cells and thalidomide to inhibit angiogenesis
- Autologous **bone marrow transplantation** isn't a cure but may put the patient into remission for a period; allogenic transplant carries a higher risk of serious complications but produces longer-lasting remissions
- Plasmapheresis is sometimes used to temporarily remove the high-myeloma protein and thereby improve symptoms
- Hypercalcemia may be managed with a diuretic, hydration, or a phosphate
- Other therapies include bortezomib (Velcade) for resistant forms and thalidomide (Thalomid)

Multiple myeloma–-

Nursing interventions

- Educate the patient and his family about diagnostic and treatment procedures and adverse effects
- Assess the patient for pain, and administer an analgesic regularly
- Instruct the patient on safety measures to prevent fractures of affected bones; direct the patient not to walk without assistance; have the patient use devices, such as splints or braces, to prevent injury and reduce pain
- Encourage the patient to drink plenty of fluids, or administer I.V. fl uids, to dilute calcium, prevent dehydration, and prevent renal precipitates
- Assess the patient for signs and symptoms of infection and bleeding, and take measures to prevent their occurrence

C/ Bleeding disorders

Platelets d/o & clotting factors d/o

1. Thrombocytopenia
2. Thrombocytosis
3. ITP (Idiopathic (Autoimmune)Thrombocytopenic Purpura)
4. TTP (Thrombotic Thrombocytopenic Purpura)
5. HUS (Hemolytic Uremic Syndrome)
6. VWD (Von Willebrand's Disease)
7. Hemophilia A & B
8. DIC (Disseminated Intravascular Coagulation)
9. Vitamin K deficiencies

Low platelets/Thrombocytopenia/

Increased bleeding time (BT)

ITP = Idiopathic Thrombocytopenic Purpura

TTP =Thrombotic Thrombocytopenic Purpura

HUS = Hemolytic Uremic Syndrome

HIT = Heparin Induced Thrombocytopenia

Normal platelets

ASA (Inhibit Cox1 --- TMboX2- decrease platelets aggregation)

Uremic, scurvy

Ethlers- Danlos syndrome (Defect in type III collagen)

Bernard Solier (defect in Gp Ib)

Glanzmann's thrombocytopenia (GP IIb & IIIa) severe bleeding

VWD (PT & platelets are normal but increase BT,PTT)

Henoch Schonlein's syndrome: is immune complex (type III) hypersensitivity

1. Thrombocytopenia

- **↓ platelets**
  Chemotherapeutic agents, neoplasms (leukemia), autoimmune
- **Increased destruction**
  Drugs (heparin), idiosyncratic, TTP
- **Sequestration in the spleen**
  Portal hypertension, splenic tumor, genetic

• **Normal platelets**: 200,000 - 400,000
< 50,000 is serious

- Formed by fragmentation of megakaryocytes in the bone marrow
- Aged platelets removed by reticuloendothelial system and spleen

Table 33-5 • Causes and Management of Thrombocytopenia

CAUSE	MANAGEMENT
Decreased Production	
Hematologic malignancy, especially acute leukemias	Treat leukemia; platelet transfusion
Myelodysplastic syndromes (MDS): metastatic involvement of bone marrow from solid tumors	Treat MDS; platelet transfusion Treat solid tumor
Aplastic anemia	Treat underlying condition
Megaloblastic anemia	Treat underlying anemia
Toxins	Remove toxin
Medications	Stop medication
Infection (esp. septicemia, viral infection, tuberculosis)	Treat infection
Alcohol	Refrain from alcohol consumption
Chemotherapy	Delay or decrease dose; growth factor; platelet transfusion
Increased Destruction	
Due to Antibodies	Treat condition
Idiopathic thrombocytopenic purpura	
Lupus erythematosus	
Malignant lymphoma	
Chronic lymphocytic leukemia (CLL)	Treat CLL and/or treat as ITP
Medications	Stop medication
Due to Infection	Treat infection
Bacteremia	
Postviral infection	
Sequestration of platelets in an enlarged spleen	If thrombocytopenia is severe, splenectomy may be needed
Increased Consumption	
Disseminated intravascular coagulation (DIC)	Treat underlying condition triggering DIC; administer heparin, EACA, blood products

Acquired Platelet Disorders

- Thrombocytopenia : platelets <150,000/mm^3
  - Inadequate production by bone marrow
  - Splenic sequestration
  - Consumption coagulopathy
  - Dilutional thrombocytopenia
  - Immunogenic destruction
- Platelet dysfunction
  - Myeloproliferative and myelodysplastic syndromes
  - Renal failure, liver disease, DIC, CPB
  - Drugs: NSAIDS, ASA

* DDAVP: tx platelet dysfunction due to uremia, liver disease, and patients on ASA for CABG

S & SXs

- Petechiae: pinpoint
- Ecchymoses: >petechiae
- Hematoma: the largest
- Gingival hemorrhage
- Systemic hemorrhage: epistaxis, hematemesis, hemoptysis, uremia

2. Thrombocytosis

Definition

- Thrombocytosis is the presence of high platelet (thrombocytes) counts in the blood ,which play an important role in blood clotting.
- A normal platelet count ranges from 150,000 and 450,000 per mm^3 or ($150–450 \times 10^9/L$)
- Counts over 750,000 (and especially over a million) are considered serious enough to warrant investigation and intervention.

Thrombocytosis–-

Types

1/ **Autonomous/primary/essential/**thrombocytosis (thrombocythemia)

Essential thrombocytosis (a form of myeloproliferative disease:- blood and bone marrow disease)

Other myeloproliferative disorders such as chronic myelogenous leukemia, polycythemia vera, myelofibrosis

2/ **Secondary/reactive thrombocytosis**(thrombocythemia)

- Inflammation
- Surgery (which leads to an inflammatory state)
- Hyposplenism (decreased breakdown due to decreased function of the spleen)
- Splenectomy
- Asplenia (absence of normal spleen function)
- Iron deficiency anemia or hemorrhage
- Over-medication with drugs that treat thrombocytopenia, such as eltrombopag or romiplostim, may also result in thrombocytosis. Other causes include the following:-
- Kawasaki disease, Soft tissue sarcoma, Osteosarcoma, Dermatitis (rarely)
- Inflammatory bowel disease, Rheumatoid arthritis, Nephritis, Nephrotic syndrome
- Bacterial diseases, including pneumonia, sepsis, meningitis, urinary tract infections, and septic arthritis.
- The vast majority of causes of thrombocytosis are acquired disorders, but in a few cases, they may be congenital, such as thrombocytosis due to congenital asplenia.

Thrombocytosis---

Signs and symptoms

- High platelet levels do not necessarily signal any clinical problems, and are picked up on a routine full blood count. However, it is important that a full medical history be elicited to ensure that the increased platelet count is not due to a secondary process. Often, it occurs in tandem with an inflammatory disease, as the principal stimulants of platelet production (e.g. thrombopoietin) are elevated in these --- Clinical states as part of the acute phase reaction.

• High platelet counts can occur in patients with polycythemia vera (high red blood cell counts), and is an additional risk factor for complications.

- A very small segment of patients report symptoms of erythromelalgia, a burning sensation and redness of the extremities that resolves with cooling and/or aspirin use.
- Scientific literature sometimes excludes thrombocytosis from the scope of thrombophilia by definition, but practically, by the definition of thrombophilia as an increased predisposition to thrombosis, thrombocytosis (especially primary thrombocytosis) is a potential cause of thrombophilia. Conversely, secondary thrombocytosis very rarely causes thrombotic complications.

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Thrombocytosis–-

Diagnosis

- Laboratory tests might include: full blood count, liver enzymes, renal function and erythrocyte sedimentation rate.
- If the cause for the high platelet count remains unclear, bone marrow biopsy is often undertaken, to differentiate whether the high platelet count is reactive or essential.

Thrombocytosis–-

Treatment

- Often, no treatment is required or necessary for reactive thrombocytosis. In cases of reactive thrombocytosis of more than 1,000x10^9/L, it may be considered to administer daily low dose aspirin (such as 65 mg) to minimize the risk of stroke or thrombosis.
- However, in primary thrombocytosis, if platelet counts are over 750,000 or 1,000,000, and especially if there are other risk factors for thrombosis, treatment may be needed. Selective use of aspirin at low doses is thought to be protective. Extremely high platelet counts in primary thrombocytosis can be treated with hydroxyurea (a cytoreducing agent) or anagrelide (Agrylin)

3. ITP (Idiopathic (Autoimmune)Thrombocytopenic Purpura)

– ITP is an autoimmune disease in which auto antibodies against factor IIb and IIIa bind to platelets & destruction of these coated platelets in the spleen. There is no direct lysis of the platelets

- Resolves spontaneously (most often)

Acute form : Occurs mainly during childhood often appears 1-6 wks after viral infections, self limiting within 6 months period

- More commonly found in children (2-6 yrs), HIV infection

Chronic form: seen in adults (20-50 yrs), does not follow viral infection, often linked with autoimmune diseases like SLE

• ITP in adults is diagnosed by exclusion of other conditions

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3. ITP (Idiopathic (Autoimmune)Thrombocytopenic Purpura)

Healty looking, Petechiae <20,000x 10/l platelets & purpura

- Ecchymosis, Bleeding <10,000x 10/l (Epistaxis ,gum bleeding, menorrhagia)
- Anemia, no spleenomegally

Complication: Intracranial bleeding

DX: Low platelets count, Platelets morphology = Slight elargement ,

BM = normal or increased megakaryoytes, prolonged BT

CBC: normal Hgb, HCT, WBCS, Anti platelet antibodies

Medical management

- Predinsolone 1mg/kg/d, Spleenectomy if predinsolone fails and in emergency cases
- Platelet transfusion is not effective b/se there is destruction of transfused platelets
- ITP in HIV pts:- Steroids & spleenectomy aggravate immunodeficency ,So ART drugs show good response
- VitK

4. TTP (Thrombotic Thrombocytopenic Purpura)

– TTP is characterized by microangiopathic anemia, thromboccytopenia, & markedly elevated LDH

- Uncommon syndrome/disease/ but often lethal (90%) within 3 months
- It is a disease of young adults
- In TTP, Platelets clump together inappropriately in the microcirculation & therefore insufficient platelets remain in the systemic circulation leading to bleeding disorder.

Cause: Unknown

Predisposing factors:- HIV, estrogen use, pregnancy, metastatic cancers, high dose chemotherapy

TTP –

S & Sxs

Anemic sxs, bleeding disorders(purpura & petechaie) ,& neurologic abnormalities (H/A, aphasia, confusion, in severe cases hemiparesis / hemiplegia),fever

P/E ; Acutely sick looking,hepatospleenomegaly, abdominal tenderness

Complications:- MI,RF, & stroke

Dx

- Microangiopatic hemolytic anemia with low Hgb & HCT, & marked leukocytosis and LDH
- Thrombocytopenia, elevated direct bilirubin (b/se of hemolysis)
- Coombs test negative, Normal PT,PTT, & Fibrinogen
- Renal failure :- high BUN & creatinine
- Peripheral smear: Presence of nucleated & fragmented RBCs, increased band neutrophils

RX

- Plasmapheresis with exchange transfusion with FFP
- ASA
- Corticosteroids or splenectomy only if the above measures are not effective

5. HUS (Hemolytic Uremic Syndrome)

- Microangiopathic hemolytic anemia ,thrombocytopenia,& renal failure are important features of HUS
- It is not distinct disease from **TTP**(thrombotic thrombocytopenic purpura)
- TTP is characterized by neurologic abnormality
- HUS is characterized by renal failure
- In HUS, there is no neurologic abnormality

Causes

Unknown

Familial tendency

In children follows diarrheal diseases

In adults it is precipitated by estrogen use,pregnancy,high dose steroids,& chemotherapy

DX: Elevated LDH,& normal coagulation test

RX: In children it is self limiting & Rx is conservative

In adults, treat like TTP

6. VWD (Von Willebrand's Disease)

- Most common congenital bleeding disorder
- Quantitative or qualitative abn. of vWF
- VWF is carrier protein for F VIII
- VWF is essential for platelet adhesion

Type 1: most common form

  - Partial quantitative deficiency of vWF
  - Autosomal dominant
  - Mucocutaneous bleeding
  - Hematology consult prior to surgery
  - Prolonged bleeding time, normal platelet

Hereditary Platelet Disorders

von Willebrand Disease (vWD)

Type 2: Qualitative alterations in the vWF structure & function

Type 3: Least common and most severe

- Complete absence of vWF in plasma or storage organelle
- Autosomal recessive

- Acquired vWD
  - Lymphoproliferative disease
  - Tumors
  - Autoimmune disease
  - cardiac/valvular disease
  - medications (valproic acid)
  - hypothyroidism

TABLE 64–4. Classification of von Willebrand Disease

Phenotype	Mechanism of Disease	% Cases	Genetic Transmission
Type 1 vWD	Partial (mild to moderate) quantitative deficiency of vWF and factor VIII	70%–80%	Autosomal dominant
Type 2 vWD	Qualitative abnormalities of vWF	10%–30%	Autosomal dominant
2A	Decreased platelet-dependent vWF function owing to lack of larger multimers		
2B	Increased platelet-dependent vWF function owing to lack of larger multimers		
2M	Defective platelet-dependent vWF functions not associated with multimer defects		
2N	Defective vWF binding to factor VIII		
Type 3 vWD	Severe quantitative deficiency of vWF	1%–5%	Autosomal recessive

Hereditary Platelet Disorders

von Willebrand Disease

DX:- Increase BT & PTT, normal platelets

Reduced vWF concentration

VWF gene is on chromosome #12

- Treatment: Desmopressin (DDAVP)

Inform to avoid ASA

  - Synthetic analog of vasopressin
  - ↑ both F VIII and vWF 3 - 5x in 30 mins
  - Preop prophylactic dose: 0.3 mcg/kg IV in 50 -100 ml NS infused 30-60 mins q 12-24 hrs PRN
  - Duration 8-10 hrs
  - Intranasal dose: 300 mcg – for home treatment, not for preop prophylaxis

von Willebrand Disease

- DDAVP
  - vasomotor effect: flushing, ↑HR, headache
  - SE: hyponatremia, seizures
  - not for children < 3 yrs old

  Treatment: Tranexzamic acid (antifibrinolytic)
- Unresponsive to DDAVP (15%)
  - Cryoprecipitate
  - FFP
  - Factor VIII / vWF concentrate

7. Hemophilia A & B

- X-linked recessive conditions (males only)
- IR = 1 : 10,000 males born Hemophilia A
- It affects only males

Type

A : F **VIII**:C deficiency (Classical Hemophilia)

B : F **IX** deficiency (Christmas disease)

C : F **XI** deficiency

- Unexplained bruising or bleeding in young males, usually ~ 1 yr of age
- Joint & muscle bleeding → arthropathy

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Hereditary Factor Deficiencies

Hemophilia

Mild: F VIII 5 -25%

Moderate: F VIII 1-5 %

Severe : life-threatening (CNS bleed) F VIII < 1%

DX: Prolonged PTT

Normal platelets ,BT ,vWF level, & PT

Low serum level of factor VIII

- No petechiae

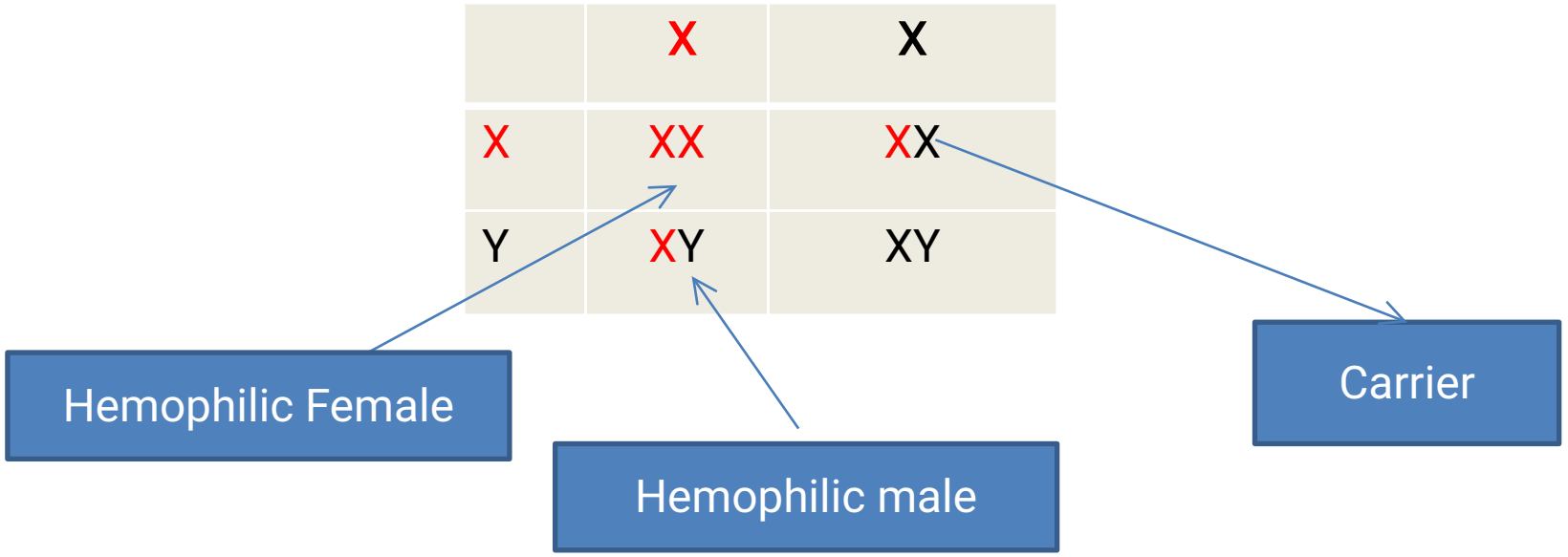
■ **Treatment**

- FFP(Fresh Frozen Plasma) or cryoprecipitate transfusion
- Factor VII precipitate (Cheap)
- Factor VIII replacement
- Desmopressin(DDAVP) – Unknown mechanism

Hemophilia––

XY X XX

- Father with Hem Carrier mother with hemophilia



Hemophilia–-

Three types

– A, factor VIII, X-linked recessive, abnormal PTT
– B, factor IX, X-linked recessive, abnormal PTT
_ C, factor XI, X-linked recessive, abnormal PTT
– von Willebrand's disease; A.D.; abnormal PTT, BT

• Hemarthrosis ⇒ arthritis and ankylosis
• Pseudo tumor of hemophilia
• Precautions: Clotting factor replacement, antifibrinolytic agent EACA (ε-aminocaproic acid)
• Hemophilia and HIV infection

Hemophilia--

- Normal bleeding time (excl. von Willebrand)
- Normal platelet number
- Normal prothrombin time
- Abnormal partial thromboplastin time
- No petechiae
- Females can have excessive bleeding (X-linked)

8. DIC

- Disseminated intravascular coagulation (DIC) is a serious blood coagulation disorder that occurs as a complication of conditions that accelerate blood clotting.
- It’s characterized by suppression of the fibrinolytic system and the development of small clots in the microcirculation, which consumes clotting factors, resulting in excessive bleeding
- The disorder can result from septicemia; obstetric complications, such as abruptio placentae and amniotic fluid embolism; cancer; blood transfusion reactions; and cirrhosis
- Tissue hypoxia and multiple organ failure can occur; the mortality rate can exceed 80%

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DIC–-

Signs and symptoms

- The main sign is abnormal bleeding, evidenced by cutaneous oozing, petechiae, ecchymoses, hematomas, GI bleeding, and bleeding from wounds and I.V. sites
- Signs of organ compromise include dyspnea, oliguria, and muscle or abdominal pain; shock can also occur

Diagnosis and treatment

- LAB :- Decreasing platelet count, Hgb and HCT
  - Increasing PT, PTT and FDP
- Medical management aims to identify and treat the underlying disorder, promote oxygenation, replace fluids and electrolytes, and provide hemodynamic support
- Treatments include clotting factor and blood replacement and I.V. heparin

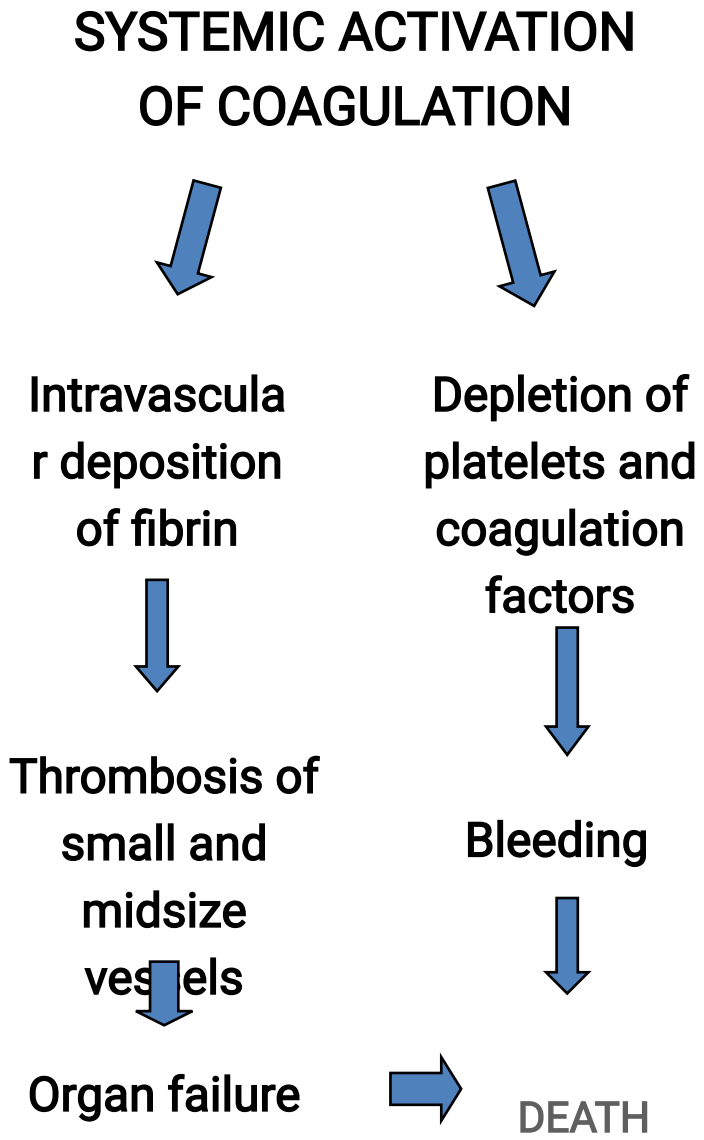
DIC––

Nursing interventions

- Early recognition of the onset of DIC improves patient outcomes, so closely monitor patients at risk, watching for signs and symptoms
- For a patient with DIC, avoid trauma to skin or wounds to minimize bleeding, protect the patient from injury, and avoid dislodging clots
- Apply pressure to puncture sites until bleeding stops
- Monitor the patient’s vital signs, and administer I.V. fluids and blood products as ordered
- Monitor the patient’s intake and output carefully and record blood loss
- Watch for signs of tissue ischemia and failure
- Provide emotional support to the patient and family

DIC––

- An acquired syndrome characterized by **systemic intravascular coagulation**
- Coagulation is always the initial event



Pathophysiology of DIC

- Activation of blood coagulation
- Suppression of physiologic anticoagulant pathways
- Impaired fibrinolysis
- Cytokines

Activation of blood coagulation

- Tissue factor/factor VIIa mediated thrombin generation via the extrinsic pathway
  - complex activates factor IX and X
- TF
  - endothelial cells
  - monocytes
  - Extravascular:
    - lung
    - kidney
    - epithelial cells

Suppression of physiologic anticoagulant pathways

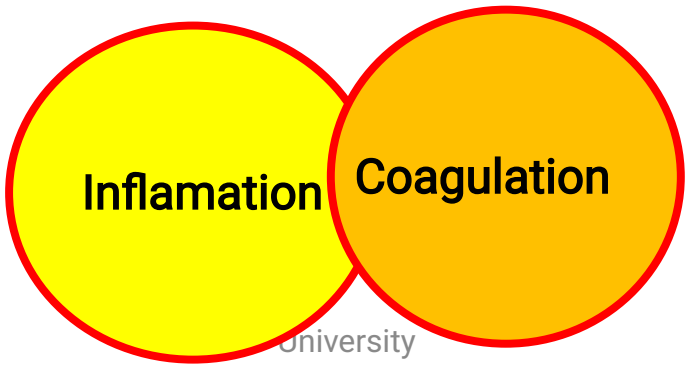
- reduced antithrombin III levels
- reduced activity of the protein C-protein S system
- Insufficient regulation of tissue factor activity by tissue factor pathway inhibitor (TFPI)
  - inhibits TF/FVIIa/Fxa complex activity

Impaired Fibrinolysis

- Relatively suppressed at time of maximal activation of coagulation due to increased plasminogen activator inhibitor type 1

Cytokines

- IL-6, and IL-1 mediates coagulation activation in DIC
- TNF-α
  - mediates dysregulation of physiologic anticoagulant pathways and fibrinolysis
  - modulates IL-6 activity
- IL-10 may modulate the activation of coagulation



University

Diagnosis of DIC

- Presence of disease associated with DIC
- Appropriate clinical setting
  - Clinical evidence of thrombosis, hemorrhage or both.
- Laboratory studies
  - No single test is accurate
  - Serial test are more helpful than single test

Table 33-6 • Laboratory Values Commonly Found in Disseminated Intravascular Coagulation (DIC)*

TEST	FUNCTION EVALUATED	NORMAL RANGE	CHANGES IN DIC
Platelet count	Platelet number	150,000–450,000/mm^3	↓
Prothrombin time (PT)	Extrinsic pathway	11–12.5 sec	↑
Partial thromboplastin time (PTT)	Intrinsic pathway	23–35 sec	↑
Thrombin time (TT)	Clot formation	8–11 sec	↑
Fibrinogen	Amount available for coagulation	170–340 mg/dL	↓
D-dimer	Local fibrinolysis	0–250 ng/mL	↑
Fibrin degradation products (FDPs)	Fibrinolysis	0–5 µg/mL	↑
Euglobulin clot lysis	Fibrinolytic activity	≥2 hours	≤1 hour

Conditions associated with DIC

- Malignancy
  - Leukemia
  - Metastatic disease
- Cardiovascular
  - Post cardiac arrest
  - Acute MI
  - Prosthetic devices
- Hypothermia/Hyperthermia
- Pulmonary
  - ARDS/RDS
  - Pulmonary embolism
- Severe acidosis
- Severe anoxia
- Collagen vascular disease
- Anaphylaxis

Conditions associated with DIC––

- Infectious/Septicemia
  - Bacterial
    - Gm - / Gm +
  - Viral
    - CMV
    - Varicella
    - Hepatitis
  - Fungal
- Intravascular hemolysis
- Acute Liver Disease
- Tissue Injury
  - trauma
  - extensive surgery
  - tissue necrosis
  - head trauma
- Obstetric
  - Amniotic fluid emboli
  - Placental abruption
  - Eclampsia
  - Missed abortion

Clinical manifestations of DIC

Ischemic Findings are earliest!

ORGAN	ISCHEMIC	HEMOR.
Skin	Pur. Fulminans	Petechiae
	Gangrene	Echymosis
	Acral cyanosis	Oozing
CNS	Delirium/Coma	Intracranial
	Infarcts	bleeding
Renal	Oliguria/Azotemia	Hematuria
	Cortical Necrosis	
Cardiovascular	Myocardial	
	Dysfxn	
Pulmonary	Dyspnea/Hypoxia	Hemorrhagic
	Infarct	lung
GI	Ulcers, Infarcts	Massive
Endocrine	Adrenal infarcts	hemorrhage.

Bleeding is the most obvious clinical finding

Clinical Manifestations of DIC

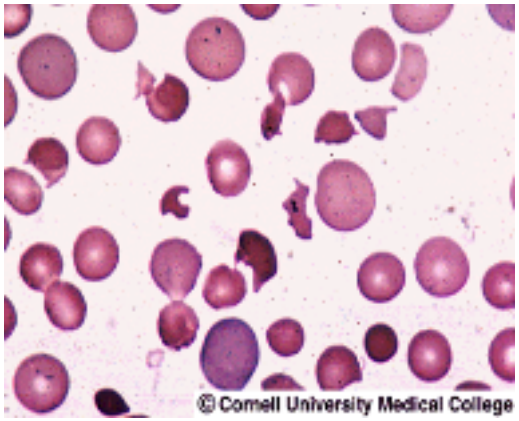








Microscopic findings in DIC



- Fragments
- Schistocytes
- Paucity of platelets

Laboratory Tests Used in DIC

- ***D-dimer****
- ***Antithrombin III****
- F. 1+2*
- Fibrinopeptide A*
- Platelet factor 4*
- ***Fibrin Degradation Prod***
- ***Platelet count***
- Protamine test
- Thrombin time
- Fibrinogen
- ***Prothrombin time***
- ***Activated PTT***
- Protamine test
- Reptilase time
- Coagulation factor levels

*Most reliable test

Laboratory diagnosis

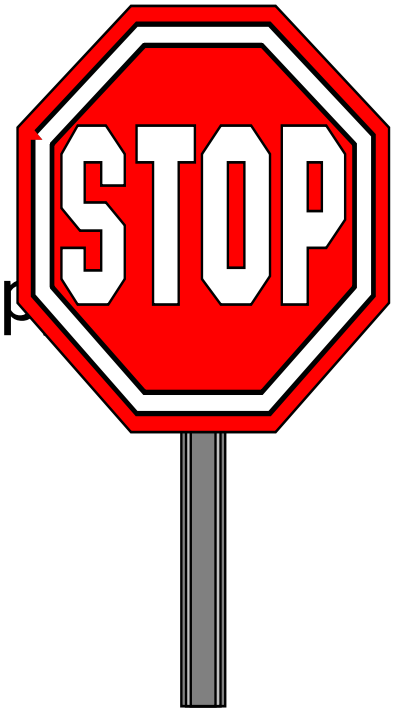
- Thrombocytopenia
  - plat count <100,000 or rapidly declining
- Prolonged clotting times (PT, APTT) 50 -60%
- Presence of Fibrin degradation products or positive D-dimer
- Low levels of coagulation inhibitors
  - Antithrombin III, protein C
- Low levels of coagulation factors
  - Factors V,VIII,X,XIII
- Fibrinogen (28%) levels not useful diagnostically

Differential diagnosis

- Severe liver failure
- Vitamin K deficiency
- Liver disease
- Thrombotic thrombocytopenic purpura
- Congenital abnormalities of fibrinogen
- HELLP syndrome

Treatment of DIC

- Stop the triggering process
  - *The only proven treatment!*
- Supportive therapy
- No specific treatments
  - Plasma and platelet substitution therap
  - Anticoagulants
  - Physiologic coagulation inhibitors



Plasma therapy

- Indications
  - Active bleeding
  - Patient requiring invasive procedures
  - Patient at high risk for bleeding complications
- *Prophylactic therapy has no proven benefit*

Cons:-

- Fresh frozen plasma(FFP):
  - provides clotting factors, fibrinogen, inhibitors, and platelets in balanced amounts.
  - Usual dose is 10-15 ml/kg

Platelet therapy

- Indications
  - Active bleeding
  - Patient requiring invasive procedures
  - Patient at high risk for bleeding complications
- Platelets
  - Approximate dose 1 unit/10kg
- Blood

Replaced as needed to maintain adequate oxygen delivery.

  - Blood loss due to bleeding
  - RBC destruction (hemolysis)

Coagulation Inhibitor Therapy

- Antithrombin III
- Protein C concentrate
- Tissue Factor Pathway Inhibitor (TFPI)
- Heparin

Antithrombin III

- The major inhibitor of the coagulation cascade
  - Levels are decreased in DIC.
  - Anticoagulant and antiinflammatory properties
- Therapeutic goal is to achieve supranormal levels of ATIII (>125-150%).
  - Experimental data indicated a beneficial effect in preventing or attenuating DIC in septic shock
    - reduced DIC scores, DIC duration, and some improvement in organ function
  - Clinical trials have shown laboratory evidence of attenuation of DIC and trends toward improved outcomes.
  - ***A clear benefit has not been established in clinical trials*** .

Protein C Concentrates

- Inhibits Factor Va, VIIa and PAI-1 in conjunction with thrombomodulin.
- Protein S is a cofactor
- Therapeutic use in DIC is experimental and is based on studies that show:
  - Patients with congenital deficiency are prone to thromboembolic disease.
  - Protein C levels are low in DIC due to sepsis.
  - Levels correlate with outcome.
  - ***Clinical trials show significantly decreased morbidity and mortality in DIC due to sepsis.***

Tissue Factor Pathway Inhibitor

- Tissue factor is expressed on endothelial cells and macrophages
- TFPI complexes with TF, Factor VIIa,and Factor Xa to inhibit generation of thrombin from prothrombin
- TF inhibition may also have antiinflammatory effects
- *Clinical studies using recombinant TFPI are promising.*

Heparin

- Use is very controversial. Data is mixed.
- May be indicated in patients with clinical evidence of fibrin deposition or significant thrombosis.
- Generally contraindicated in patients with significant bleeding and CNS insults.
- Dosing and route of administration varies.
- Requires normal levels of ATIII.

Antifibrinolytic Therapy

- Rarely indicated in DIC
  - Fibrinolysis is needed to clear thrombi from the micro circulation.
  - Use can lead to fatal disseminated thrombosis.
- May be indicated for life threatening bleeding under the following conditions:
  - bleeding has not responded to other therapies and:
  - laboratory evidence of overwhelming fibrinolysis.
  - evidence that the intravascular coagulation has ceased.
- Agents: tranexamic acid, EACA

Summary

- DIC is a syndrome characterized systemic intravascular coagulation.
- Coagulation is the initial event and the extent of intravascular thrombosis has the greatest impact on morbidity and mortality.
- Important link between inflammation and coagulation
- Morbidity and mortality remain high
- *The only proven treatment is reversal or control of the underlying cause*

9. Vitamin K deficiencies

- Vit K activates Factor X,IX,VII,II (Mnemonic: 1972---10,9,7,2)

Causes

Poor diet, Malabsorption syndromes

Broad spectrum antibiotics (Supress normal flora)

Pancreatic insufficiency, Biliary obstruction

GI obstruction, Excess Warfarin

DX : Increase PT more than PTT

Normal fibrinogen, thrombin time & platelet count

Treatment: Vitamin K (Leafy vegetable, Intestinal bacteria)

Blood transfusion

- American Association of Blood Banks Guidelines
  - Hgb <6 – Transfusion recommended
  - Hgb 6-7 – Transfusion likely recommended
  - Hgb 7-8 – **Restrictive Transfusion Strategy** for stable patients (Strong recommendation).
- Consider transfusion only if post-operative or symptomatic (chest pain, orthostatic hypotension or tachycardia unresponsive to fluid resuscitation, or congestive heart failure)
  - Hgb 8 – 10 – transfusion generally not indicated
    - Can consider special circumstances (ie active ischemia, symptomatic anemia, active bleeding, critical ill septic shock).
- Hgb >10 – transfusion not indicated

Compatibility testing

- Compatibility testing involves three separate procedures involving both donor and recipient blood.
  1. ABO & Rh blood type identification both the donor and recipient
  2. Antibody screening of donor plasma
  3. Donor/recipient cross-match
- Most of the fatal hemolytic transfusion reactions result from the transfusion of ABO incompatible blood
- Blood types are defined by the antigens present on the surface of the RBCs
  - Type A has A antigens on the surface of their red cells
  - Type B has B antigens
  - Type AB has both A and B antigens
  - Type O has neither antigen

Compatibility testing—

- The serum contains antibodies to the AB antigens that are lacking on the RBC.
  - Type A has antibodies against the B antigen
  - Type B has antibodies against the A antigen
  - Type AB has no antibodies
  - Type O has both anti-A and anti-B antibodies
- To determine which types are compatible you need to focus on which antibodies will be present in the recipient plasma.
- It is the reaction of the antibodies with donor RBC antigens that can activate the complement system and lead to intravascular hemolysis of the red cell.
  - Type O- is the universal donor
  - Type AB+ is the universal recipient

Rhesus (D) antigen

- Patients with the Rhesus (D) antigen are said to be Rh+ and those without are Rh-
- Anti-D antibodies are not constitutively present in the serum of an Rh-negative patient.
- 60-70% of Rh- patients exposed to Rh+ RBCs will develop anti-D antibodies
- There is a latency period before the antibodies are synthesized.

Plasma compatibility table

Recipient	Donor			
	O	A	B	A B
AB				✓
A		✓		✓
B			✓	✓
O	✓	✓	✓	✓

Red blood cell compatibility table

Recipient	Donor							
	O-	O+	A-	A+	B-	B+	AB-	AB+
O-	✓							
O+	✓	✓						
A-	✓		✓					
A+	✓	✓	✓	✓				
B-	✓				✓			
B+	✓	✓			✓	✓		
AB-	✓		✓		✓		✓	
AB+	✓	✓	✓	✓	✓	✓	✓	✓

Blood Products

- Whole Blood
- Red Blood Cells
- Platelets
- Fresh Frozen Plasma
- Cryoprecipitate

Complications of blood transfusion

- **Infection**
  - Hepatitis B and C, HIV, CMV,Human T-cell lymphotropic viruses
- **Immunological**
  - **Early**: anaphylaxis, acute lung injury, alloimmunization, urticaria, acute haemolysis
  - **Delayed**: delayed haemolysis, immunosuppression
- **Metabolic**
  - Hyperkalaemia, hypocalcaemia, acid–base disturbance, coagulopathy,
- **Physical**
  - Hypothermia, microemboli, air embolus, circulatory overload